Crown & Calculation: The Crownmaster’s Manual

Six-Square Chess, Tactics, Strategy, Calculation and Mastery

Crown & Calculation · Rebuilt Academy Edition

6 November 2026

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About this rebuilt edition

This is the Crownmaster's Manual, rebuilt from the game's checked instructional sources after the earlier workspace became unavailable. It is **not a byte-identical recovery of the earlier 217-page, 61,614-word file**. It contains an edited, reorganised selection of the owner's Sovereign's Grimoire, new practical guidance and 100 newly composed, exhaustively verified exercises. The actual page and word counts accompany the download. No game code, saves, rating system, market or deployment was changed.

The rules are crown-la6-v1: six files a–f and ranks 1–6, RNQKNR back ranks, no bishops, castling, double pawn steps or en passant, and mandatory Q/R/N promotion. Every diagram is White-at-bottom; its caption names the side to move. The workbook uses White to move throughout. All composed problems have zero halfmoves and no prior repetition history.

A verified mate means every legal defence was tested within the stated horizon. A finite engine preference is not a proof. Practical advice is counsel, not a promised Elo gain. Difficulty labels are editorial. DTM model results ignore repetition and draw counters; the official referee still controls the game's actual outcome. No claim of human or superhuman strength follows from a product setting.

Worked examples reveal their answers. The 100 workbook problems have a **separate answer key**. On iPhone use the EPUB in Apple Books; its text reflows. The HTML is a single offline file. Save the PDF for printing or fixed-page reading.

The safety scan

Before choosing: am I in check; what changed last move; what can be captured; which defender moved; what is the opponent's strongest reply? Repeat the scan **after your intended move**. Keep a candidate only when you have examined resistance, not merely cooperation.

PART I · The Foundations

The Board and Its Coordinates

Explanation

The board has six **files** (columns), lettered **a** to **f** from White's left to White's right, and six **ranks** (rows), numbered **1** to **6** from White's side to Black's side. A square is named by its file then its rank: `a1` is White's left-hand corner, `f6` is Black's left-hand corner (White's far right). There are 6 × 6 = **36 squares**, 18 light and 18 dark. As in the older game, the corner square on each player's right-hand side is light, so `f1` and `a6` are light and `a1` and `f6` are dark.

Each side has twelve pieces: one king, one queen, two rooks, two knights and six pawns. **There are no bishops.** White's back rank, from a1 to f1, reads **R N Q K N R**: rooks on a1 and f1, knights on b1 and e1, the queen on c1 and the king on d1. White's six pawns stand on rank 2. Black mirrors this on ranks 6 and 5: Black's queen is on c6 and king on d6, directly opposite White's.

In the standard notation for positions (FEN), the starting position is written `rnqknr/pppppp/6/6/PPPPPP/RNQKNR w - - 0 1`. Capital letters are White, small letters are Black, digits count empty squares, and the ranks are listed from 6 down to 1.

Notice the shape: only **two empty ranks** (3 and 4) separate the armies. In the older eight-by-eight game there are four. Everything in Crown & Calculation happens sooner: contact, tactics and promotion.

Illustrated example

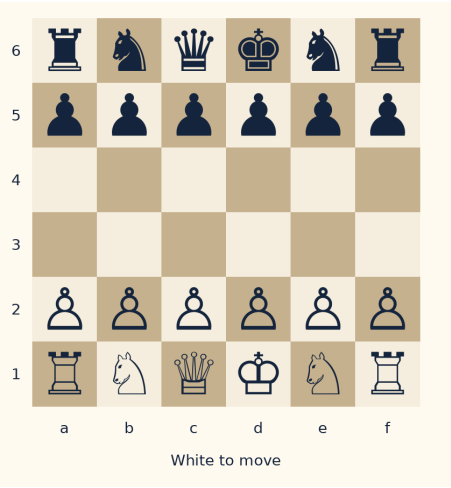


Figure 1: The starting position. Kings face each other on d1 and d6, queens on c1 and c6. Only ranks 3 and 4 lie between the pawns.

Step-by-step reasoning

How do we read a move? Take the line below: White plays a knight to d3, and Black answers with a knight to d4.

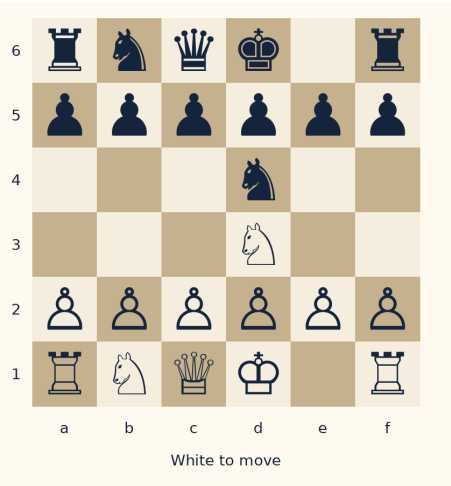


Figure 2: After 1. Nd3 Nd4. White's e1-knight jumped to d3; Black's e6-knight jumped to d4.

Verified line: Nd3 Nd4

1. `Nd3` — the letter N says "a knight", `d3` says where it lands. Only one White knight can reach d3 (the one on e1), so nothing more is needed.
2. `Nd4` — Black's e6-knight lands on d4. Again, only one knight can get there.
3. If two identical pieces could reach the same square, we add the starting file or rank, for example `Nbd3` or `R1a3`. The checker insists on the shortest correct spelling.

Common beginner mistakes

- Reading the board from the wrong side. White's a1 is at White's bottom-left; from Black's chair it is at the top-right.
- Placing the king and queen the "old" way. Here the queen is on **c1** and the king on **d1** for White, and the queen on **c6** and king on **d6** for Black.
- Looking for bishops, castling or the pawn's two-square first move. None of them exist in this game.

Advanced insight

With 36 squares instead of 64, each piece is a larger share of the whole army, and every square nearer to the enemy. A knight on d3 already touches squares on Black's half of the board (c5 and e5). In this game "the opening" and "the middlegame" overlap from the first move.

Worked example

Question: In the starting position, which White pieces can move at all?

Answer: Only the two knights and the six pawns. The rooks, queen and king are hemmed in by their own pieces. Each pawn can step forward one square (a3, b3, c3, d3, e3, f3), the b1-knight can go to a3 or c3, and the e1-knight to d3 or f3. That is 6 + 2 + 2 = **10 legal first moves**, as the rules engine confirms.

Summary

Files a–f, ranks 1–6, 36 squares. White: R N Q K N R on rank 1 with the king on d1; Black mirrors on rank 6 with the king on d6. Two empty ranks between the pawns. No bishops, no castling, no two-square pawn step.

The Pieces I: Rook, Queen and King

Explanation

- The **rook** moves any number of squares along a rank or a file, but cannot jump over pieces.
- The **queen** moves like a rook *or* diagonally any number of squares; it cannot jump either.
- The **king** moves one square in any direction. It may never move to a square attacked by an enemy piece.

A piece **controls** (or **attacks**) every square it could move to, and it can capture an enemy piece standing on such a square.

On an empty 6 × 6 board the numbers are small and exact:

Piece	Corner	Other edge square	Second ring (b2–e2–e5–b5)	Centre (c3, d3, c4, d4)
Rook	10	10	10	10
Queen	15	15	17	19
King	3	5	8	8

A rook always reaches **10 squares** on an empty board: 5 along its file and 5 along its rank, wherever it stands. The queen adds the diagonals: 5 on any edge square, 7 on the second ring, 9 in the centre. (These counts were made with the rules engine’s own move generator.)

Illustrated example

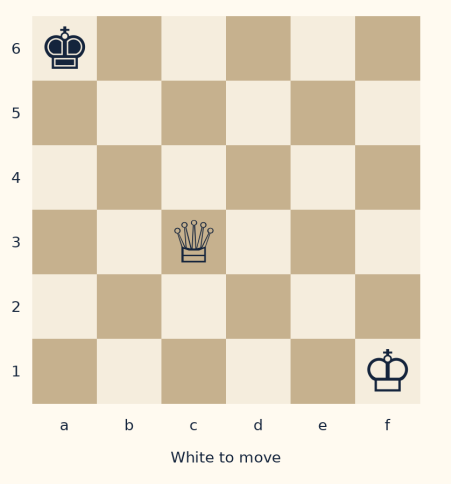


Figure 3: The queen on c3 has 19 moves: 10 along the c-file and 3rd rank, and 9 along the two diagonals.

Step-by-step reasoning

Count the queen’s moves in the diagram one direction at a time: 1. Up the c-file: c4, c5, c6 (3). Down: c2, c1 (2). 2. Along rank 3: d3, e3, f3 (3) and b3, a3 (2). Rook-like total: 10. 3. Diagonal up-right: d4, e5, f6 (3). Down-left: b2, a1 (2). 4. Diagonal up-left: b4, a5 (2). Down-right: d2, e1 (2). Diagonal total: 9. 5. 10 + 9 = **19**. Several of these moves give check to the Black king on a6 (for example `Qa3+`, `Qc6+` and `Qa5+`).

Common beginner mistakes

- Letting a rook or queen “jump”. If a piece is in the way, the line stops there (on it, if it is an enemy piece you can capture; before it, if it is your own).
- Moving the king next to the enemy king. Two kings may never stand on neighbouring squares, because each would be attacking the other.

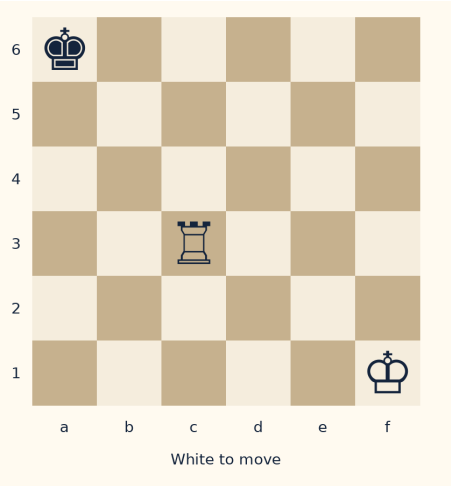


Figure 4: A rook on c3: 10 moves, exactly as on any other square of an empty board.

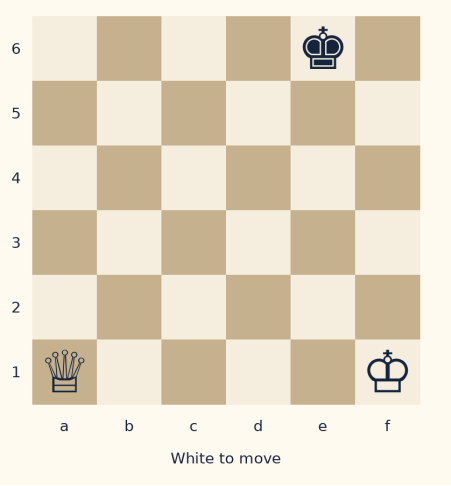


Figure 5: A queen in the corner a1 reaches 14 squares here (15 if her own king were not standing on f1).

- Forgetting that the king cannot step onto a square that is attacked, even by a piece that seems far away.

Advanced insight

On the big board a queen in the centre controls 27 squares, 42% of 64. On this board a central queen controls 19 of 36, more than half. A queen can cross the whole board in one move, so **no square is far from her.** This is why queen checks and queen forks (Book II) are so dangerous here. **Guidance:** never leave your queen where an enemy rook or knight can attack it with tempo.

The engine that runs the Academy measures piece values by statistical fitting (`crown-eval-4`). Its middlegame values, in pawns, are roughly: pawn 0.9, rook 4.1, knight 4.4, queen 9.3; in the endgame: pawn 0.9, knight 2.6, rook 4.6, queen 7.8. **Guidance:** these are engine estimates, not laws. The surprise is the knight: in a crowded middlegame on this small board the engine values a knight about as highly as a rook.

Worked example

In the position below, how many moves does the White rook on c2 have?

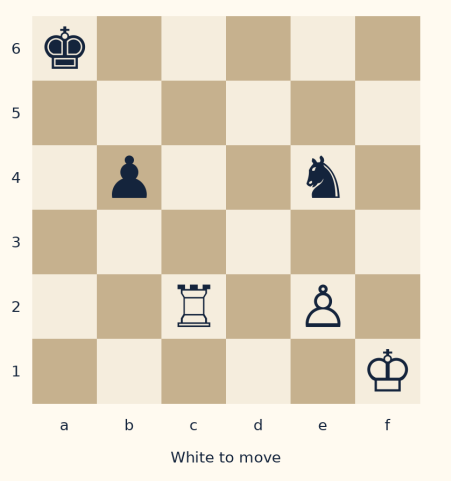


Figure 6: Worked example — count the rook's moves.

Solution. Along rank 2 to the right: d2, then e2 is White's own pawn, so the rook stops (1). To the left: b2, a2 (2; `Ra2+` is even check). Up the c-file: c3, c4, c5, c6 (4). Down: c1 (1). Total **8**. The Black pawn on b4 and knight on e4 are not on the rook's lines, so it cannot capture them.

Summary

Rook: 10 squares anywhere on an empty board. Queen: 15 in a corner, 19 in the centre. King: 3, 5 or 8. Lines are blocked by pieces in the way. A king can never move into check or next to the other king.

The Pieces II: Knight and Pawn

Explanation

The knight moves in an "L": two squares in one direction and one square to the side. It is the only piece that **jumps** over others. A knight always lands on a square of the *opposite colour* from where it started.

The knight's reach depends strongly on where it stands. On an empty board:

```
rank 6:  2 3 4 4 3 2
rank 5:  3 4 6 6 4 3
```

```
rank 4:  4 6 8 8 6 4
rank 3:  4 6 8 8 6 4
rank 2:  3 4 6 6 4 3
rank 1:  2 3 4 4 3 2
         a b c d e f
```

Only the four centre squares (c3, d3, c4, d4) give a knight all 8 moves, which is 4 of 36 squares, compared with 16 of 64 on the big board. Averaged over the board, a knight here has about 4.4 moves.

The pawn moves straight forward **one square**, never backwards, and only onto an empty square. **There is no two-square first move and no en passant.** A pawn captures one square diagonally forward. White pawns move up the board (towards rank 6), Black pawns move down (towards rank 1).

When a pawn reaches the last rank (rank 6 for White, rank 1 for Black) it **must** be promoted, in the same move, to a **queen, rook or knight** of its own colour. There is no bishop to choose and no option to stay a pawn.

Illustrated example

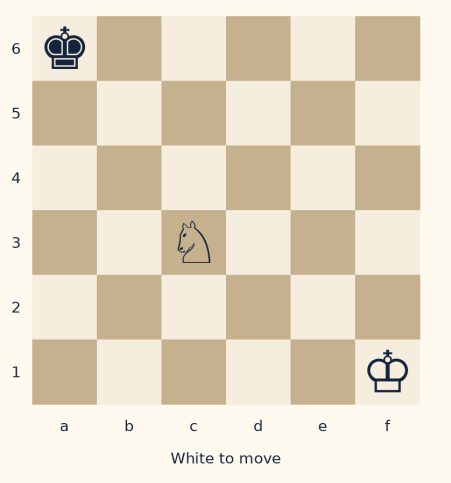


Figure 7: A knight on c3 has 8 moves: b1, d1, a2, e2, a4, e4, b5, d5. All are light squares; c3 is dark.

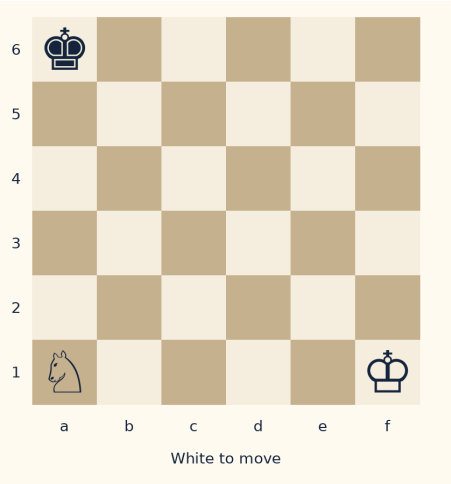


Figure 8: The same knight in the corner a1 has only 2 moves: b3 and c2.

Step-by-step reasoning

Here are pawns in contact. White to move.

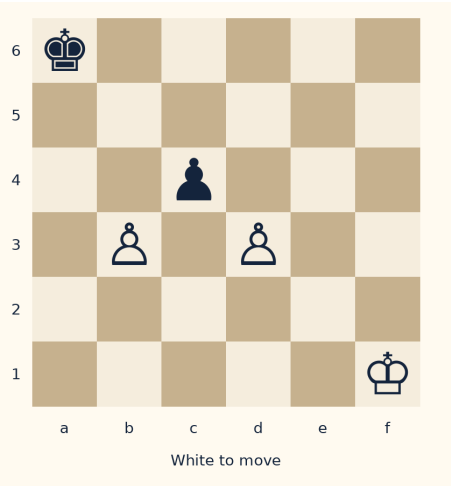


Figure 9: White pawns on b3 and d3, a Black pawn on c4.

1. The b3-pawn may step to b4 (empty) or capture diagonally on c4: `bxc4`.
2. The d3-pawn may step to d4 or capture `dxc4`.
3. The Black pawn on c4 also attacks b3 and d3 (diagonally *downwards*): if White does nothing, Black can capture.
4. Neither White pawn may move two squares, and neither may capture straight ahead. Together with the king’s three moves, White has exactly 7 legal moves.

Now a promotion. White’s pawn on d5 must promote when it moves. Which piece?

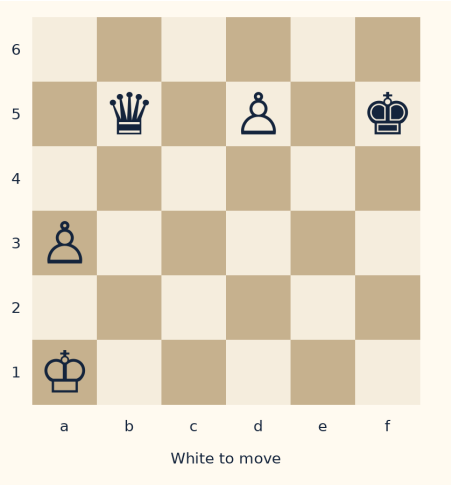


Figure 10: The engine’s choice is to promote to a knight with check, forking the king on f5 and the queen on b5.

Verified line: d6=N+

5. A new queen on d6 would only give queen against queen; the engine rates it far lower.
6. A new knight on d6 checks the king on f5 *and* attacks the queen on b5. After the king moves, `Nxb5` wins the queen. White is left with knight and pawn against a bare king, which the engine judges winning (a lone knight could not mate, so the a3-pawn matters).

Common beginner mistakes

- Trying to move a pawn two squares on its first move. Illegal here.
- Trying to capture *en passant*. That rule does not exist here.
- Forgetting that promotion is compulsory and that a bishop cannot be chosen.
- Moving a knight and landing on a square of the same colour. If you did, the move was wrong.

Advanced insight

Because knights always change colour, **a knight can never attack two squares of different colours at once.** The White king (d1, light) and queen (c1, dark) stand on different colours, so no single knight can attack both at the start. But a knight on e3 attacks d1 *and* f1 (both light), and a knight on b3 attacks a1 *and* c1 (both dark). Book II builds on this.

The longest knight journey on the 6 × 6 board takes 4 moves. For example a1 to f6 goes a1–c2–e3–d5–f6. Curiously, reaching the diagonal neighbour b2 from a1 *also* needs 4 moves.

Worked example

How many moves does the White knight on d3 have, and which of them are special?

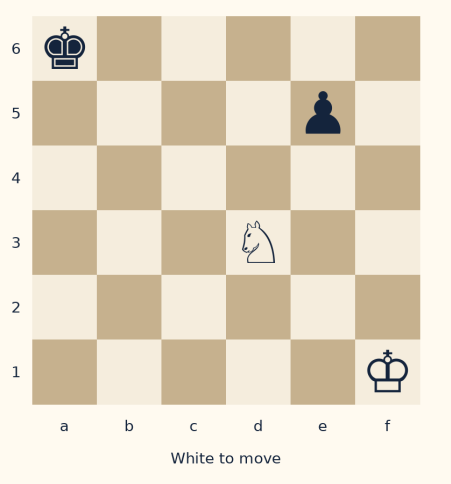


Figure 11: Worked example — the knight on d3.

Solution. d3 is a centre square, so 8: b2, b4, c1, c5, e1, e5, f2, f4. On e5 stands a Black pawn, so that move is the capture `Nxe5`. Two moves give check to the king on a6: `Nb4+` and `Nc5+`.

Summary

Knights jump in an L, change square colour every move, and are strongest on the four centre squares (8 moves) and weakest in the corners (2). Pawns step one square, capture diagonally, have no double step and no en passant, and must promote to Q, R or N on the last rank.

Capturing and Piece Safety

Explanation

You capture by moving onto a square occupied by an enemy piece, which is removed from the board. A piece is: - **attacked** if an enemy piece could capture it next move; - **defended** (protected) if one of your own pieces could recapture on its square; - **hanging** if it is attacked and not defended. A hanging piece can be taken for free.

Before capturing on a square, **count**: how many of your pieces attack it, and how many enemy pieces defend it? Then imagine the exchanges in order, each side recapturing with its *cheapest* piece first. A useful scale of values (**Guidance**, see Chapter 2 for the engine's figures): pawn 1, knight 3 to 4, rook 5, queen 9. The king can never be "traded"; it may only capture a piece that is not defended.

Illustrated example

The Black rook on a6 is attacked by the queen along the diagonal d3–c4–b5–a6. Is it defended? The pawn on a5 cannot defend it, because pawns capture forwards (for Black, downwards), and

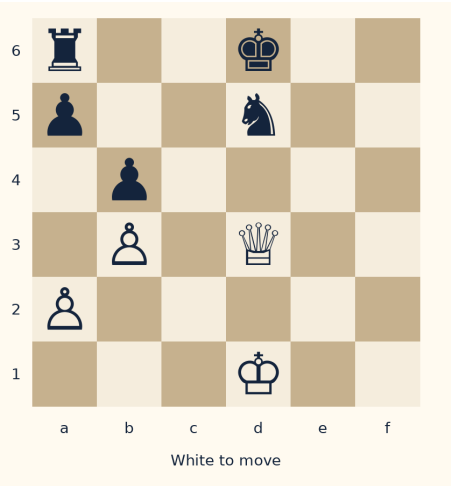


Figure 12: Academy puzzle pz-cfbmy9. White to move. Which Black piece is hanging?

the knight on d5 does not reach a6. The rook is **hanging**: `Qxa6+` wins a whole rook, with check along the 6th rank.

Step-by-step reasoning

Counting matters most when a piece *is* defended. Here White can capture the Black knight on d4 in two ways.

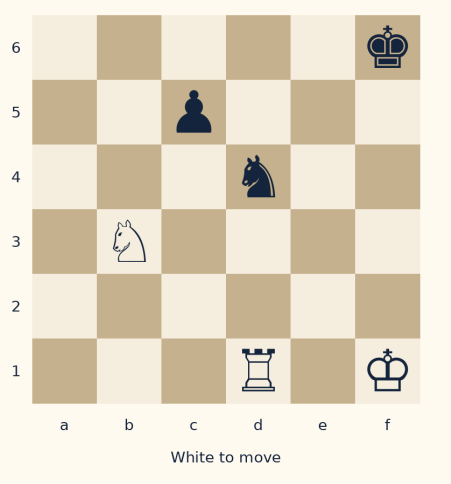


Figure 13: The knight on d4 is attacked twice (Nb3, Rd1) and defended once (pawn c5).

1. Attackers: White's knight on b3 and rook on d1. Defenders: Black's pawn on c5 (it captures diagonally down onto d4).
2. Two attackers against one defender: White will end up owning d4. But *which piece captures first* decides the profit.
3. Capture first with the cheaper piece, the knight:

Verified line: Nxd4 cxd4 Rxd4

4. The wrong order loses material:

Verified line: Rxd4 cxd4 Nxd4

Common beginner mistakes

- Seeing only the piece you want to take and not the piece that will take back.
- Capturing first with the queen. If the target is defended, the queen is usually lost for less.

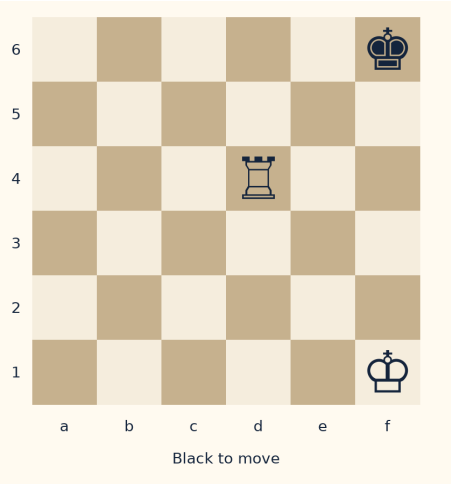


Figure 14: Correct order: knight takes, pawn takes back, rook takes. White gave a knight and won a knight and a pawn.

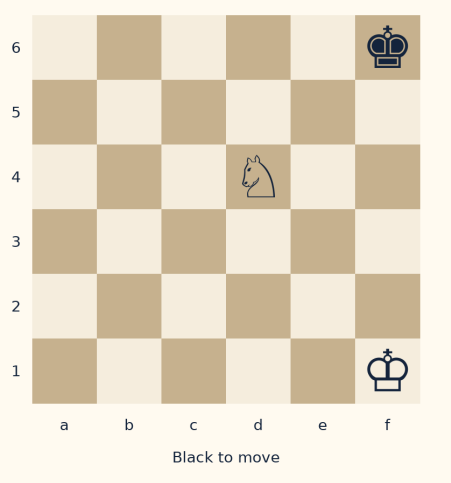


Figure 15: Wrong order: White gave a rook (5) and won only a knight and a pawn (about 4).

- Forgetting that your own king can be a defender of nearby squares, and that the enemy king can capture an undefended piece next to it.
- Leaving your own piece hanging after making a capture. Always ask: "after my move, what is attacked?"

Advanced insight

On a 6 × 6 board pieces stand close together, so almost every piece is attacked or defended by several others at once. A capture often opens a line for *another* piece, which is the root of the tactics in Book II. **Guidance:** after every enemy move, ask two questions: *What does it attack now? What did it stop defending?*

Worked example

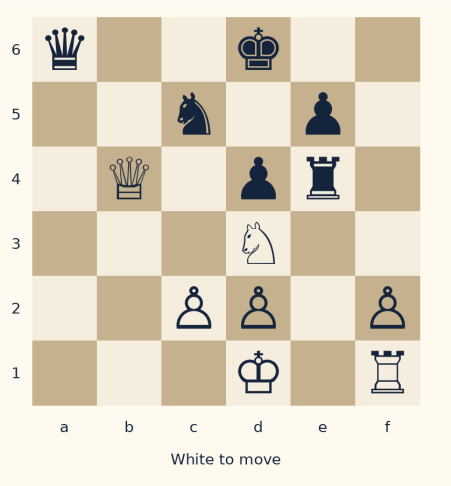


Figure 16: Academy puzzle pz-kqfffd. White to move and win material.

Solution. The Black knight on c5 is attacked by the White queen on b4. Its only possible defender is the Black king on d6, which stands next to c5. If White plays `Qxc5+`, the queen gives check, and can the king take back? No: the queen on c5 is defended by the White knight on d3. Count: one attacker (the queen) against one "defender" that cannot legally recapture, because a king may never capture a defended piece. **`Qxc5+` wins a knight for nothing.**

Summary

Before you capture, count attackers and defenders, and capture with the cheapest piece first. A hanging piece is a gift; taking it is the first winning pattern every Apprentice learns. Never leave your own pieces hanging.

Check and Escaping Check

Explanation

A king is in **check** when an enemy piece attacks its square. You must get out of check *at once*; no other move is legal. There are three ways: 1. **Move** the king to a square that is not attacked. 2. **Block**: put one of your pieces between the king and a checking rook or queen. (A knight or pawn check cannot be blocked: there is no line.) 3. **Capture** the checking piece, with any piece, including the king if the checker is undefended.

If none of these is possible, the king is **checkmated** and the game is over (Chapter 6).

Illustrated example

Verified line: Rd5+

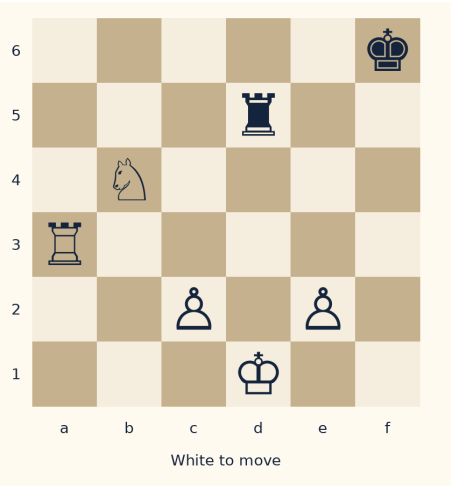


Figure 17: Black has just played Rd5+. White is in check from the rook on d5, and all three escapes are available.

White’s legal moves are exactly five: - **Move:** `Kc1` or `Ke1`. (Not `Kd2`: that square is still on the d-file, attacked by the rook. Not `Kc2` or `Ke2`: White’s own pawns stand there.) - **Block:** `Rd3` (the a3-rook steps into the line) or `Nd3` (the knight jumps into the line). - **Capture:** `Nxd5+`, which removes the rook and gives check to the king on f6.

Step-by-step reasoning

How do you find every escape without missing one? Use the same order each time. 1. **Who gives check, and from where?** The rook on d5, along the d-file. 2. **Can I capture it?** List every piece that attacks d5: the knight on b4 does. `Nxd5+`. 3. **Can I block?** The squares between d5 and d1 are d4, d3, d2. Which pieces can reach them? The rook (a3 to d3) and the knight (b4 to d3). 4. **Where can the king go?** Check each neighbouring square: c1 (safe), c2 and e2 (own pawns), d2 (on the d-file), e1 (safe). 5. **Choose the best.** Here capturing a free rook is best. In general, check all three before choosing.

Common beginner mistakes

- Moving the king *along* the line of the check (here `Kd2`). The rook still attacks that square.
- Trying to block a knight check. It is impossible.
- Making a “good” move elsewhere and ignoring the check. Illegal.
- Capturing the checker with the king when it is defended.

Advanced insight

On the crowded 6 × 6 board the king often has very few flight squares. In the starting position White’s king has none at all: c1 holds the queen, e1 a knight, and c2, d2, e2 pawns. **A check in the opening is therefore dangerous by itself**, because the only defences are blocking or capturing. **Guidance:** early in the game, keep a piece ready to block on the d-file and the a4–d1 and d1–f3 diagonals, and do not open your king’s lines carelessly.

Worked example

Verified line: Qd2+

Solution. The queen attacks c1, e1, c2 and e2 as well as d1, so every flight square is covered. Nothing can block a piece that is adjacent. But the queen is **undefended**, so the king may capture it: `Kxd2` is White’s only legal move, and it wins the queen.

Now give Black one more piece:

Verified line: Qd2#

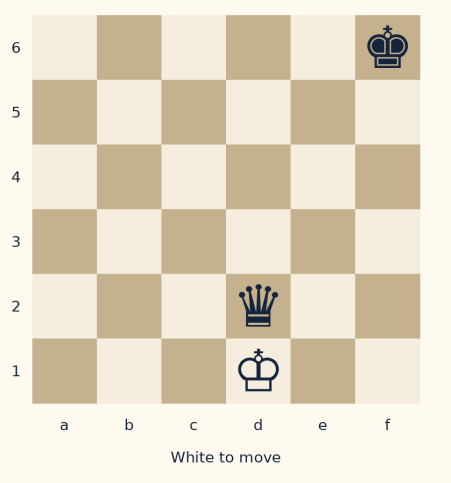


Figure 18: Black’s queen has just checked from d2, right next to White’s king.

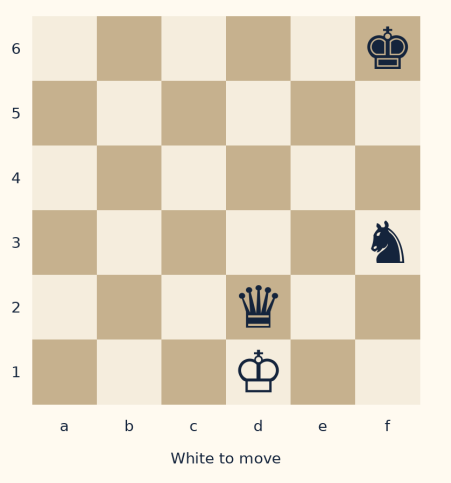


Figure 19: The same queen check, but the knight on f3 now defends d2. The king cannot capture, cannot flee and cannot block. Checkmate.

Summary

Check must be answered at once: move the king (not along the line!), block a line check, or capture the checker. Always look at all three before you choose.

Checkmate: King and Queen, King and Rook

Explanation

Checkmate is check with no escape: the king cannot move, the check cannot be blocked, and the checker cannot be captured. Checkmate ends the game at once and wins it, even on a move that would otherwise complete a repetition or the move-count limit.

A lone queen or rook cannot mate by itself: **the attacking king must help.** On the 6 × 6 board the process is short, because the enemy king never stands more than two squares from an edge. A typical mate has the defending king on the edge, the attacking king two squares in front of it (in "opposition", one square between them), and the queen or rook checking along the edge.

Illustrated example

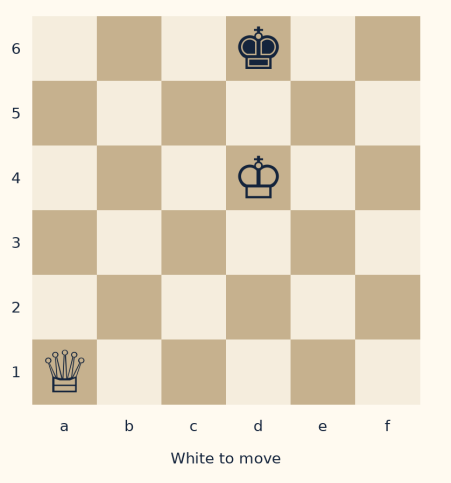


Figure 20: King and queen mate. The kings face each other on d6 and d4; the queen checks along the 6th rank.

Verified line: Qa6#

After `Qa6#` the Black king cannot go to c5, d5 or e5 (all next to White's king) nor to c6 or e6 (on the queen's rank). Nothing can capture or block. Checkmate.

Verified line: Rd6#

Step-by-step reasoning: the queen's box

The queen should stand a **knight's move away** from the enemy king. That leaves the king a small "box" without ever checking it pointlessly. Then bring your own king up.

Verified line: Qd5

1. `Qd5` takes away rank 5 and the d-file. The Black king on b4 is confined to a3, a4, b3 and the squares around them.
2. Black's king shuffles (for example `Ka4`). White's king comes closer: `Kc3`.
3. Now the box is tiny, and the queen delivers mate on the edge.

The position after `Qd5 Ka4` shows the second step:

Verified line: Kc3

And after `Kc3 Ka3` there are two different mates:

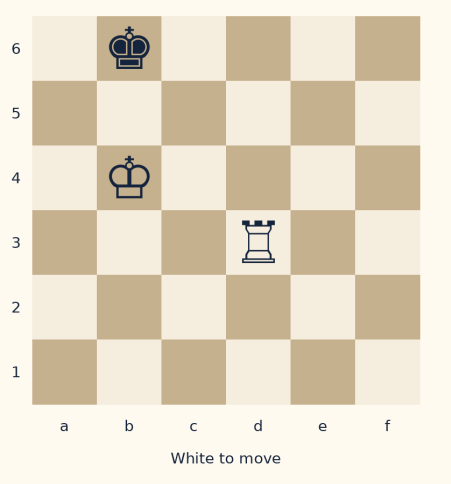


Figure 21: King and rook mate, the same picture: the kings face each other on b6 and b4, and the rook checks along the edge.

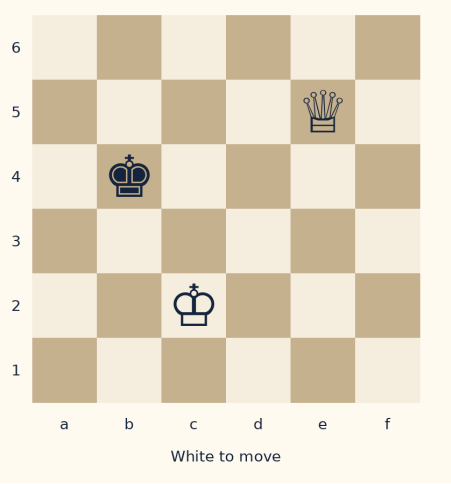


Figure 22: White mates in 3, and Qd5 (a knight’s move from b4) is the only first move that does it. Proven by trying every defence.

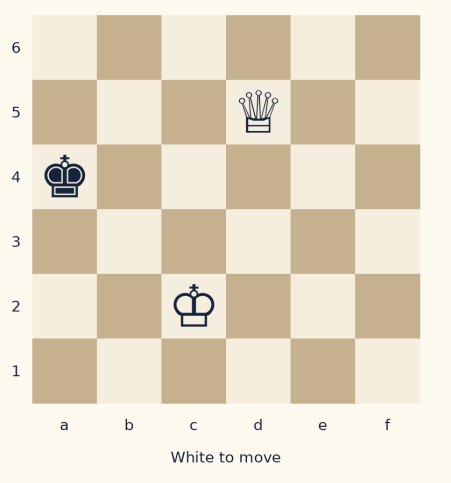


Figure 23: Mate in 2. The quiet king move Kc3 is the only move that forces it.

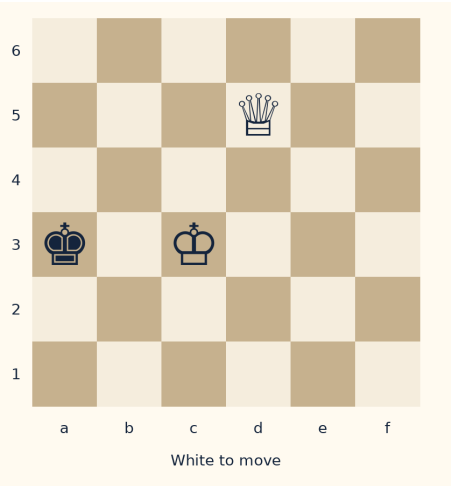


Figure 24: Mate in 1. Both Qa5# and Qb3# work here.

Verified line: Qa5#

Here is a whole king-and-queen mate from a central start, as the Academy engine plays it. (This is one good line; the defence is the engine's choice, not a proof that nothing lasts longer.)

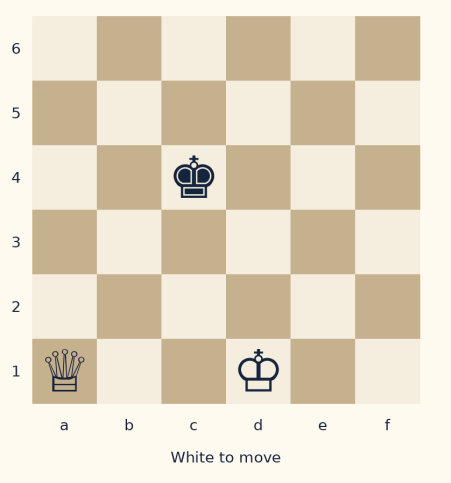


Figure 25: From the centre to mate in eleven half-moves: king up, queen boxes, king closes in, mate on the a-file.

Verified line: Kc2 Kb4 Qc3+ Ka4 Qe5 Kb4 Qd5 Ka4 Kc3 Ka3 Qa5#

Step-by-step reasoning: the rook's method

The rook cuts the king off along a rank or file; the attacking king walks up to face the enemy king; then the rook checks along the edge. When the kings are not yet facing each other, a **waiting move** with the rook hands the move to the defender.

Verified line: Rd3

After `Rd3` the Black king on c6 cannot go to d5 or d6 (the rook's file) or to b5 or c5 (next to White's king on b4). It must go to b6, and `Rd6#` follows.

Verified line: Kc2 Kd4 Ra4+ Kc5 Kc3 Kd5 Rd4+ Kc5 Rd3 Kb5 Rd5+ Kc6 Kc4 Kb6 Kb4 Kc6 Rd3 Kb6 Rd6#

Common beginner mistakes

- Checking again and again with the queen, chasing the king around. Checks are not progress; **shrinking the box** is.

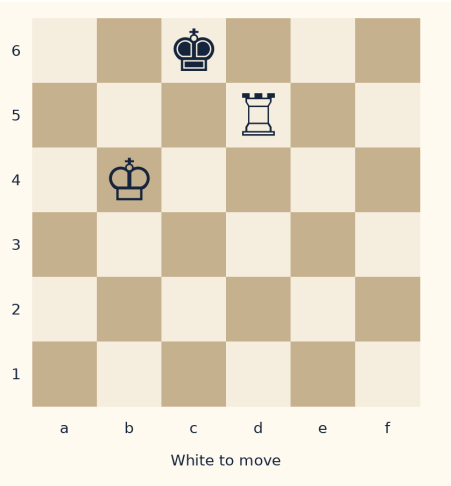


Figure 26: Mate in 2. The rook confines the king to the 6th rank; a waiting move along the d-file (Rd1, Rd2, Rd3 or Rd4) forces the king to b6, where Rd6 is mate.

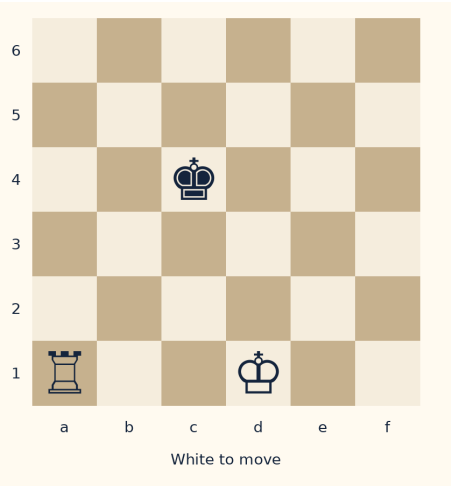


Figure 27: A whole king-and-rook mate as the engine plays it: cut off, approach, check the king back, wait, and mate on the 6th rank.

- Forgetting the king. Without the attacking king's help no mate is possible.
- Putting the queen next to the enemy king without protection: it is simply captured.
- Taking away the king's last square without giving check: that is **stalemate**, a draw (Chapter 7).

Advanced insight

The 6 × 6 board helps the attacker. From any starting position the engine finds king-and-rook mates in a handful of moves, because the defending king is never more than two steps from an edge. Even so, the **50-move rule exists here in the form of 100 halfmoves (claimable) and 150 halfmoves (automatic)**: a careless attacker can still run out of moves. **Guidance:** practise these mates until you can do them without thinking.

Worked example

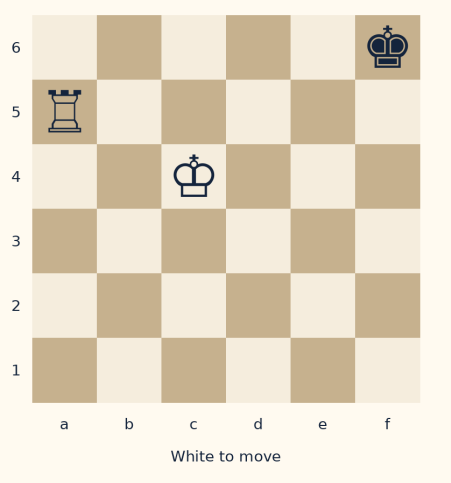


Figure 28: White mates in 3, and only the king move Kd5 achieves it. Proven exhaustively; the line shows the forced play.

Verified line: Kd5 Kf5 Ra4 Kf6 Rf4#

Reasoning. 1. `Kd5`. The king steps towards the Black king. Notice that it now stands *on the rook's rank*, so the rook no longer covers e5 and f5, but the king itself covers e5 and e6. Black's only legal move is `Kf5`. 2. `Ra4`. The rook drops one rank and seals the 4th rank. On f5 the Black king has only one square left: f6 (e4 and f4 are on the rook's rank, e5 and e6 are next to White's king). 3. `Kf6` is forced, and `Rf4#` checks up the f-file. f5 is on the file; e5, e6 and e… every square on the e-file next to the Black king is covered by White's king on d5. There is no g-file to run to. Checkmate.

Notice how the **edge of the board** did half the work: on the 6 × 6 board the f-file *is* the edge.

Summary

Checkmate needs teamwork: the queen or rook builds a box or cuts off the king; your king walks up to face the enemy king; then the major piece mates on the edge. Do not chase with checks. Watch out for stalemate.

Stalemate and the Other Draws

Explanation

If the player to move has **no legal move** and is **not in check**, the game ends at once in a draw by **stalemate**. The side that was "winning" gets only half a point. This is the commonest way Apprentices throw away won endings.

Crown & Calculation (adjudication policy `crown-adj-1`) also draws a game: - **automatically**, when only the two kings remain (bare kings), after the same position occurs for the fifth time, or after 150 halfmoves (75 moves each) with no capture and no pawn move; - **on a claim** by the player to move, after the third occurrence of the same position or after 100 halfmoves without a capture or pawn move; - **by agreement**, but only after each side has made at least ten moves.

Note one Crown difference: only *bare kings* is an automatic "dead position". King and knight against king, for example, is **not** drawn automatically; play continues until a repetition or move-count draw (or agreement). And checkmate on the final move always beats every draw rule.

Illustrated example

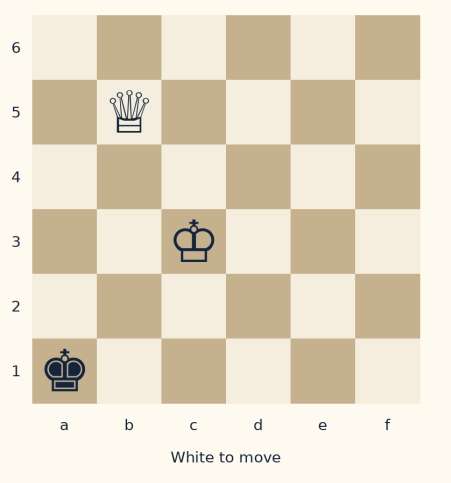


Figure 29: The trap. Qb3 looks like it tightens the net, but the Black king on a1 has no legal move and is not in check: stalemate, a draw.

Verified line: Qb3

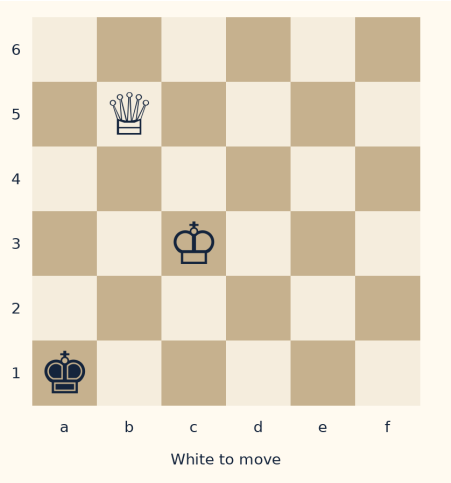


Figure 30: The same position. Qb2# is the only mate: the queen stands next to the king, protected by White's king on c3.

Verified line: Qb2#

Step-by-step reasoning

Before every move in a winning ending, ask: **"After my move, does my opponent have a legal move?"** 1. List the defending king's squares: in the diagram a2, b1 and b2. 2. After `Qb3`: a2 is covered by the queen (diagonal), b1 and b2 by the queen (b-file) and b2 also by White's king. Zero squares. 3. Is the king in check after `Qb3`? No: b3 does not attack a1. Zero moves and no

check means **stalemate**. 4. After `Qb2#`: the same three squares are covered, *and* the king is in check. That is mate.

Common beginner mistakes

- Grabbing the last enemy pawn or piece and leaving the king with no moves.
- Making a "quiet" queen move a knight's distance away when the enemy king is already in the corner. In the corner, a knight's-move box means stalemate.
- Thinking a big material lead makes the result safe. Stalemate ignores material.

Advanced insight

When **you** are losing, look for stalemate. If your king has no legal moves, every one of your other pieces becomes a "desperado": you would be glad to lose it. A rook that checks again and again, and is stalemate if captured, can save a hopeless game.

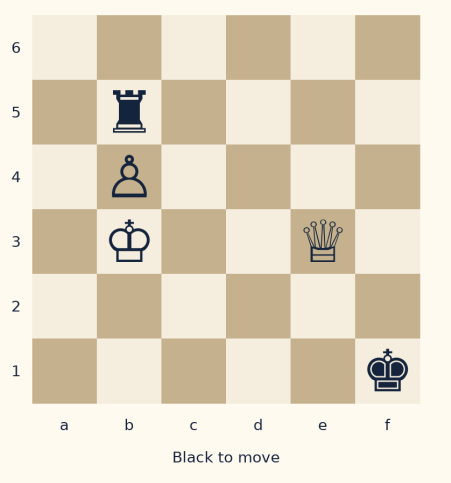


Figure 31: Black is a queen down, but the king on f1 has no legal move. The engine's choice: Rxb4+, offering the rook.

Verified line: Rxb4+

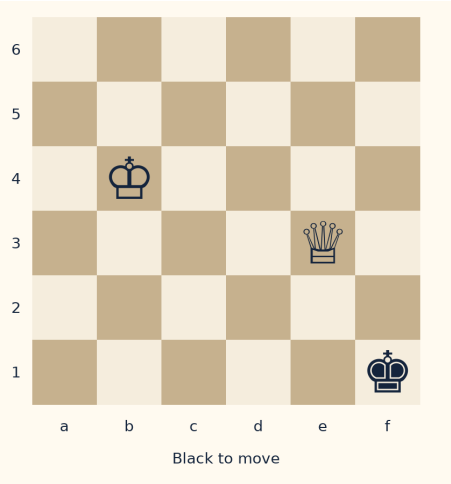


Figure 32: If White's king takes the rook, Black has no legal move and is not in check: stalemate. If the king walks away instead, the rook keeps checking.

Verified line: Rxb4+ Kxb4

Worked example

Why is this position a draw, with Black to move?

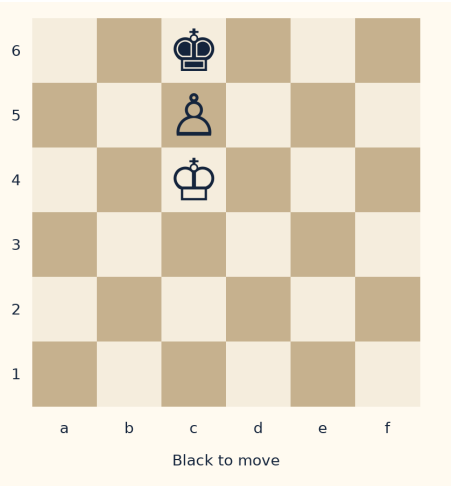


Figure 33: Black to move has no legal move. Stalemate.

Solution. The Black king on c6 could go to b6 or d6, but both are attacked by the White pawn on c5 (a White pawn attacks diagonally *forwards*, here b6 and d6). b5 and d5 are next to White's king on c4. c5 itself is occupied by the pawn, which is defended by the king. Black is not in check. Stalemate. **Guidance:** in king-and-pawn endings, a defending king that blocks the square in front of the pawn very often reaches a stalemate like this one.

Summary

Stalemate (no legal move, no check) is a draw. When winning, always check that the opponent will have a legal move. When losing, look for stalemate tricks. Bare kings, fivefold repetition and 150 halfmoves draw automatically; threefold repetition and 100 halfmoves may be claimed.

Unequal Scales: Material Imbalances and Trading Decisions

Explanation

The engine's price list (method D). Pawn 89 / 89; knight **441** / 262; rook 407 / **463**; queen 934 / 775 (middlegame / endgame). Two surprises:

1. In the middlegame the engine values a **knight slightly above a rook**. On a 36-square board a central knight reaches 8 squares — more than a fifth of the board — and rooks are hemmed in by pawns until files open.
2. In the endgame the rook is worth almost **twice** the knight. Open board, few pawns: the rook's range wins.

Engine judgement (method C, depth 12, tempo-neutral). Removing material from the starting position:

Imbalance (first side has it)	Value for the first side
Queen vs rook + knight	+134
Queen vs two rooks	+118
Queen vs two knights	+183
Queen vs rook + knight + pawn	+17 (≈ equal)
Rook vs knight	+24
Rook vs knight + pawn	−67
Knight vs one pawn	+222
Knight vs two pawns	+102
Knight vs three pawns	−36 (≈ equal)
Rook vs two pawns	+125
Rook vs three pawns	−24 (≈ equal)

So in the opening phase: **queen ≈ rook + knight + pawn**, **knight ≈ rook** (the rook is slightly better), **knight ≈ three pawns**, **rook ≈ three pawns**.

Evidence (method A, balanced games). How did these imbalances actually score when they lasted at least 8 plies (score for the side with the first-named material, pawns equal unless stated)?

Imbalance	Games	Score
Rook vs knight	509	67.7%
Rook vs knight + pawn	319	45.8%
Queen vs rook + knight	60	53.3%
Queen vs rook + knight, all pawn counts	141	56.7%
Queen vs two rooks	41	43.9%
Queen vs two knights	39	79.5%
Knight vs one pawn	913	75.2%
Knight vs two pawns	354	58.2%
Knight vs three pawns	45	43.3%
Rook vs two pawns	118	76.3%

Where the experiment and the games agree, and where they do not: - **Rook against knight is worth about one pawn**, no more: give the rook side one pawn fewer and the score falls below 50% in the games (45.8%), and below zero in the experiment (−67). - **Queen against rook and knight** favours the queen a little in both (+134 in the opening experiment, 53–57% in games), and adding one pawn to the rook-and-knight side balances it. - **Queen against two rooks** is the open question: the opening experiment prefers the queen (+118), but in the 41 games that reached it the rooks scored 56.1%. Open files, which arrive later in the game, favour the rooks. Treat it as roughly balanced and judge by the files and by the safety of the kings.

Trading decisions: which ending? Evidence (method A). For the four "pure" endings (each side has exactly the same single piece, or only pawns), first reached in the balanced games:

Ending	Games	Drawn when pawns are equal	Side with more pawns: wins / draws
Pawn ending	286	30.8%	**77.4%** / 17.3%
Rook ending	1,511	53.5%	62.2% / 28.7%
Queen ending	125	55.4%	63.8% / 18.8%
Knight ending	428	**81.6%**	49.0% / 46.2%

The extra pawn wins most often in a **pawn ending** and least often in a **knight ending**. (These are associations from engine games: the extra pawns differ in quality, and the counts for queen endings are small.)

Illustrated example

Step-by-step reasoning

How to think about an imbalance:

1. **Count with the right price list.** Middlegame: knight ≈ rook. Endgame: rook ≈ knight + pawn and more.
2. **Ask about pawns.** Pawns decide the endgame. Rook vs knight with equal pawns: the rook side scored 67.7%; one pawn less, 45.8%.
3. **Ask about targets.** A queen needs targets (loose pieces, an exposed king) to beat rook and knight; two rooks need an open file.
4. **Ask which ending you are heading for.** If you are a pawn up, aim for a pawn ending; if you are a pawn down, aim for a knight ending.

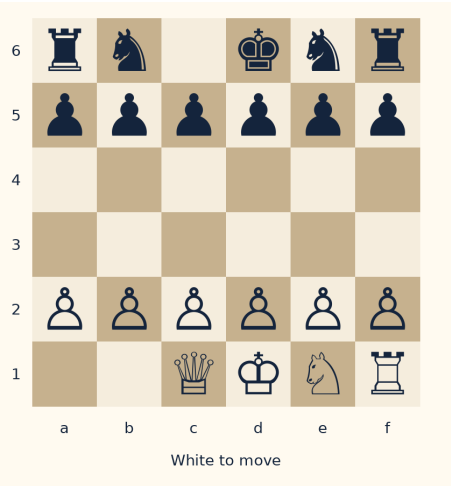


Figure 34: Method C, queen against rook and knight. White has kept the queen but lost the a1-rook and b1-knight; Black has lost the queen. Depth 12: +166 for White here (+134 with the first move’s value removed).

Common mistakes

- Using 8×8 values (knight 3, rook 5). Here, in the middlegame, a knight is worth as much as a rook.
- Trading into a knight ending to “simplify” an extra pawn. Knight endings were drawn 46% of the time even with the extra pawn.
- Assuming “the queen always beats two rooks”. In the archive the rooks did slightly better (56.1% in 41 games); on open files they are formidable.

Advanced insight

The engine also has a **drawish-material rule** (method D): when the side ahead has *no pawns* and is ahead by at most a knight’s worth of piece material, its whole evaluation is multiplied by 0.25. Without pawns, small material advantages rarely win. You will see this proven exactly for rook against knight in Book IV: in quiet positions with the defender to move, it is a draw 88% of the time.

Worked example

When to trade queens.

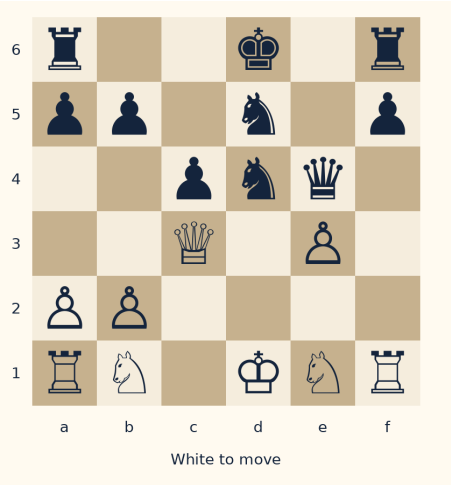


Figure 35: Academy puzzle pz-epqhdh. White to move. Black’s knight on d4 and queen on e4 press against White’s king; White’s queen is on c3.

White's king is the more exposed one. The solution trades the attacker off: `8. Qxd4! Qxd4+ 9. exd4`. **Engine judgement (depth 10):** +36 for White after the exchange; the "greedy" `8. exd4` is −228, because after `8...Nxc3+` Black keeps queen against knight. **Guidance:** trade queens when your king is the one in danger; keep them when theirs is.

Summary

- Engine prices: knight 441 / 262, rook 407 / 463: equal in the middlegame, rook far ahead in the endgame.
- Queen ≈ rook + knight + pawn; queen vs two rooks is roughly balanced; knight ≈ three pawns; rook vs knight ≈ one pawn.
- With an extra pawn, pawn endings win most (77%), knight endings least (49%).

The Bones of the Position: Pawn Structures

Explanation

The engine's definitions (method D, `eval4.ts`) are precise; learn them as they are written:

- **Doubled:** two or more own pawns on one file. Each pays −15 (middlegame) / −6 (endgame).
- **Isolated:** no own pawn on either neighbouring file. −8 / **−18**: isolated pawns become weaker as pieces come off.
- **Supported:** defended by an own pawn diagonally behind it. +13 / −4.
- **Phalanx:** an own pawn on the square beside it. +16 / +15: side-by-side pawns are good in every phase.
- **Passed:** no enemy pawn ahead of it on its own file or the two neighbouring files. Worth little in the middlegame but large in the endgame, rising with every step: on its 4th rank −12 / **+90**, on its 5th rank +37 / **+109**, plus up to +56 more if the square in front is empty, plus bonuses when the enemy king is far away (+26 per step of distance, scaled by rank).

The six-rank fact. A pawn on its starting rank needs **four single steps** to crown: three quiet ones and the promoting step. There is no two-step first move. A passed pawn in the middle of the board is never more than two or three moves from a new queen.

Evidence (method A, balanced games). Promotion is not a rare event here: - **51.3%** of games (2,859 of 5,568) contained at least one promotion; the median first promotion came on ply 64. - In **67.2%** of decisive games (2,480 of 3,688) the winner promoted at least once. - Of 3,855 promotions, 3,682 were to a queen, **121 to a knight** (3.1%) and 52 to a rook. Under-promotion to a knight, usually to give check or fork, is a real weapon.

Illustrated example

Step-by-step reasoning

Let us read the gallery exactly as the engine does (its term trace for this position confirms each item):

1. **a4.** No White pawn on the b-file: isolated. No Black pawn on the a- or b-file ahead of it: passed, on its 4th rank, with a5 empty: the endgame bonus is +90 +24, plus a bonus because Black's king on d6 is far from a5.
2. **c2, c3.** Two pawns on the c-file: both doubled. No White pawn on b or d: both isolated. They are also blocked from becoming passed by Black's c5 and d5.
3. **e3, f3.** Side by side: both are phalanx pawns. Neither is passed (Black's f5 stands ahead of them on the f-file and controls e4).
4. **Black's c5, d5.** A phalanx. d5 is not passed (White's c3 and e3 guard c4 and e4).
5. **Black's f5.** No Black pawn on e: isolated.

Engine judgement (static evaluation): +295 cp for White, mostly because the passed a-pawn is outside the Black king's reach: Black's king needs three moves to reach a6, the pawn needs only two to crown (a5, a6=Q).

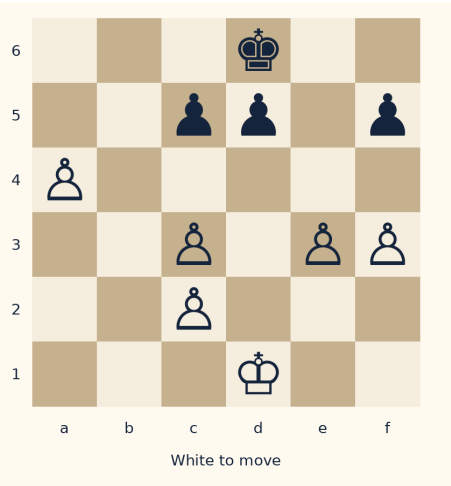


Figure 36: A gallery of structures. White: a4 is isolated AND passed; c2 and c3 are doubled and isolated; e3 and f3 form a phalanx. Black: c5 and d5 form a phalanx; f5 is isolated.

Common mistakes

- Thinking doubled pawns are a disaster. In the middlegame the engine charges only −15 each; they matter when they cannot become passed or defend each other.
- Ignoring isolated pawns because "they are only −8". In the endgame the charge more than doubles (−18), and a lone pawn needs a piece to guard it.
- Underestimating a passed pawn on the fourth rank in the middlegame. It is two steps from crowning; when the pieces come off it is suddenly worth a whole extra pawn to the engine.

Advanced insight

Why do phalanx pawns keep their value in the endgame while supported pawns lose it (+13 → −4)? A phalanx can advance together and each pawn can protect the other's next square; a supported pawn has a supporter that stands *behind*, which is slower and blocks nothing. When you choose how to recapture, prefer the recapture that makes a phalanx.

Worked example

A passed pawn changes everything, even when its owner is short of material.

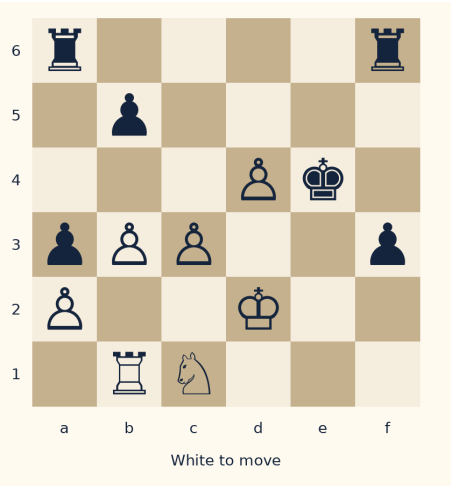


Figure 37: Academy puzzle pz-1h5x471. White to move. Black's f3-pawn is two steps from f1; White's d4-pawn is passed too.

White must deal with `...f2` and `...f1=Q`. The answer is `21. Nd3!`, covering f2 and preparing `Nc5+`, a fork of the king on e4 and the rook on a6. **Engine judgement (depth 10):** the main

line `21...f2 22. Nc5+ Kd5 23. Nxa6 f1=Q 24. Rxf1 Rxf1` gives White the better of it (+130); every other move lets the pawn crown with a large advantage for Black (−60 to −200 and worse).

Summary

- Five structures, each with a precise engine definition and weight.
- Isolated pawns grow weaker in the endgame; phalanxes stay strong; passed pawns become huge (+90 to +165 on the 4th–5th rank).
- Four steps to crown; half of all engine games see a promotion, and two thirds of winners promote.

PART II · Tactics

Double Attacks and Forks

Explanation

A **fork** is a move by one piece that attacks two (or more) enemy targets at once. The opponent can usually save only one. The strongest forks include a **check**, because the defender must answer the check first and then loses the second target.

How the small board changes it. On 6 × 6 the pieces start only two ranks apart, and every square is near the action, so forks appear from the first moves. Three facts matter: 1. **Knights change colour every move**, so a knight only ever attacks squares of one colour. Two targets on opposite colours can never be forked by one knight. White's king (d1, light) and queen (c1, dark) begin on *different* colours: no knight can fork them in the starting position. But the squares **e3** and **b3** (for Black's knights) and **e4** and **b4** (for White's) are deadly: a knight on e3 hits c2, d1 and f1; a knight on b3 hits a1, c1 and d2. 2. In the starting position those fork squares are guarded by two pawns each (e3 by d2 and f2; b3 by a2 and c2). **Guidance:** be careful about letting both guards disappear. 3. The queen reaches more than half the board from the centre (Book I, Chapter 2), so **queen forks** with check are everywhere.

Illustrated example

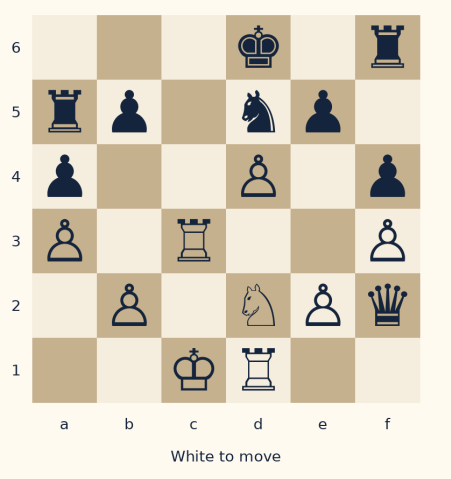


Figure 38: Academy puzzle pz-11z7k92. White to move. One knight jump attacks three pieces.

`Ne4+!` The knight lands on e4 and attacks d6 (the king), f6 (a rook) and f2 (the queen). Nothing Black owns can capture on e4. Black must move the king, and then White collects the queen on f2.

Step-by-step reasoning

How do you *find* a fork? Use a checklist: 1. **List loose and valuable targets**: the king, the queen, undefended pieces. Here: king d6, rook f6, queen f2. 2. **Ask which squares attack two of them.** For a knight, check the squares a knight's jump away from each target and look for

overlap. d6, f6 and f2 are all dark squares, so a knight on a light square can hit them together; e4 hits all three. 3. **Can my piece reach that square, and is it safe there?** The d2-knight reaches e4 in one jump, and no Black piece attacks e4. 4. **Is the first threat forcing?** `Ne4+` is check, so Black has no time to save both the rook and the queen.

Here is the colour rule at work in an opening-like position. Black's knight jumps to e3, a light square, and hits three light squares at once:

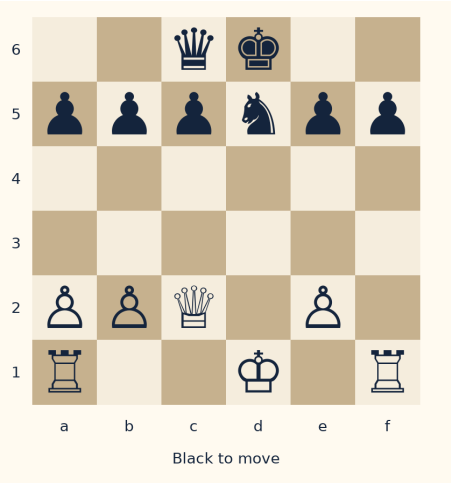


Figure 39: The engine's choice. With White's d- and f-pawns gone, e3 is unguarded: Ne3+ forks the king on d1, the queen on c2 and the rook on f1.

Verified line: Ne3+

Common beginner mistakes

- Looking only at captures and missing quiet squares from which a knight attacks two pieces.
- Forgetting that a fork only works if the forking piece cannot simply be captured.
- Defending one forked piece and forgetting the check comes first.
- Trying to fork two pieces of different colours with one knight.

Advanced insight

On this board a knight that reaches the third rank from the enemy's side (e3/b3 for Black, e4/b4 for White) is often a fork in waiting. A queen fork works best with check, because a queen that attacks two pieces without check can simply be met by a counter-attack on the queen. **Guidance:** when your king and queen stand on the same rank or diagonal with one square between them, look for enemy queen checks that also hit the second piece.

Worked example

Solution. Black's knight on b3 is under attack by the queen on c2, so it must do something. The best "something" is `Nd4+`: from d4 it checks the king on e2 and attacks the queen on c2. The check must be answered (`Kd2`, `Kd3`, `Ke3` or `Kf2`), and then `...Nxc2` wins the queen for the knight. Notice the pattern: **when a piece is attacked, look for a move that saves it *with* a threat.**

Summary

A fork is one move, two threats. Look for squares that attack two targets of the **same colour** (for knights), make the first threat a check when possible, and make sure the forking piece is safe. On 6 × 6, the fork squares e3/b3 and e4/b4 are always worth a look.

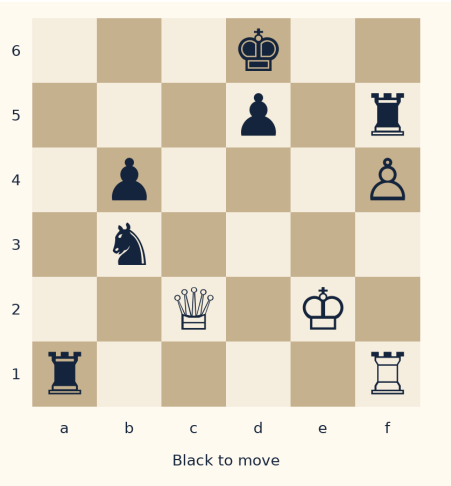


Figure 40: Academy puzzle pz-6vlr5l. Black to move. The knight on b3 is attacked by the queen.

Pins

Explanation

A **pin** happens when a rook or queen attacks an enemy piece that cannot move without exposing a more valuable piece behind it on the same line. - If the piece behind is the **king**, the pin is **absolute**: moving the pinned piece off the line is illegal. It may still move *along* the line, and it may capture the pinning piece if that is along the line. - If the piece behind is a queen or rook, the pin is **relative**: the pinned piece may move, but the piece behind will be lost.

A pinned piece is a poor defender and an easy target: attack it with something cheaper (often a pawn), and it cannot run away.

How the small board changes it. There are no bishops in Crown & Calculation, so every pin is made by a **rook or queen**, along a rank, a file, or (queen only) a diagonal. The kings start on the **d-file**, facing each other across it, and cannot castle away. The d-file and the back ranks are therefore the natural pinning lines. **Guidance:** once the d-pawns are exchanged, any piece standing on the d-file between your king and an enemy rook or queen is a pin waiting to happen.

Illustrated example

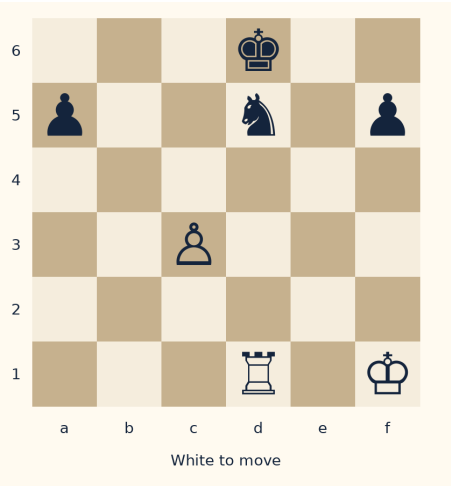


Figure 41: The engine’s choice. The knight on d5 is pinned to the king on d6 by the rook on d1. c3-c4 attacks it with a pawn.

Verified line: c4

Step-by-step reasoning

1. **Find the line.** Rook d1, then d2, d3, d4 empty, then Black's knight on d5, then Black's king on d6. The knight is pinned absolutely.
2. **What can the pinned piece still do?** It may not leave the d-file. It cannot even capture the pawn on c3, which it attacks, because `Nxc3` would expose the king.
3. **Attack it again.** `c4` attacks d5 with a pawn. The knight cannot move away, and the king defends it only once.
4. **Count.** After `cxd5`, White wins a knight for a pawn. Black can unpin by moving the king off the d-file, but that does not save the knight: it is still attacked by the c4-pawn.

Common beginner mistakes

- Moving a pinned piece "to safety" and losing the king's protection (or making an illegal move).
- Thinking a pinned piece defends the squares it attacks. An absolutely pinned piece cannot capture off the line, so it defends nothing there.
- Forgetting that the pinned piece can sometimes capture **the pinning piece itself**, if it moves along the line.
- Lining up king and queen on the same open file.

Advanced insight

Pins and forks combine. A pin freezes one defender; a fork then hits two targets that the frozen piece was supposed to cover. And because there are no bishops, **diagonal pins can only come from a queen**: if the enemy queen is gone, your pieces on diagonals are pin-proof. **Guidance:** in the opening, move the king off the d-file or keep a pawn on d2/d5 until the danger is past.

Worked example

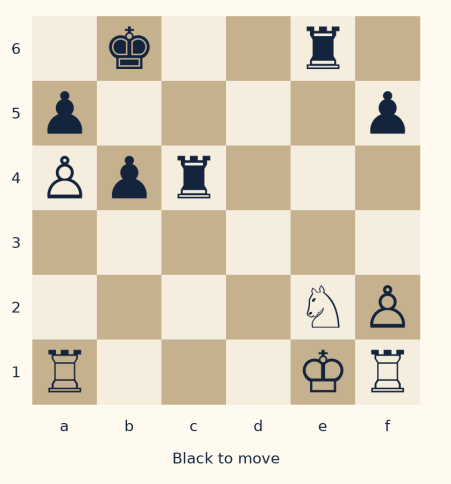


Figure 42: Academy puzzle pz-1tnry0q. Black to move and win material.

Solution. The White knight on e2 stands between Black's rook on e6 and White's king on e1: it is **absolutely pinned** along the e-file (e5, e4 and e3 are empty). It is defended only by the king. Black plays `Rc2!`: the second rook swings to c2 and attacks the knight along the 2nd rank (d2 is empty). Now the knight is attacked twice and defended once, and it cannot move. White cannot add a second defender in time, so Black wins the knight.

Summary

A pinned piece is frozen. Pile up on it, preferably with a pawn. Pins come only from rooks and queens here, and the d-file, where the kings start, is the most dangerous line on the board.

Skewers

Explanation

In a pin, the *less* valuable piece is in front. In a **skewer**, the *more* valuable piece is in front: usually the king, in check. When it steps off the line, the piece behind it is captured.

How the small board changes it. The board is only six squares wide, so a rook or queen on any rank or file covers the whole line, and kings in the endgame often share a rank or file with their own rooks. The back ranks (1 and 6) are short highways: a check along the back rank frequently skewers the king and a rook standing at the other end. And with no bishops, **only a queen can skewer along a diagonal.**

Illustrated example

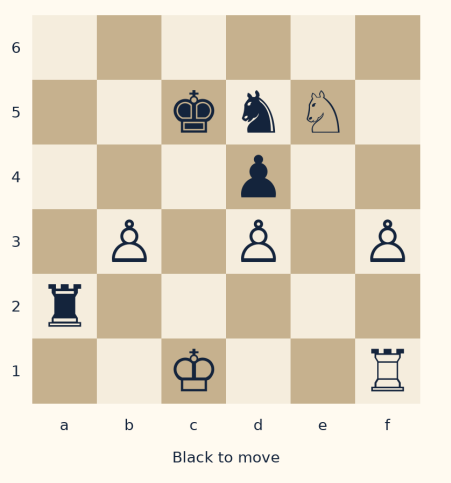


Figure 43: Academy puzzle pz-qn6wq1. Black to move. King and rook share the 1st rank.

`Ra1+!` The rook checks the king on c1 along the 1st rank, and behind the king stands the rook on f1 (d1 and e1 are empty). The king must step off the rank (its legal moves are `Kb2`, `Kc2` and `Kd2`), and then `...Rxf1` wins the rook. After `Kb2` the king attacks the Black rook on a1, but Black moves first and the rook escapes by capturing.

Step-by-step reasoning

1. **Find two enemy pieces on one line** with the more valuable in front: king c1 and rook f1 on the 1st rank.
2. **Find a square on that line from which your piece checks the front piece**: a1 (b1 is empty between).
3. **Check that your piece is safe**, or that it does not matter: after `Ra1+` the king cannot capture at once.
4. **Check that nothing can block.** A White piece would have to land on b1; none can.

Common beginner mistakes

- Giving the "skewer" check when the piece behind is defended. Then you only trade.
- Forgetting that the king can sometimes step *towards* your checking piece and attack it.
- Missing that the opponent may block the check with a piece, which can turn the skewer into a pin.

Advanced insight

Because the board is short, a rook skewer from one corner to the other is common in endgames. **Guidance:** in rook endgames, keep your king off the rank or file where your own rook stands, unless the rook is defended.

Worked example

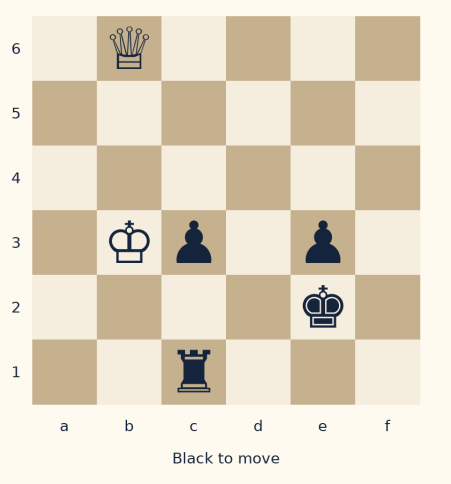


Figure 44: Academy puzzle pz-1jx5lpm. Black to move. Black has only a rook and two pawns against a queen.

Solution. White's king on b3 and queen on b6 share the b-file (b4 and b5 are empty). `Rb1+` checks along the file. All six king moves (`Kxc3`, `Kc4`, `Ka4`, `Ka3`, `Ka2`, `Kc2`) leave the b-file, and then `...Rxb6` wins the queen. The Academy line is `1...Rb1+ 2. Kxc3 Rxb6`, after which Black's pawns decide.

Summary

A skewer attacks the front piece and wins the one behind it. Look for an enemy king that shares a line with an undefended rook or queen. On 6 × 6, the back ranks and the d-file are the usual skewer lines, and only queens skewer diagonally.

Discovered Attacks and Discovered Check

Explanation

A rook or queen can stand behind one of your own pieces, aimed at an enemy target. When the front piece moves, the line opens and the back piece attacks. If the front piece *also* makes a threat, there are two threats at once, and the opponent can usually meet only one. When the uncovered attack is a check, it is a **discovered check**, and the moving piece is free to capture anything it likes.

How the small board changes it. Lines are short and crowded: a rook on the 1st rank sees the whole rank as soon as one knight jumps off it. In the starting position the rooks on a1 and f1 look along the back rank through the knights on b1 and e1. **Every knight move off the back rank opens a rook line.** On 6 × 6, the "front piece" is very often a knight, and its jump is likely to land with check or on a fork square at the same time.

Illustrated example

`Nc3+!` The knight leaves b1 with check (c3 attacks b5), and at the same moment the rook on f1 sees all the way down the 1st rank to the Black rook on a1. Black must deal with the check (`Kxc5`, `Kc6`, `Ka6` or `Ka5`), and then `Rxa1` wins the rook.

Step-by-step reasoning

1. **Look for your long-range pieces aimed at enemy pieces through your own pieces.** Rook f1, then e1, d1, c1 empty, then White's own knight b1, then Black's rook a1.
2. **Ask where the front piece can go with a threat of its own.** The knight's best square is c3: it gives check.

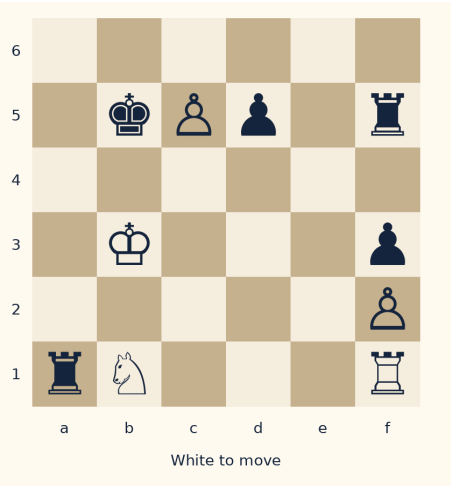


Figure 45: Academy puzzle pz-ephpsb. White to move. The knight on b1 blocks White's own rook.

3. **Count the threats.** One check plus one attack on an undefended rook. Black can answer only the check.

Common beginner mistakes

- Not noticing your own discovered attacks because the back piece "isn't doing anything" yet.
- Forgetting the opponent's discovered attacks: a piece standing in front of an enemy rook is a loaded spring.
- Moving the front piece to a square where it is simply captured, when a check was available.

Advanced insight

A discovered check is the strongest tactic in this book: the moving piece may go anywhere, even capture a defended queen, because the opponent must answer the check. **Guidance:** when the enemy king and your rook or queen share a line with one of your own pieces between them, stop and examine every move of that piece.

Worked example

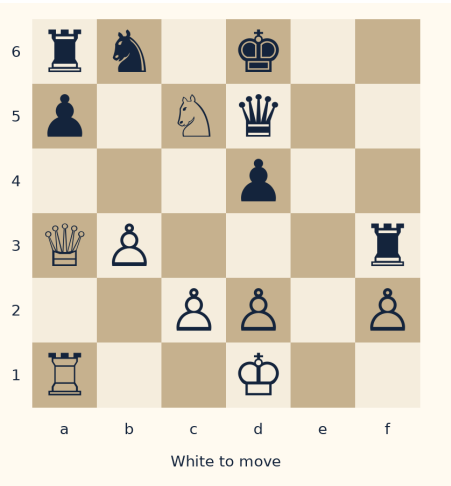


Figure 46: Academy puzzle pz-10qk93h. White to move. The knight on c5 is attacked by the queen on d5.

Solution. White's queen on a3 aims along the diagonal a3–b4–c5–d6 at Black's king, but White's

own knight on c5 is in the way. The knight is attacked, so it must move anyway, and it moves with a capture: `Nxa6+`. It takes the rook on a6 and uncovers **discovered check** from the queen. Black must answer the check (`Qc5` to block, or a king move), and White has won a rook.

Summary

Look behind your own pieces: a rook or queen aimed through them is a hidden weapon. Move the front piece with a threat of its own, ideally a check, and the opponent cannot meet both threats.

Removing the Defender and Overloading

Explanation

A defended piece is not safe for ever. Ask: *who defends it?* If you can eliminate or distract that defender, the piece becomes hanging. - **Removing the defender**: capture it (often trading something), chase it with a check, or cut its line. - **Overloading**: a single piece defends two things. Force it to do one job, and the other collapses.

How the small board changes it. Pieces stand close together, so one piece very often defends two or three neighbours at once, and the king is frequently one of the defenders. Kings are easy to chase with checks on 6 × 6, because they have few squares. **Guidance:** when a piece is defended only by its king, look for a check that drives the king away.

Illustrated example

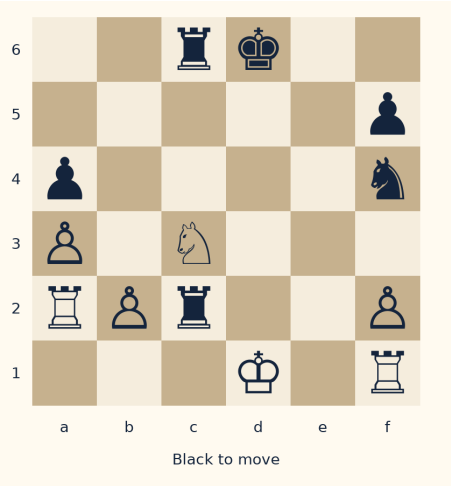


Figure 47: Academy puzzle pz-19qx3hv. Black to move. The rook on a2 is defended by the knight on c3.

`R6xc3!` Black gives a rook for the knight that guards a2. After `bxc3` the recapture opens the 2nd rank, and `...Rxa2` takes the undefended rook. Black has won a knight. (`R6xc3` means "the rook on the 6th rank takes on c3"; both Black rooks could capture there.)

Step-by-step reasoning

1. **Pick a target**: the White rook on a2.
2. **List its defenders**: only the knight on c3 (c3 attacks a2).
3. **Attack the defender**: the c3-knight is attacked twice (rooks c6 and c2) and defended once (pawn b2).
4. **Remove it**: `R6xc3 bxc3`. Now a2 is undefended, and the b2-pawn has moved out of the c2-rook's way.
5. **Collect**: `...Rxa2`. Count: Black gave a rook (5) and won a knight and a rook.

Overloading

Here one Black rook has two jobs.

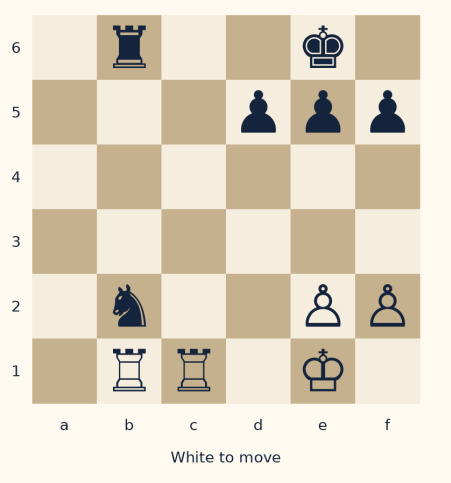


Figure 48: The engine's choice. The rook on b6 defends the knight on b2 and also guards the 6th rank against a back-rank mate.

Verified line: Rxb2

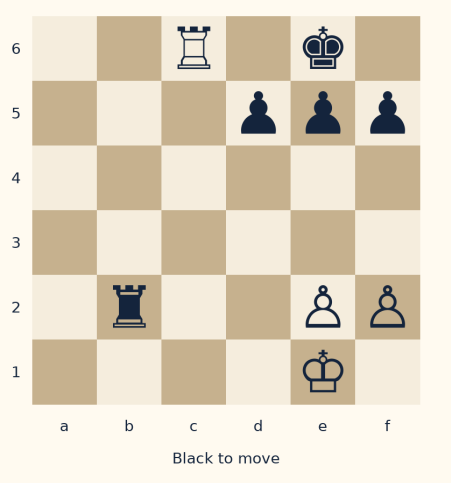


Figure 49: If Black recaptures, the 6th rank is unguarded and Rc6 is mate: the king on e6 is walled in by its own pawns.

Verified line: Rxb2 Rxb2 Rc6#

The Black rook cannot both recapture on b2 and guard c6. Black's best is not to recapture, and White has simply won a knight.

Common beginner mistakes

- Attacking a defended piece again and again without asking who defends it.
- Forgetting that the defender can be **driven away with check**: kings are defenders too.
- Missing your *own* overloaded pieces: if one piece defends two things, the opponent will find it.

Advanced insight

Removing the defender often costs material first (an exchange sacrifice, or a check that gives up a pawn). Count to the end of the sequence, not the first move. **Guidance:** in the middlegame

on 6 × 6, the most common overloaded piece is a rook on the back rank that must also defend something on its file.

Worked example

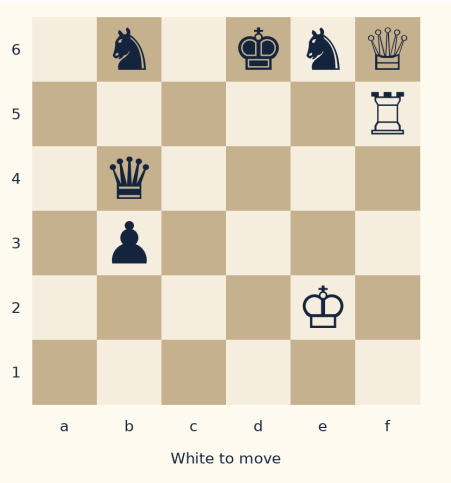


Figure 50: Academy puzzle pz-e6r29g. White to move. The knight on e6 is defended only by the king.

Solution. `Qxe6+?` at once fails to `Kxe6`. First drive the defender away: `Qe5+!` The queen checks from e5. The king's only legal move is `Kc6`: c5 and d5 are covered by the queen, the queen itself is protected by the rook on f5, and e6 holds Black's own knight. Now the knight on e6 has lost its only defender, and `Qxe6+` wins it with check along the 6th rank.

Summary

Every defended piece has a defender. Capture it, chase it (often with check), or overload it, and the piece falls. Count the whole sequence.

Deflection and Decoy

Explanation

- **Deflection** forces a piece to leave the square or line it guards. It is a cousin of removing the defender: the defender is not captured but dragged away.
- **Decoy** forces an enemy piece to *come to* a particular square, usually by offering a piece there. The decoyed piece then falls to a fork, a pin, a skewer or a mate.

Both usually begin with a sacrifice: you offer material that the opponent must, or wants to, take.

How the small board changes it. Forks are so common on 6 × 6 that the most frequent decoy is simply **"lure two pieces onto one knight fork."** Kings and queens start on adjacent squares and often stay close to each other; a sacrifice that drags one of them one square sideways can put both on the same knight's colour.

Illustrated example

`Rxe1+!` The rook takes the knight on e1 with check. White must answer with `Qxe1` (the Academy line) or block with `Qd1` (then Black simply takes again). After `Qxe1` the queen has been **decoyed** to e1, and `...Nxd3+` forks the king on c1 and the queen on e1 (the knight on d3 also attacks the rook on f2). The d3-knight cannot be captured, and Black wins the queen.

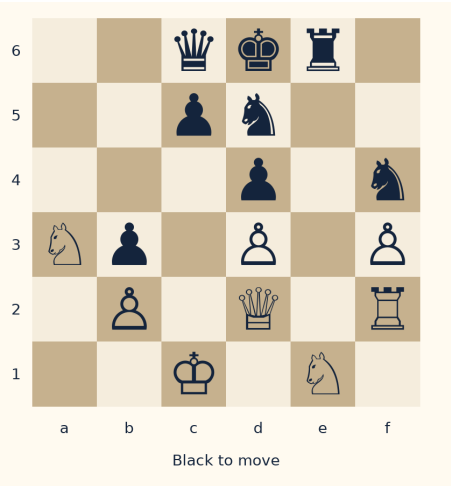


Figure 51: Academy puzzle pz-uyo7nn. Black to move. A knight on d3 would attack the king on c1, but the White queen on d2 is not on one of its squares.

Step-by-step reasoning

1. **See the fork square.** A knight on d3 attacks c1, e1, b2, f2, b4, f4, c5 and e5. The White king is on c1. What else could stand on e1?
2. **Find a forcing way to put a valuable piece there.** `Rxe1+` captures on e1 *with check*. The natural recapture brings the queen to e1.
3. **Check the fork is safe.** After `Qxe1 Nxd3+`, no White piece attacks d3.
4. **Count.** Black gave a rook for a knight, then the knight captures the d3-pawn with a fork and wins the queen. A large profit.

Common beginner mistakes

- Playing the sacrifice without checking that the follow-up really works against *every* reply.
- Forgetting that the opponent may decline the sacrifice.
- Using deflection on a piece that has a second defender behind it.

Advanced insight

A decoy works best when the sacrifice is a **check** or a **capture**, because the reply is then nearly forced. **Guidance:** when you see "the fork would work if only their king (or queen) were one square over", look for a check or capture on that square.

Worked example

Solution. `Rxb6!` White's rook takes the knight on b6; it is defended by the rook on a6, so `Rxb6` recaptures (the Academy line). The Black rook has been **decoyed** to b6, a knight's jump from c4. Now `Nc4+` checks the king on e5 and attacks the rook on b6. Whatever the king does, `Nxb6` follows, and White is a knight up.

Summary

Deflection drags a defender away; decoy drags a target in. Both usually start with a sacrifice that is a check or capture, and both most often end, on this board, in a knight fork.

Trapped Pieces

Explanation

A piece is **trapped** when every square it can move to is attacked (or occupied by its own side), so that once it is attacked it cannot escape. A trapped piece is often lost even though it is attacked

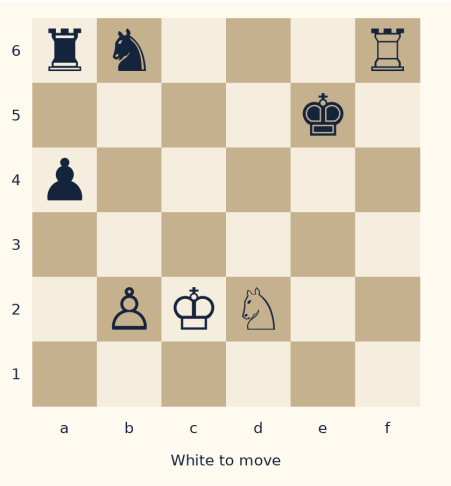


Figure 52: Academy puzzle pz-1ng565a. White to move. Rook and knight against rook and knight.

only once. The winning move is frequently **quiet**: no check, no capture, just a move that takes away the last square or adds the attack.

How the small board changes it. On 6 × 6 the edges are close and crowded. - A knight in a corner has only **2** moves, on an edge square next to the corner **3**, against 8 in the centre (Book I, Chapter 3). A knight that captures a rook in the corner (a1, f1, a6, f6) is very often trapped there. - The rooks begin in the corners behind a wall of pawns and knights, with nowhere to run if a pawn arrives next to them. - Pawns are always close to promotion, so a pawn that attacks a trapped piece may also threaten to queen.

Illustrated example

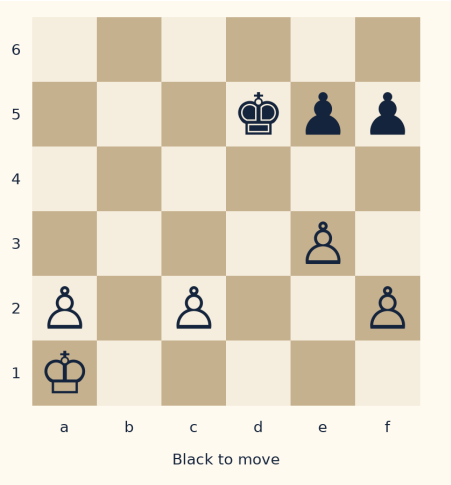


Figure 53: The Black knight has captured on a1 and is trapped there. Its two squares are b3 (attacked by the a2- and c2-pawns) and c2 (a pawn defended by White's king). White's king simply walks over and takes it.

Verified line: Kc1 Ke6 Kb1 Kd5 Kxa1

Check the knight's moves one by one. `...Nb3+` is met by `axb3` or `cxb3`; `...Nxc2` is met by `Kxc2`. Either way the knight dies, so Black leaves it, and White's king collects it with `Kc1`, `Kb1` and `Kxa1`. **Guidance:** a trapped piece will usually still be there in a few moves, so there is no hurry. The Academy engine actually prefers to push the a-pawn first in this position, because the knight cannot run and the a-pawn is only four steps from a6.

Step-by-step reasoning

1. **Find the enemy pieces with few squares.** Corners and edges first, then pieces deep in your camp.
2. **List each of their moves and ask whether the landing square is attacked.** If every move loses the piece, it is trapped.
3. **Add an attacker, or remove the last flight square,** with a quiet move if necessary.
4. **Check the opponent's counterplay** while you collect: a trapped piece can still be sold dearly, for example by capturing a defended pawn with check.

Common beginner mistakes

- Grabbing a corner rook with a knight without checking how the knight will get out.
- Chasing a trapped piece with checks instead of quietly cutting off its last square.
- Forgetting your own trapped pieces, especially a rook in the corner behind its own pawns.

Advanced insight

Pins and traps are relatives: an absolutely pinned piece has *no* legal squares off the line, so it is "trapped" in place (Chapter 2). And a piece can be trapped by **pawns alone**: two pawns side by side control four squares in front of them, which on a six-file board is two thirds of a rank. **Guidance:** in the opening, do not send a knight to the edge of the enemy camp unless you have checked its way back.

Worked example

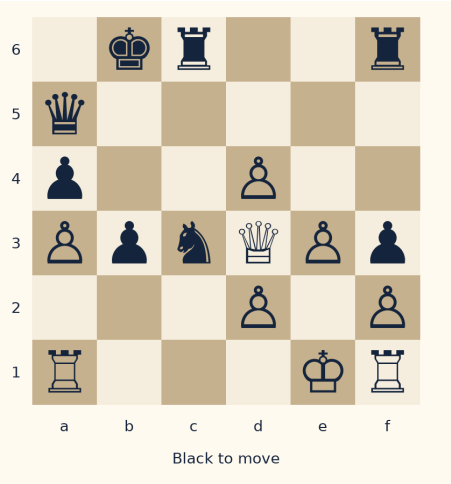


Figure 54: Academy puzzle pz-1k2b7wn. Black to move. Black is under pressure; find the only move that holds. Look at White's rook on a1.

Solution. `1...b2!` The pawn steps from b3 to b2 and attacks the rook on a1 (and threatens to promote on b1). Now list the rook's moves: `Rb1`, `Rc1`, `Rd1` and `Ra2`. Every one of those squares is attacked by the Black knight on c3 (or by the b2-pawn, which also covers c1). The rook is **trapped**, and Black will win it, or promote, next move. The Academy verification marks this as the only move that holds Black's position.

Summary

Count the squares of enemy pieces on the edge and in your camp. If every square is covered, attack the piece, quietly if necessary. Corners are cages on a 6 × 6 board, especially for knights.

The Zwischenzug (the In-Between Move)

Explanation

In a sequence of exchanges, both players expect the "obvious" next move: you took, so I take back. A **zwischenzug** interrupts that sequence with a more forcing move. The recapture is still there afterwards, but in the meantime you have gained something: a tempo, a better square, or extra material.

The in-between move must be **at least as forcing** as the threat it ignores. A check is ideal, because the opponent must answer it first.

How the small board changes it. On 6 × 6 almost every capture is answered by another attack, and kings are close to the action, so in-between checks are everywhere. Equally, the opponent's knight that has just captured is often *still attacking something*, and recapturing blindly lets it do a second job.

Illustrated example

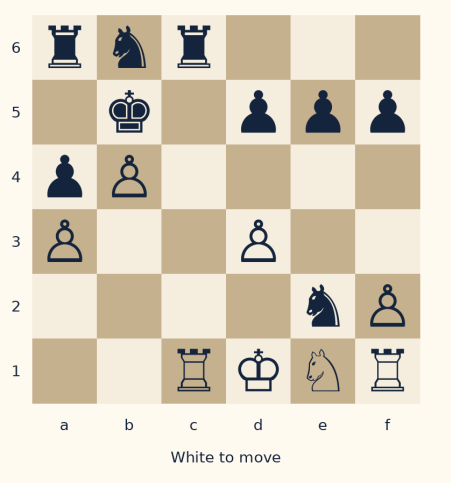


Figure 55: The engine's choice. Black's knight has just captured on e2. The natural recapture Kxe2? loses a rook. First Rxc6!

Verified line: Rxc6

The Black knight on e2 attacks the White rook on c1, and Black's rook on c6 attacks it too. If `1. Kxe2?`, then `1...Rxc1` (the c-file is open from c6 to c1), and White has lost a rook. Instead `1. Rxc6!` first exchanges the attacking rook. After `1...Kxc6` (the engine's choice for Black), `2. Kxe2` recaptures the knight, and White has come through the exchanges without losing the rook.

Step-by-step reasoning

1. **Before recapturing, ask: what does the capturing piece attack now?** The knight on e2 attacks c1 (and c3, d4, f4).
2. **What else is attacking my pieces?** The rook on c6 also hits c1. So c1 is attacked twice and not defended at all.
3. **Is there a forcing move that solves this first?** `Rxc6`: a capture that removes one attacker. The knight on e2 is not going anywhere useful in the meantime.
4. **Then recapture.**

Common beginner mistakes

- Recapturing automatically.
- Playing an "in-between" move that is not forcing, so the opponent simply saves the piece.
- Forgetting that the opponent can play a zwischenzug too.

Advanced insight

A zwischenzug is the natural partner of the discovered attack and the fork: the in-between check often lands on a square from which the checking piece *also* attacks something. **Guidance:** whenever you are about to recapture, spend two seconds on every check you have.

Worked example

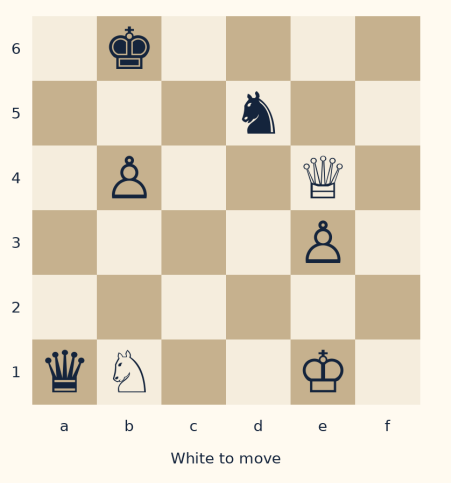


Figure 56: Academy puzzle pz-tef4fw. White to move. The knight on d5 is undefended. So is White's knight on b1.

Solution. `Qxd5?` wins a knight but takes the queen off the diagonal e4–b1, and `...Qxb1+` wins White's knight back with check. Instead `Qe6+!` first: the queen checks the king on b6 along the 6th rank *and still attacks d5*. After `Kb5` (the Academy line) comes `Qxd5+`, again with check, so Black never gets time for `...Qxb1+`. White then saves the b1-knight and is a piece up.

Summary

Before you recapture, look for a check or a stronger capture. The recapture will usually still be there afterwards; the opponent's chance to defend may not.

Mate Threats and Sacrifices

Explanation

A **mate threat** is almost as forcing as a check: the opponent must stop the mate, so he has no time for anything else. If your move threatens mate *and* attacks a piece, you usually win the piece.

A **sacrifice** gives up material to force a result: mate, or a larger gain later. The sacrifice is only sound if **every** reply loses. For mating sacrifices on this board the checker below proves it by brute force.

How the small board changes it. Kings on 6 × 6 have very few squares. The back rank sits one step behind the pawn line, there is no castling to tuck the king into a corner, and the edge of the board is never more than two squares from the king. So mate threats appear early and often, and pieces crowd around their own king, which makes **smothered mates** (a knight mating a king that is boxed in by its own pieces) much more common than on the big board.

Illustrated example

`Rxf4!` The rook takes the f4-pawn and opens the f-file towards f6. Now White threatens **two** mates on f6: `Rf6#` and `Qf6#` (the queen on b2 sees f6 along the long diagonal b2–c3–d4–e5–f6). Black must spend the next move stopping both threats. Only a handful of replies do (an

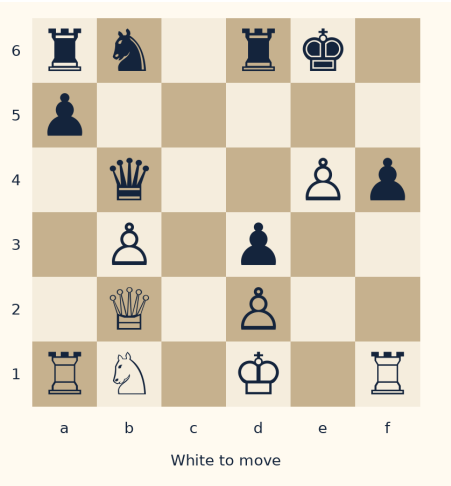


Figure 57: Academy puzzle pz-lvixyc. White to move. The Black king on e6 is short of squares.

exhaustive search finds `...Qd4`, `...Qc3`, `...Rd4`, `...Nd5`, `...Qxe4`, `...Qxb3+` and `...Qxd2+`); every other move allows mate on f6. The Academy engine rates White's position as decisive.

Step-by-step reasoning: a smothered mate

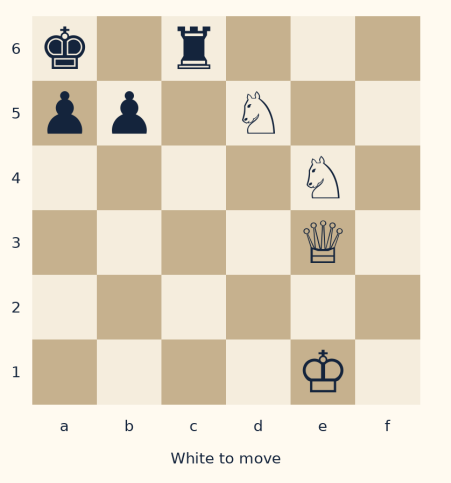


Figure 58: White mates in 2, and only the queen sacrifice Qb6+ does it. Proven by trying every defence.

Verified line: Qb6+ Rxb6 Nc5#

1. **The box.** The Black king on a6 is surrounded by its own pawns on a5 and b5. Its only free square is b6.
2. **The mating square.** A knight on c5 would check a6. Right now `Nc5+?` fails because Black's rook on c6 simply captures it, and in any case the king could step to b6.
3. **The sacrifice.** `Qb6+!` The queen checks from b6, protected by the knight on d5, so the king cannot take it. Black's only legal reply is `...Rxb6`.
4. **The mate.** The rook has left the c-file and now blocks the king's last square. `Nc5#`: the king is smothered by its own pieces.

Common beginner mistakes

- Making a mate threat that the opponent can parry *with a gain of tempo* (for example by attacking your queen).
- Sacrificing without checking every defence.
- Forgetting your own king: before launching an attack, check the opponent's checks.

Advanced insight

A move that threatens mate on two squares (like `Rxf4` above, threatening both `Rf6#` and `Qf6#`) is hard to meet, because one defensive move must cover both. **Guidance:** when the enemy king has only one or two free squares, look at every check and every sacrifice that takes those squares away.

Worked example

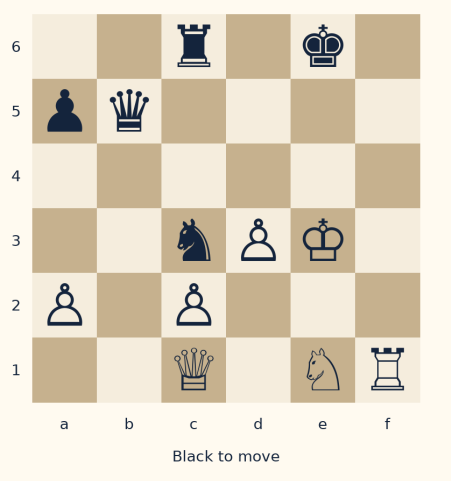


Figure 59: Academy puzzle pz-10qaias. Black to move and mate in 3.

Solution. `1...Qe5+!` The queen centralises with check on the e-file (e4 is empty). White has three replies: - `2. Kd2` or `2. Kf2`: `2...Qe2#`. The queen is protected by the knight on c3. - `2. Kf3 Qe2+ 3. Kf4 Nd5#`: the knight jumps from c3 to d5, checking f4, and every flight square is covered.

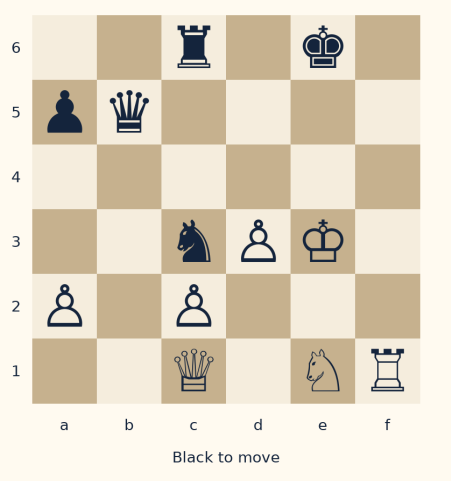


Figure 60: Proof by exhaustive search: Black mates in 3 against every defence (the longest line shown).

Verified line: Qe5+ Kf3 Qe2+ Kf4 Nd5#

Summary

Mate threats are forcing moves. Combine them with attacks on pieces. On a small board kings are easy to box in, so look for sacrifices that take away the last square, and remember the smothered mate.

Promotion Tactics

Explanation

A pawn one step from the last rank (rank 5 for White, rank 2 for Black) threatens to become a queen next move. That threat is as strong as a threat to win a queen, so it can be used like a check: to gain time, to deflect defenders, or to distract a piece from something else.

How the small board changes it. Pawns start only **four steps** from promotion, and after one step they are three steps away; many pawn captures move a pawn both sideways and forwards. In Crown & Calculation promotion is therefore a middlegame weapon, not only an endgame one. The promotion square is often next to the enemy king's starting area, so a new queen frequently arrives **with check**. And because promotion to a **knight** is allowed (but not to a bishop), underpromotion forks are a real resource.

Illustrated example

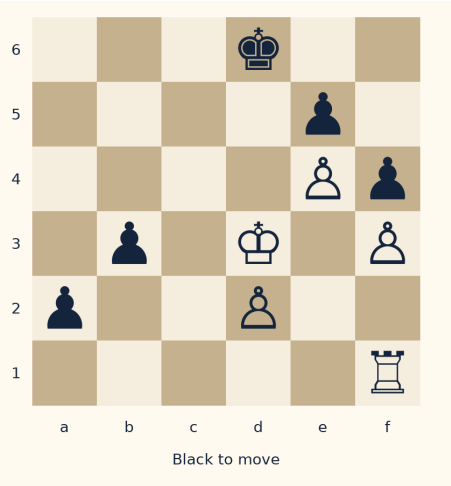


Figure 61: Academy puzzle pz-l1g2cx. Black to move. Black has two pawns on the a- and b-files; White has a rook.

`1...b2!` Now both Black pawns stand on the 2nd rank, side by side, each one step from promotion. The rook on f1 cannot stop both: if it attacks one promotion square, the other pawn promotes. After `2. Kc2` (the Academy line; the king attacks b2), `2...a1=Q` makes a queen that also protects the b2-pawn. **Two connected pawns on the rank before promotion beat a rook.**

Step-by-step reasoning

1. **Count steps, not material.** Black's a2-pawn is one step from a1; after `...b2`, the b-pawn is one step from b1.
2. **List what can stop each pawn.** The rook would like to stand on a1, b1 or c1. But the pawns guard each other's promotion squares: the a2-pawn attacks b1, and the b2-pawn attacks a1 and c1. So `Ra1` is met by `...bxa1=Q`, `Rb1` by `...axb1=Q+` and `Rc1` by `...bxc1=Q`: each time a pawn captures the rook *and* promotes.
3. **Check the king.** White's king can reach c2, attacking b2, but then `...a1=Q` makes a queen that defends b2. The engine prefers Black clearly in every line.

Common beginner mistakes

- Counting the material ("I have a rook against two pawns") instead of counting the steps.
- Promoting to a queen automatically when a knight would check and fork.
- Forgetting that promotion is compulsory: you cannot "keep the pawn" on the last rank.
- Letting a pawn reach the rank before promotion with no blockader in front of it.

Advanced insight

The new queen arrives on a back-rank square: a1–f1 for Black, a6–f6 for White. Those squares are often on the same rank, file or diagonal as the enemy king or rooks. **Guidance:** before promoting, ask what the new piece will attack *on the move it appears*. A promotion with check or with an attack on a rook is a free tempo.

Worked example

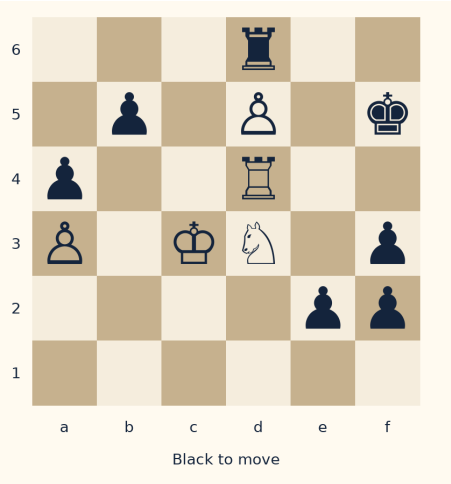


Figure 62: Academy puzzle pz-oy1e4. Black to move. Black has two pawns on the 2nd rank, e2 and f2.

Solution. `1...e1=Q+!` The e-pawn promotes with check along the diagonal e1–d2–c3. White's legal replies are `Kb2`, `Kc2` and `Nxe1`. If `2. Nxe1`, then `2...fxe1=Q+` brings a second queen, again with check. So White must move the king, and Black is a queen up (the Academy engine judges the position decisive).

Summary

Count the steps to promotion, not the material. Two connected pawns one step from the end beat a rook. Promote with check when you can, consider the knight when it forks, and always ask what the new piece attacks the moment it appears.

Forcing-Move Calculation

Explanation

A forcing move limits the opponent's choices. Checks are the most forcing (the reply must deal with the check), then captures (the reply is usually a recapture), then threats (the reply must parry the threat). Calculate them **in that order**, for both sides: 1. **List your candidate forcing moves**: all checks, all captures, all serious threats. 2. **For each candidate, list the opponent's forcing replies first**, then his best quiet defence. 3. **Go one step deeper** only along forcing lines. Stop when the position is quiet (no checks, no hanging pieces) and count the material. 4. **Compare** the end positions and choose.

How the small board changes it. On 6 × 6 the number of legal moves is small (there are only 10 in the starting position), and in tactical positions often only one to six replies to a check exist. This makes calculation *easier* than on the big board, and it means that a well-ordered search really can be complete. The flip side: the opponent's forcing moves arrive just as fast. **Guidance:** in every position, look at your opponent's checks before your own.

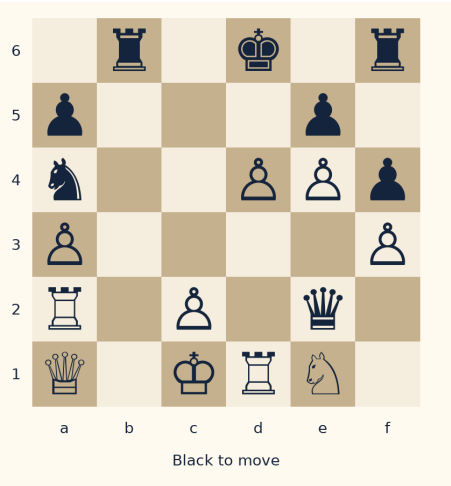


Figure 63: Academy puzzle pz-1vh7cy4. Black to move. The knight on e1 is guarded only by the rook on d1.

Illustrated example

`1...Qe3+!` checks along the diagonal e3–d2–c1. White's only legal move is `2. Rd2`. Now the rook has left d1, so `2...Qxe1+` takes the knight with check along the 1st rank, and again White has a single legal move, `3. Rd1`. Then `3...Qe3+` repeats the check. Black has won a knight with a chain of checks that left White no choice at all.

Step-by-step reasoning: the tree

For the illustrated position, a disciplined search looks like this: 1. **Black's checks:** write them all down. There are five: `...Qd2+`, `...Qxc2+`, `...Qxd1+`, `...Rb1+` and `...Qe3+`. The first four simply lose the checking piece (`Rxd2` or `Kxd2`; `Kxc2`; `Kxd1`; `Kxb1` or `Qxb1`). Only `...Qe3+` is safe. 2. **White's replies to `...Qe3+`:** only `Rd2` (the king has no square; nothing else can block). A tree with one branch. 3. **Black's forcing moves now:** `...Qxe1+` (capture with check). White's replies: only `Rd1`. 4. **Stop and assess:** a quiet moment after `...Qe3+` again; count: Black has won a knight.

Four plies, two branches with exactly one reply each. On a 6 × 6 board, such "only move" chains are common, and they are exactly what you should look for first.

Common beginner mistakes

- Calculating quiet moves first and running out of time before seeing the check.
- Stopping the calculation in the middle of an exchange, with a piece still hanging.
- Forgetting the opponent's forcing replies (a counter-check or a zwischenzug).
- Calculating one line very deeply and not looking at the alternatives at move one.

Advanced insight

Strong players stop calculating **as soon as the position becomes quiet and clear**, not after a fixed number of moves. **Guidance:** a good rule of thumb is "calculate until no checks, no captures and no threats remain, then evaluate". The Academy engine works in exactly this spirit: its quiescence search keeps looking at captures and promotions until the position is quiet.

Worked example

Solution, by the method. 1. **Inventory.** Black's queen on b5 is attacked by the rook on b1 (b-file). White's queen on a3 is attacked by the knight on c4. Both knights on c4 and e2 are Black's. 2. **Black's captures:** `...Nxa3` (takes the queen), `...Nxd2` and others. `...Nxa3` is the biggest. 3. **White's forcing replies to `...Nxa3`:** `Rxb5` (takes the queen back). Now material is level. 4. **Black's forcing moves now:** `...Nc3+!` The e2-knight checks the king on d1 and attacks the

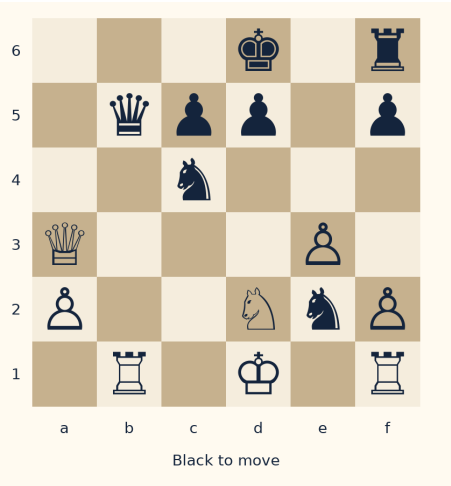


Figure 64: Academy puzzle pz-z3smsm. Black to move. Both queens are attacked.

rook on b5. White has only two legal replies (`Ke1`, `Kc1`), and then `...Naxb5` or `...Ncxb5` wins the rook. 5. **Assess:** queens traded, Black a rook up. The Academy line is `1...Nxa3 2. Rxb5 Nc3+`.

Summary

Checks, captures, threats, for both sides, in that order. Build a small tree, follow only forcing lines, stop at a quiet position and count. On 6 × 6 the trees are small and often complete: use that.

Interference and the line defender

Interference blocks a defender’s route without necessarily moving the defender. On this bishop-free board, trace rook and queen lines square by square. Before trusting a rook to defend a knight, ask whether a pawn or piece can enter between them with tempo. A deflection asks a defender to leave; interference can leave it exactly where it stands. Check whether the blocker can be captured with check or whether a second defender survives. Do not label the idea a win before examining those replies.

PART III · Advanced Tactics and Calculation

The Branching Path: Candidate Moves and Calculation

Explanation

The tree is narrow. Evidence (archive, 640-game match): a position offers on average **12.9** legal moves (max 44); in the endgame (after ply 60) only **10.5**. One fifth of all moves played were captures (21.4%) and one in seven gave check (14.1%). With so few branches, a human can calculate three or four moves deep in most positions — if the candidates are chosen well.

The candidate order. 1. **Checks** — every one, even the absurd ones. On a small board, a check often forces the only reply. 2. **Captures** — every one, including captures by the king. 3. **Threats** — moves that attack something or threaten mate. 4. **Quiet moves that take squares away** — especially king moves and rook moves that cut the enemy king off. These are the moves players miss.

The defence check. For each candidate, list every legal reply, not only the one you hope for. Small boards make this possible: often there are fewer than five replies.

Stop at quiet positions. Calculate until nothing hangs and no check is pending, then evaluate with the strategy of Book III.

Illustrated example

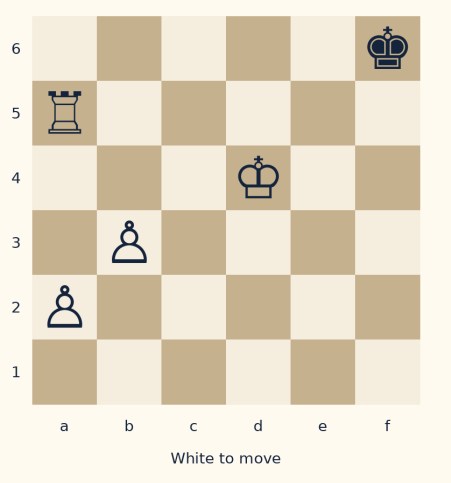


Figure 65: From an engine game. White to move: the only mate in two is the quiet king move 1. Ke4. No check, no capture — the king takes e5 and f5 away, and Black's king must walk into the rook's sixth-rank mate.

Verified line: Ke4 Ke6 Ra6#

Step-by-step reasoning

1. **Checks:** `1. Ra6+` and `1. Rf5+`. After `1. Ra6+` the king escapes to f5 (the rook has left the fifth rank); after `1. Rf5+` it escapes to e6. Both checks give the king air.
2. **Captures:** none.
3. **Threats:** none forcing.
4. **Quiet moves that take squares:** `1. Ke4` takes e5 and f5 (and d5). Now Black's king on f6 has one move: `1...Ke6` (e5 and f5 are covered by the king, the fifth rank by the rook). Then `2. Ra6#`: the rook covers d6–f6 along the sixth rank, the king covers d5, e5 and f5.

Proven: this is the only mate in two.

Common mistakes

- Looking only at checks. Here every check lets the king out.
- Stopping the calculation at the move you like ("then I check and win") without listing the defender's replies.
- Forgetting that the king itself is an attacking piece in the endgame.

Advanced insight

Because rooks and queens cut whole files and ranks on a six-square edge, the defending king usually has only one or two squares. A quiet move that takes one of them away is often stronger than any check. Train your eye to ask "how many squares does their king have?" before "what checks do I have?".

Worked example

A real engine position with a capture-check, a queen check and a quiet finish.

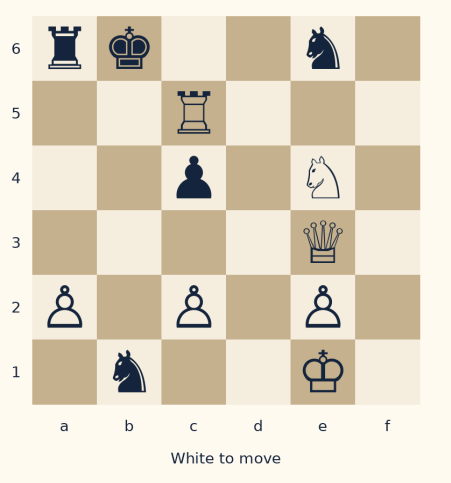


Figure 66: White to move mates in three, and only by 1. Rxc4+. The line played: 1. Rxc4+ Kb5 2. Qb3+ Ka5 3. Qb4#.

Verified line: Rxc4+ Kb5 Qb3+ Ka5 Qb4#

1. **Candidates.** Black's king is on b6, White's rook on c5 and queen on e3 — all on one diagonal, e3–d4–c5–b6. So any move of the c5-rook is a **discovered check** from the queen. The best of them also captures: `1. Rxc4+` takes a pawn, and on c4 the rook will later protect the queen.
2. **Replies.** The king must answer the queen's check. The game went `1...Kb5`; the checker tried every reply and found a mate by move 3 in each.
3. **The finish.** `2. Qb3+` checks along the b-file, `2...Ka5` is forced, and `3. Qb4#`: the queen, protected by the rook on c4, covers a4, b5 and b6, and Black's own rook blocks a6.

Summary

- 12.9 legal moves per position on average: calculate deeper than you would on a larger board.
- Candidate order: checks, captures, threats, then quiet square-takers (especially king moves).
- List every defence; stop at quiet positions and evaluate.

The Chain of Compulsion: Evaluating Forcing Lines

Explanation

A **forcing line** is a sequence in which the opponent's replies are limited to a few legal or sensible moves. Checks are the most forcing (the reply must deal with check), then captures that must be recaptured, then threats of mate or promotion.

How to evaluate the end of a forcing line: 1. **Count material** — but count the pawns about to crown as nearly queens. 2. **Check king safety** — after a sacrifice, which king is in the open? 3. **Look one move further** — is there a check or capture *right after* your line ends? That is the "horizon" trap: the position looks good until the next move.

Engine mate scores. When the engine finds a forced mate it reports a score near 30,000 (for example 29,991 means mate in a few moves from the root). Puzzle analyses in this book quote such scores as "the engine announces mate".

Illustrated example

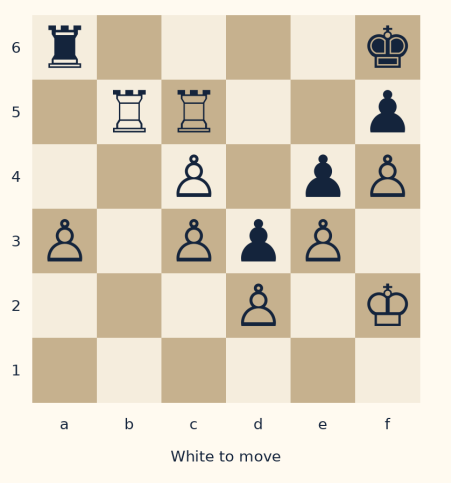


Figure 67: From an engine game. White mates in three, and only by 1. Rxf5+. Two rooks drive the king along the sixth rank: 1. Rxf5+ Ke6 2. Rbe5+ Kd6 3. Rf6#.

Verified line: Rxf5+ Ke6 Rbe5+ Kd6 Rf6#

Step-by-step reasoning

1. **The forcing first move.** `1. Rxf5+` captures with check (the c5-rook slides through d5 and e5; the b5-rook is blocked). Black's king on f6 must move, and e6 is its only square: e5 is covered by the rook on f5 and by the f4-pawn.
2. **The second check.** `2. Rbe5+`: the b5-rook joins on e5 (the `b` says which rook), protected by the f4-pawn. The king must go to d6 (d5 is covered by the rook and by the c4-pawn).
3. **The finish.** `3. Rf6#`: the f5-rook rises to f6 and checks along the sixth rank; the e5-rook covers c5 and d5; the king has nowhere to go. The checker confirms it is the only mate in three.

Common mistakes

- Evaluating the end of a line by material only, when an enemy pawn is one step from crowning.
- Stopping one move early: "I win the queen" — but the opponent then mates or forks.
- Trusting a "forcing" line that contains a quiet move by the opponent you did not consider.

Advanced insight

Promotion makes horizons treacherous on a six-rank board: a pawn on its 5th rank crowns next move, so a line that "wins a rook" may end with the opponent queening. Always add "and then, can a pawn crown?" to the last position of every line.

Worked example

A forcing line that the engine sees to mate.

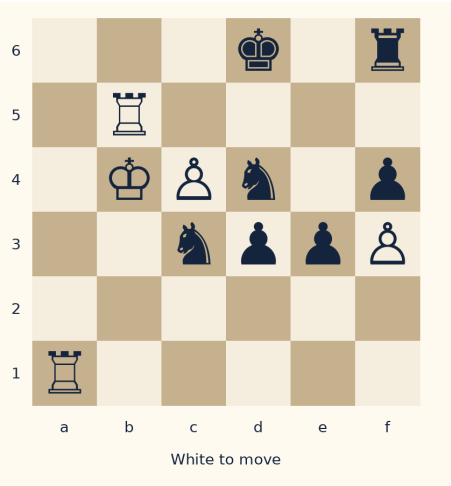


Figure 68: Academy puzzle pz-culn07. White to move. The engine announces mate.

The solution begins `35. Ra6+ Nc6+ 36. Kxc3`. The rook check from a6 drives the king; Black's best is to block with a knight check, and White's king calmly captures on c3. **Engine judgement (depth 10):** a forced mate (score 29,991; its line continues `36...Rf5 37. Rxf5 Ke6 38. Rd5 d2 39. Rxc6#`). The calmer `35. c5+` is only +271. When a line is all checks and captures, calculate it to the end before choosing a "safe" move.

Summary

- Forcing moves: checks, captures, mate and promotion threats.
- Evaluate the final position by material, king safety and pawns about to crown; look one move further.
- Engine mate scores are near 30,000.

The Unbroken Ward: Defensive Resources

Explanation

Defence is common and often works. Evidence (archive): with a one-pawn deficit at the first quiet moment after move 20, the defender lost only 48.0% of games (Book III, Chapter III.9); 22.9% of all balanced games ended in a threefold-repetition claim, 4.9% by bare kings, 4.9% by the 100-halfmove claim and 1.0% by stalemate.

The defender's toolkit: 1. **Perpetual check.** If you can check forever, you draw. Knights and queens are the usual instruments. 2. **Counter-sacrifice / returning material.** Give back what you are behind to break the attack. 3. **The bare-kings trade.** If you can trade your last piece

for the opponent's last piece, the game is **automatically drawn** (rule: only two kings remain). 4. **Stalemate.** A defender with few moves can sometimes run out of them (Chapter IV.7). 5. **Fortresses.** Some material advantages cannot win at all (Chapter IV.6, IV.7).

Illustrated example

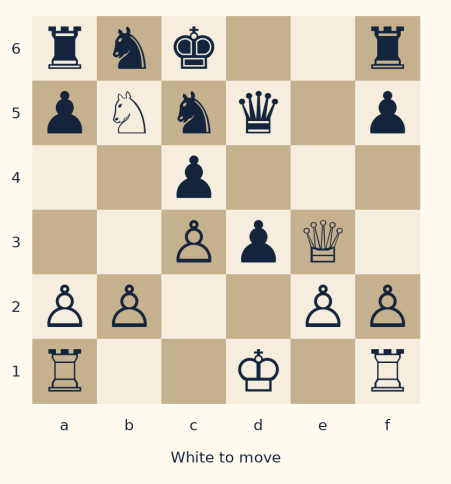


Figure 69: Academy puzzle pz-1uhg8w8. White to move. White's knight on b5 is attacked by Black's king, and Black's pawn on d3 eyes e2. Find the only move that holds.

Step-by-step reasoning

1. **The danger.** White's knight on b5 is attacked by the king on c6, and Black's pawns on d3 and c4 tear at White's king. Quiet defence (`Re1`, `Qd4`) loses: **engine judgement (depth 10)** −593.
2. **The resource.** `11. Nd4+` — the knight checks from d4. After `11...Kd6 12. Nb5+ Kc6 13. Nd4+` the same position returns.
3. **The verdict. Engine judgement:** 0.00, a draw by repetition. The defender who is losing should look for a sequence that the attacker cannot escape.

Common mistakes

- Defending passively when a perpetual check is available.
- Refusing to give back material. A pawn returned is better than a king lost.
- Forgetting that trading the last pieces into bare kings is an automatic draw.

Advanced insight

Why are perpetual checks so frequent here? Because the kings are close to every piece on a 6×6 board, and with only 10–13 legal moves per position the attacker's king often has just one or two squares to oscillate between. When you are worse, count the checks before you count the pawns.

Worked example

A queen perpetual from the opening.

`11. Qf4+! Kd5 12. Qf5+ Kd6 13. Qf4+` and the king cannot escape the checks on the f4–d6 and f5–d5 lines. **Engine judgement (depth 10):** 0.00; every other move is −1,134 or worse. The losing side found a draw in one move.

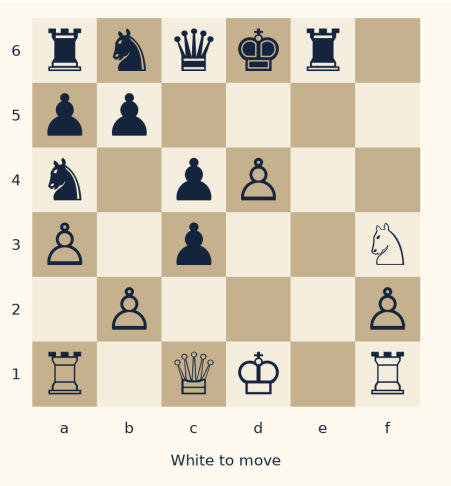


Figure 70: Academy puzzle pz-bus7cm. White to move. Black's pawn on c3 threatens to take on b2 and crown; White's position is collapsing.

Summary

- Perpetual check, counter-sacrifice, bare kings, stalemate and fortresses are the defender's tools.
- 22.9% of balanced engine games ended in a repetition claim: draws are a real result to fight for.
- When worse, count the checks before you count the pawns.

The Pruned Tree: calculation and candidate discipline

Explanation

A **calculation tree** is every line you consider: your candidate moves, the opponent's replies to each, your answers, and so on. Even on a 6×6 board it grows fast. A position with 25 legal moves and 25 replies already gives 625 two-ply lines. Strong players do not search harder. They **prune**: they stop examining a branch as soon as one good reply refutes it.

The discipline has three parts.

1. **Order the candidates.** First checks, then captures, then threats (moves that create a new attack), then quiet moves that take squares from the enemy king or improve your worst piece. The order affects only *where you start*. It never allows you to skip the quiet moves.
2. **Refute, then drop.** To reject a candidate you need **one** defence that holds. You do not need to see every defence. To *accept* a candidate you must answer **every** reasonable defence. That asymmetry is the whole art of pruning.
3. **Name the refutation.** Write it (in your head or on paper) as "Qe5+? Kxe5". A named refutation stops you coming back to a dead branch, and that wastes more clock time in practice than anything else ([Counsel]).

On this board the edges matter more than in orthodox chess. A king on its sixth rank (or White's on its first) has no squares behind it, and a king on the a- or f-file has no squares to one side. Many Crown mates come from a single rook or queen sealing the one remaining line. ([Demonstrated] in every board below.)

Illustrated example

Verified line: Qd4+ Kc6 Rf6#

White's king on a5 is exposed, and Black's e2-pawn is one step from promoting. White cannot afford a slow move. Every check is a candidate. **[Demonstrated]** The checker proved by exhaustive search that **1.Qd4+** is the only move that forces mate in two.

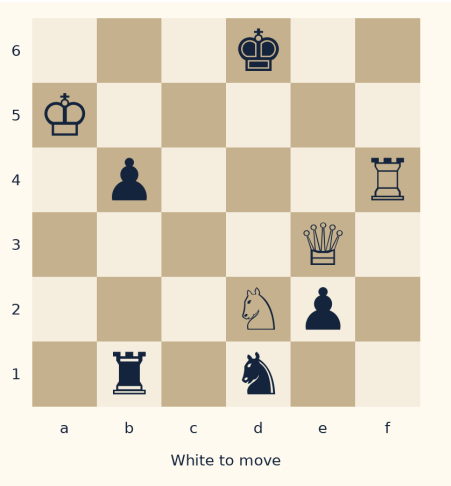


Figure 71: White to move. Ten checks are available; exactly one mates in two.

Step-by-step reasoning

List the ten checks and refute each one with a single defence. The refutations below are exact: in each case, after that defence White has no mate on the next move.

Candidate	One defence that refutes it	Why
Ne4+	...Ke6 (also ...Kd5, ...Ke5)	the king walks off the edge
Nc4+	...Kc6 (or ...Kd5)	same
Qd3+	...Ke6 or ...Ke5	no rook check covers both e-squares
Qe5+	...Kxe5	the queen is simply lost
Qe6+	...Kxe6	same
Qc5+	...Kxc5 or ...Ke6	the queen is lost or the king escapes
Qb6+	...Ke5	the king leaves the sixth rank
Rd4+	...Kc6 or ...Kc5	the king slips onto the c-file
Rf6+	...Kd5	the king steps down into open space
Qd4+	none	see below

After **1.Qd4+** the queen controls the whole d-file and the squares c5 and e5. The king's only moves are **...Kc6** and **...Ke6**, and both stay on the sixth rank. Then **2.Rf6#** closes the rank: the rook seals the sixth rank, and the queen and White's own king on a5 cover every square on the fifth. The tree has ten branches, nine of them are cut after one reply, and the tenth is proven against both replies. That is what a pruned tree looks like.

Common mistakes

- **Testing the first forcing move to the end before looking at the others.** Read all ten names first and then cut them. Seeing the whole list often shows you which one is special (here, the queen check that covers c5 and e5 at once).
- **Calling a candidate refuted because of one strong reply you only half saw.** A refutation must be a reply after which you have *nothing*. Check it.
- **Forgetting the board is six files wide.** There is no g-file. A king on f4 has no squares to its right.

Advanced insight

Pruning is safe only if your refutations are correct. If they are, a quiet first move is no harder to find than a check. It is simply later in the list. The worked example below shows a position where every check fails and the win is a quiet rook move that leaves Black **in zugzwang**: Black has a move, but every move makes things worse.

Worked example

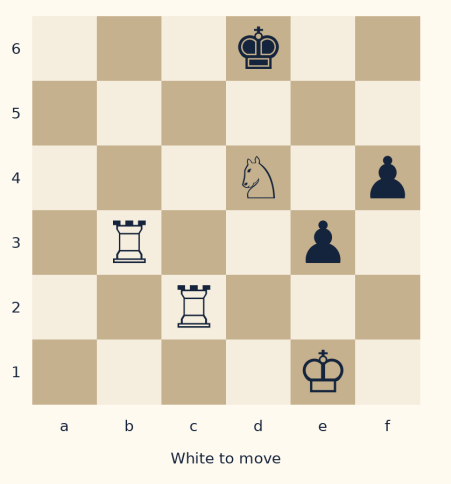


Figure 72: White to move and mate in two. Four checks are available, and all of them fail.

Verified line: Rb5 e2 Rc6#

1. **Checks:** Rc6+, Rb6+, Nf5+ and Nb5+. Each one lets the king leave the sixth rank (...Kd5 or ...Ke5 answers the rook checks; ...Ke6, ...Kd5 or ...Ke5 answers the knight checks). All four are refuted. **[Demonstrated]**
2. **Captures:** none.
3. **Threats and quiet moves:** look at the king's squares. c6 is covered by the rook on c2, and e6 by the knight. c5, d5 and e5 are free. **1.Rb5!** covers all three along the fifth rank.
4. **Black's replies:** the king has no legal move, so Black must move a pawn: ...e2 or ...f3. After either one, **2.Rc6#**. **[Demonstrated]** The checker confirms Rb5 is the only mate in two.

Notice how close this is to stalemate. If Black had no pawn moves, 1.Rb5 would be stalemate and only a draw. Always ask "does my quiet move leave them a legal move?" before you play it.

Summary

Order your candidates (checks, captures, threats, quiet moves), refute each one with a single named defence, and accept a move only when every defence fails. On this board the sixth rank and the a/f files are walls. Many mates need only one piece to seal the last line. Quiet moves live at the bottom of the list, not outside it.

Complexity, Imbalance and the Art of Simplifying

Explanation

Complexity is the number of plausible moves and the depth of calculation they need. Complexity is a weapon for the side that calculates better, or that has more time on the clock. It is a danger for the side that is already winning.

An **imbalance** is any lasting difference between the two sides. On the 6×6 board the common ones are:

- **Material:** with only 6 pawns and 5 pieces each, every pawn is a large share of the army.
- **Passed pawns:** promotion is only four steps from the starting rank. That is why the Academy coach has a topic called "Promotion races (promotion is only four steps away)" ([Demonstrated], src/academy/coach.ts).
- **King exposure:** there is no castling, and the king starts on d1/d6 in the middle of the back rank.
- **Activity:** a rook on an open file or a queen near the enemy king.

The rule of simplification is simple. **Trade when the imbalance that remains after the trade is decisive in your favour and easy to play. Keep pieces on when the imbalance is unclear or favours your opponent's skill.** A queen trade that leaves you a running passed pawn is usually the cleanest win on this board ([Counsel], illustrated below with a verified position).

Illustrated example

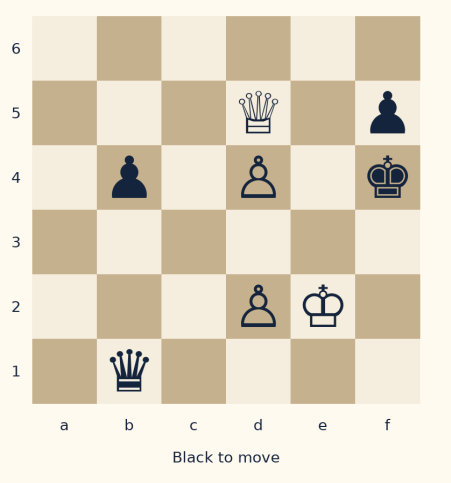


Figure 73: Black to move. The queens can come off, but is the pawn ending won?

Verified line: Qe4+ Qxe4+ Kxe4

[Demonstrated] Academy puzzle pz-am6uu4 ("Win material or a decisive advantage", difficulty 3): **1…Qe4+! 2.Qxe4+ Kxe4.** Black forces the queen trade with check. In the pawn ending that remains, Black's king is in the centre and the b-pawn is far from White's king.

Step-by-step reasoning

1. **Name the imbalance before the trade:** queens on, Black's king active on f4, Black's passed b-pawn and White's d-pawns.
2. **Name it after the trade:** no pieces at all. Black's king stands on e4 next to White's d-pawns, and Black's b-pawn has three steps to b1.
3. **Ask who is faster.** White's king on e2 must deal with the b-pawn *and* defend the d-pawns. It cannot do both.
4. **Check the forcing nature.** 1…Qe4+ is check, and 2.Qxe4+ is White's best reply according to the verified line. A forcing simplification leaves the opponent no time to complicate.

Common mistakes

- **Trading because "trades help the side ahead" without checking the ending.** Pawn endings are calculable, and calculation sometimes shows they are drawn.
- **Keeping queens on out of greed.** If you are winning, every extra tactical possibility is a chance for *your* opponent to swindle you.
- **Forgetting the timeout rule.** Under the Crown timeout rule you lose on time if your opponent has *any* non-king piece or pawn left, even one that could never mate. Simplifying into a technical ending with little time left is only safe if you can make the moves quickly. ([Demonstrated], src/rules/adjudication.ts, policy crown-adj-1.)

Advanced insight

Complexity is relative to the clock. In Blitz 3+2 a position needing a twelve-move calculation is a coin toss for both players. In Classical 15+10 the better calculator usually wins it ([Counsel]). The Lab's *Time management* dimension records whether you spend at least your median thinking time on critical decisions. Use it to see whether you rush the moments that matter ([Demonstrated], src/academy/lab.ts).

Worked example

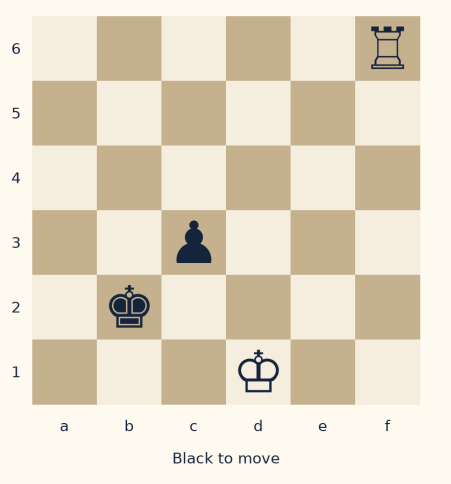


Figure 74: Black to move. A rook against a pawn that is two steps from promotion.

Verified line: c2+ Ke2 c1=Q

The material says White is better (a rook against a pawn). The *imbalance* says otherwise: White's rook is far away on f6, and the pawn advances with check, so White never has time to stop it. **1...c2+! 2.Ke2** (the reply in the verified line) **2...c1=Q** promotes. Black has turned "rook against pawn" into "queen against rook". **[Demonstrated]** puzzle pz-c0qvzm. The lesson: count tempi, not just material.

Summary

Name the imbalance before and after a trade. Simplify when what remains is decisive and easy to play, especially when you are short of time and the timeout rule applies. Keep the game complex when you are worse, have more time, or calculate better. Count tempi, not just material.

Visualise the changed board

After a capture, remove the victim from your mental board. After promotion, replace the pawn rather than imagining both pieces. Recompute the king's escape squares after every king move. For a drill, study a problem for thirty seconds, cover it, name every occupied square, then announce two legal plies and reconstruct the result. Accuracy comes before line length.

Stop a branch at a terminal result, a stable evaluation with a concrete reason, or a refutation. Do not stop because the opponent has made the move you wanted. Include a quiet candidate when checks and captures do not solve the problem. In the expert exercises, accepting a first move requires answering every legal defence.

PART IV · Opening Theory and Strategy

The First Ten Doors: Development on Six Files

Explanation

In the starting position White has exactly **ten** legal moves: six pawn steps (a3, b3, c3, d3, e3, f3) and four knight jumps (Na3, Nc3, Nd3, Nf3). The queen, rooks and king are walled in. So "development" here does not mean "bring out the minor pieces"; it means **open doors**. Each pawn step unlocks squares on the second rank:

- `c3` gives the queen c2 and, along the diagonal, b2–a3 after `b3`;
- `d3` gives the king and the queen d2, and the b1-knight d2;
- `b3` and `e3` open b2 and e2 for queen and king;
- `f3` and `a3` give the rooks a second-rank square (Rf2, Ra2) and later a lift.

Now the crucial six-file fact: **only two ranks separate the armies.** A White knight on c3 stands next door to rank 4, and every Black pawn on rank 5 needs just *one* step to attack rank 4 — and from rank 4 it attacks rank 3. A knight on a3 or c3 can be hit by `...b4`; a knight on c3 or e3 by `...d4`; and so on. A developed knight is a target from move one.

Engine judgement (method B, depth 14). Every White first move, scored for White:

Move	c3	f3	d3	b3	Nf3	a3	e3	Na3	Nc3	Nd3
Score (cp)	+54	+50	+35	+27	−9	−51	−51	−60	−101	−101

At depth 12 the order is f3 +54, c3 +46, d3 +37 and again Nc3/Nd3 last at −108. The engine's own six-ply opening tree (method E, depth 7) prefers d3, b3 and c3. All three sources agree on the shape: **central pawn steps first, knight jumps last.** The engine's square tables say the same thing in another voice: a knight on c3 costs it 38 cp and on d3 31 cp in the middlegame, while d2 (+34) and a3 (+21) are its favourite home squares for a knight that is not yet ready to jump forward.

Evidence (method A). In the balanced archive, games opening with `c3` scored 64.7% for White (440 games) and `f3` 63.5% (544), while `Nc3` scored 49.5% (596) and `a3` 49.0% (768). Treat this gently: the book chose the openings and Black's reply varied, so these are *associations*, not experiments. But the direction matches the engine.

Evidence (method E). Fairy-Stockfish's balanced book contains all ten first moves, so none of them simply loses. Strategy here is a matter of tens of centipawns, not of blunders.

Evidence (method A). Contact comes quickly: the median game sees its **first capture on ply 7**, i.e. Black's fourth move.

Illustrated example

Step-by-step reasoning

Let us test the most "natural" developing move of the older game, the knight to c3.

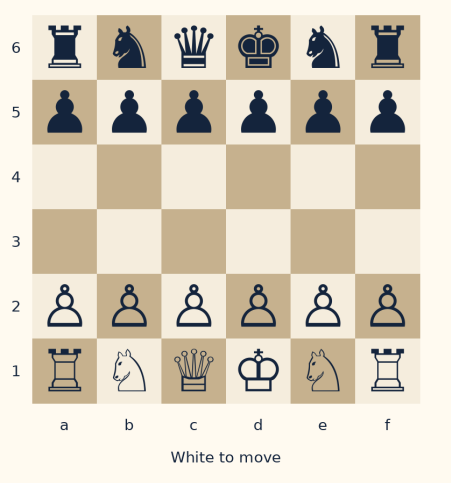


Figure 75: The starting position. Ten legal moves: six pawn steps and four knight jumps. The rooks, queen and king are locked in, and both rooks are undefended.

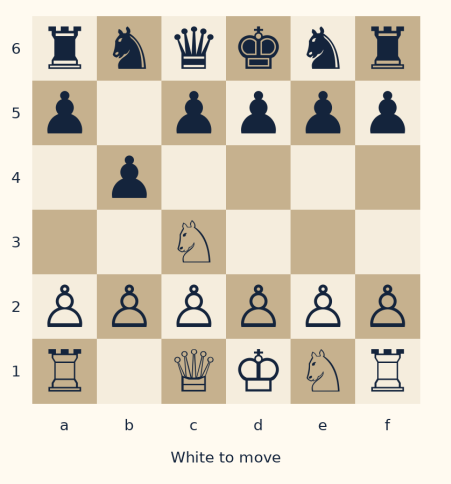


Figure 76: After 1. Nc3 b4. One pawn step and the knight must move again. The engine’s main line continues 2. Nb1, back home.

Verified line: Nc3 b4

1. `1. Nc3` puts the knight on a square that two Black pawns can attack in a single step: `...b4` and `...d4`.
2. `1...b4` attacks it at once (the engine's own line at depth 14 is 1...c4 2.Nb1 d4, with the same idea).
3. White has spent a move and must spend another. Black's b-pawn now stands on b4, controlling a3 and c3 — *space* gained for free.
4. Compare `1. c3`: the pawn takes b4 and d4 away from Black's pawns and knights, gives the queen c2, and cannot be chased.

Common mistakes

- Developing knights first "because that is what one does". Here a knight on rank 3 is the most kickable piece on the board.
- Pushing an edge pawn (`a3`, `e3`) with no plan. The engine rates `a3` and `e3` among the weaker moves: they unlock little.
- Forgetting that both rooks begin undefended. An early opened a- or f-file can lose a rook (Chapter III.6).

Advanced insight

The engine is not against knights; it is against knights *that can be hit*. Once the enemy pawn that could attack a square has advanced past it or been exchanged, that square becomes a fine home. That is why the order matters: first the pawn steps that fix the pawn structure, then the knight jumps to squares that no pawn can reach.

Worked example

Here is the engine playing itself from the start at depth 11 (method B), eight moves of real contact:

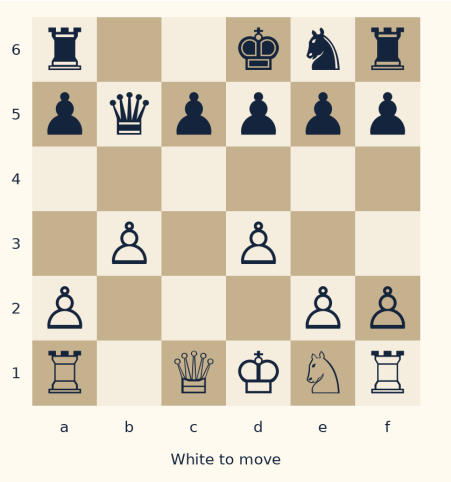


Figure 77: Engine self-play, depth 11: 1. c3 b4 2. d3 bxc3 3. Nxc3 Na4 4. Nxa4 Qxa4+ 5. b3 Qb5. Pawns first, knights only when they can land safely or trade.

Verified line: c3 b4 d3 bxc3 Nxc3 Na4 Nxa4 Qxa4+ b3 Qb5

Follow the logic. `1. c3` opens c2 and controls b4/d4; Black challenges at once with `1...b4`. `2. d3` opens d2 and prepares to recapture toward the centre. After `2...bxc3 3. Nxc3` the White knight *does* reach c3 — but now Black's b-pawn is gone, so nothing can kick it. Black trades it off with `3...Na4 4. Nxa4 Qxa4+`, and the queen's check on the a4–d1 diagonal (open because the c-pawn left c2) is answered by `5. b3`. Six of the first ten moves were pawn moves or captures.

Summary

- Ten first moves; only knights and pawns can move.
- Pawn steps are the real developing moves: they open the second rank for queen, king and rooks.
- A knight on rank 3 can be attacked by one enemy pawn step. The engine rates the knight jumps as the worst first moves, by about one pawn.
- First capture typically comes on ply 7: there is no quiet opening phase.

The Crown Without a Castle: King Safety

Explanation

The shield. The engine counts the own pawns on the three squares directly in front of the king (for a king on d1: c2, d2 and e2) and pays **+18 cp each in the middlegame but −14 cp each in the endgame** (method D). Read that sign change carefully: the same pawns that protect the king early become a cage later, when the king should be walking toward the action. The engine's king square table agrees: in the middlegame its favourite squares are on the first two ranks (f2 +19, b1 +7, f1 +6), in the endgame the centre and beyond (c4 +18, d4 +21, d5 +18).

The d-file. The kings start face to face on d1 and d6. If both d-pawns leave the d-file, every rook or queen that reaches it gives check. The queens begin on c1 and c6, one diagonal step from the d-file. This makes the d-pawns the most delicate pawns on the board.

The king walks. There is no castling, so a king that wants safety must walk. **Evidence (method A):** in 61.1% of balanced games at least one king moved within the first 20 plies (3,400 of 5,568). The walk usually has two purposes at once: to step away from the d-file and to connect the rooks, which cannot see each other while the king stands on d1.

Evidence (method A). Both queens were off the board by ply 40 in 64.5% of balanced games (3,592 of 5,568). Early queen trades are normal here, and after them king safety gives way to king activity.

Illustrated example

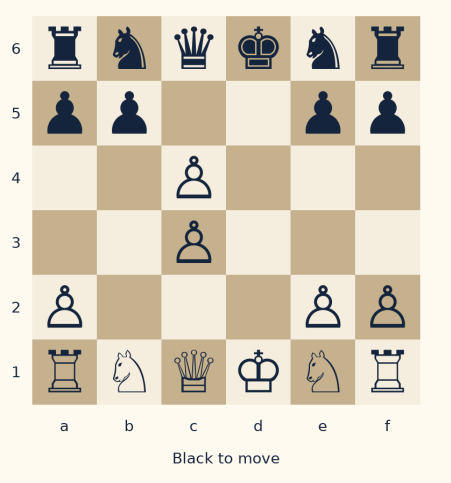


Figure 78: An illustrative sequence: 1. d3 c4 2. dxc4 d4 3. c3 dxc3 4. bxc3. Both d-pawns have left the d-file; the kings now face each other down an open file, and either queen can reach it in one move.

Verified line: d3 c4 dxc4 d4 c3 dxc3 bxc3

Step-by-step reasoning

Read the position after `4. bxc3` like a strategist:

1. **Count the shields.** White's king on d1 has only the e2-pawn in front of it (c2 and d2 are empty): shield 1. Black's king on d6 has only the e5-pawn (c5 and d5 are empty): shield 1. Both kings are half-exposed.
2. **Look along the d-file.** d2, d3, d4 and d5 are empty. A queen or rook arriving on the file gives check.
3. **Ask what each side wants. Engine judgement (depth 12):** Black's best is `4...Nxc4` (+26 for Black), regaining the pawn; in the engine's main line Black's queen later checks on d5. The safer king is the one whose side controls the open file first.
4. **Plan the escape.** Each king has one sideways step off the open file: White's to c2, Black's to c5 (the other neighbouring squares are occupied by their own pieces and pawns).

Common mistakes

- Trading the d-pawn off the d-file without asking who will control the file afterwards.
- Keeping the king on d1 "because it is safe there" after the queens are traded. With the queens gone, the shield is worth *less than nothing* to the engine (−14 per pawn).
- Moving the e2- or c2-pawn and then leaving the king on d1 with an open diagonal toward the enemy queen. Only the queen can exploit a diagonal, but she does it very well.

Advanced insight

Because there are no bishops, a king's diagonal danger comes from one piece only: the enemy queen. When the queens leave the board, the king can walk through diagonals that would be suicidal in the older game. That is why so many games here end with kings in the middle of the board escorting pawns.

Worked example

A real Academy position (from engine play) shows what an exposed king invites.

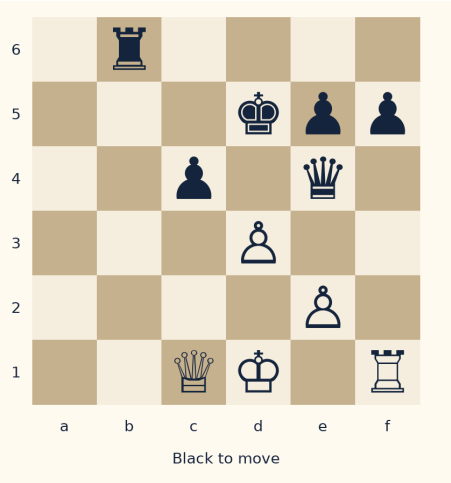


Figure 79: Academy puzzle pz-y788ac. Black to move. White's king on d1 has the queen on c1 beside it and a pawn on e2 in front. Black's queen is on e4.

Black plays `16...cxd3!` — the c4-pawn captures on d3 and lands **next to** the White king. Now the pawn on d3 covers c2 and e2, so `...Qxe2#` is threatened: the queen would take the e2-pawn protected by the d3-pawn, and the king could not escape to c2. **Engine judgement (depth 10):** `16...cxd3` scores +691 for Black; the next best, `16...Qd4`, only +44. One pawn wedge beside the king was worth more than six pawns.

Summary

- The shield (pawns in front of the king) is good early (+18 each) and a cage late (−14 each).
- The d-file between the two kings is the most dangerous line on the board; trade the d-pawns only when you will control it.

- Kings walk early in most games (61% within 20 plies) — to leave the d-file and to connect the rooks.
- Without bishops, only the queen attacks along diagonals; when queens go, kings come out.

The Heart of the Board: Centre and Space

Explanation

The board has no single centre square; it has a **central block** of four: c3, d3, c4, d4. Every piece is stronger there: a knight on c3 or d4 reaches 8 squares (on a6 only 2), a queen on c3 reaches 19 on an empty board.

Evidence (method C). How much is one pawn worth, by file? Removing one Black pawn from the starting position and searching to depth 12 (colours swapped and averaged) gives White the following advantage:

Missing pawn	a	b	c	d	e	f
Value (cp)	+13	+182	+119	+134	+178	−4

The four inner pawns are worth well over a pawn each; the edge pawns are worth almost nothing in the opening. (Removing an edge pawn also half-opens a file for the enemy rook, which partly explains the small numbers — that is the honest limit of the experiment.)

Evidence (method D). The engine's middlegame square table for White pawns is decisive about *where* the inner pawns want to be:

Rank	a	b	c	d	e	f
5 (one step from crowning)	+33	+33	+39	+36	+39	+28
4	+8	+7	**+78**	**+52**	**+44**	+18
3	−20	−14	+16	+29	+9	−6
2 (start)	−12	−28	−16	−3	−2	−23

A pawn on c4 or d4 is worth most — when it can *stay* there. But a White pawn on rank 4 is in contact with Black's starting pawns at once (c4 is attacked from b5 and d5). So the fight for the centre is a fight over **who can support a pawn on the fourth rank**, and the usual first step is a pawn on the third rank that controls the fourth.

Space. On this board, space is the territory behind your pawn front. A pawn on d3 controls c4 and e4: enemy knights cannot land there and enemy pawns cannot settle there. A pawn on d4 controls c5 and e5 — the starting squares of the enemy pawns — so it is an immediate *threat*, not territory. **Guidance:** gain space with third-rank pawns; advance to the fourth rank when you can recapture there with a pawn.

Illustrated example

Verified line: d3 c4

Step-by-step reasoning

1. **What does d3 do?** It attacks c4 and e4 and opens d2 for the king, queen and b1-knight. It also *removes* one pawn from the king's shield (the shield counts c2, d2 and e2). Every pawn move has a price.
2. **What does …c4 do?** It challenges d3 at once and controls b3 and d3.
3. **Who should capture? Engine judgement (depth 12, method B):** after `1. d3 c4` the engine does *not* capture; it plays `2. c3`, inviting `2...cxd3 3. Nxd3`, so that White recaptures toward the centre with a piece and keeps the c3-pawn controlling b4 and d4.

Verified line: d3 c4 c3 cxd3 Nxd3

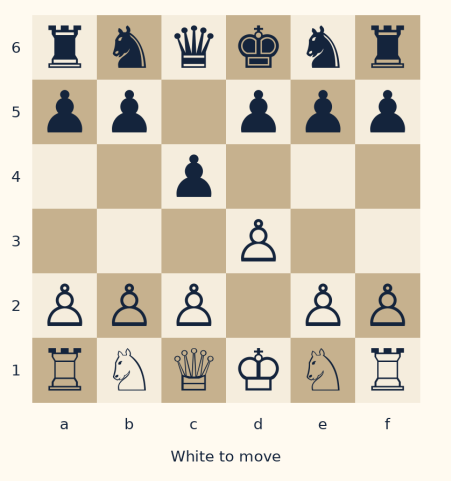


Figure 80: After 1. d3 c4. The two pawns attack each other: White can play dxc4, Black can play …cxd3. The question of the centre is asked on move one.

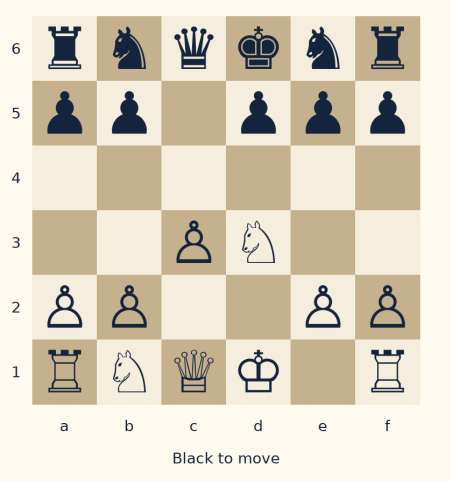


Figure 81: 1. d3 c4 2. c3 cxd3 3. Nxd3. White’s pawn on c3 controls b4 and d4; the knight on d3 eyes b4, c5, e5 and f4. Black’s c-pawn has been traded for White’s d-pawn.

4. **Read the new centre.** White has a pawn on c3 and a knight on d3; Black still has the d5 and e5 pawns. **Engine judgement (depth 12):** roughly level (Black's best `3...e4` scores −29 for Black). The knight on d3 can be hit by `...e4` — every central piece must ask which pawn can reach it.

Common mistakes

- Treating a fourth-rank pawn as "space". It is contact, and often a target.
- Pushing the a- and f-pawns to "gain space". The engine values those pawns least of all in the opening.
- Recapturing away from the centre (for example with a wing pawn) when a central recapture is available.

Advanced insight

Because pawns move only one step and there are just two empty ranks, *every* pawn advance hands squares to the opponent: the squares the pawn used to guard can never be guarded by it again. Strategists count not only the squares a move attacks but also the squares it abandons.

Worked example

Evidence (method C), the inner pawns compared. Suppose Black has lost a pawn in the opening. Which pawn would you rather Black had lost? The experiment's answer: b- or e-pawn (about +180 cp for White), then d (+134) and c (+119), and an edge pawn hardly matters (a: +13, f: −4). Why are b and e so high? Because the b- and e-pawns guard the c- and d-files' neighbouring squares (a4/c4 and d4/f4) and shield the knights' natural squares; without them, White's pawns reach the fourth rank unchallenged. **Guidance:** in the opening, trade a wing pawn for a central one whenever you can.

Summary

- The central block is c3, d3, c4, d4; inner pawns are worth more than a pawn each in the opening, edge pawns very little (method C).
- The engine's favourite pawn squares are c4, d4 and e4 — but a fourth-rank pawn is in contact at once.
- Gain space with third-rank pawns; advance when you can support the advance with a pawn.

The Double-Edged Hourglass: Tempo, Initiative and Coordination

Explanation

A tempo is real. The engine adds **+65 cp** in the middlegame simply for being the side to move (method D). The starting position is +37 to +54 cp for White at depths 12–14 (method B). **Evidence (method A):** White scored **55.9%** in the 5,566 finished balanced games. On this board the first move is a genuine advantage.

Threats buy time. The largest middlegame terms after material are the threat terms: a pawn attacking a knight, rook or queen (+128), a rook attacking the queen (+110), a knight attacking a rook or queen (+47). A threat forces a reply; if your threatening move also improves your position, you have *gained* a tempo. If the opponent's reply also improves *their* position, nobody gained anything.

Initiative is a chain of such moves: you make threats, the opponent answers, and the answers are worse moves than they would have chosen freely.

Coordination without bishops. Your long-range pieces are the queen and two rooks; your short-range pieces are the knights and king. Good teams: - **Queen + knight.** The knight's jumps cover exactly the squares a queen cannot reach in one move from the same spot, so the two complete each other around an enemy king. - **Rook + rook.** Doubled on an open file, they

are the board's strongest battering ram. - **Rook + pawn.** A rook behind a passed pawn (Book IV).

Evidence (method A, the 640-game confirmation match). On average a position offers only **12.9** legal moves (14.2 in the first 20 plies, 10.5 after ply 60). 21.4% of all moves played were captures and 14.1% gave check. With so few choices, each move's purpose must be sharp.

Illustrated example

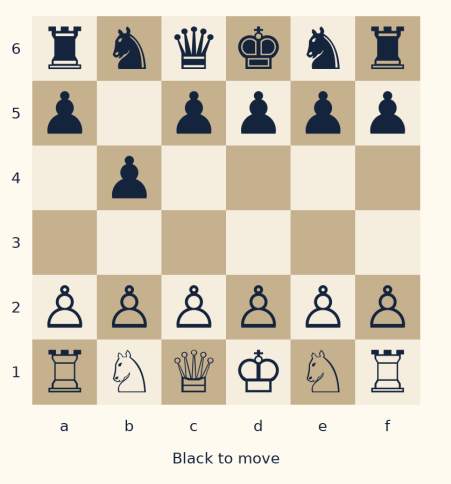


Figure 82: 1. Na3 b4 2. Nb1. White has spent two moves to return to the start; Black has a pawn on b4 for free and it is Black's move. That is a tempo gained by a threat.

Verified line: Na3 b4 Nb1

Step-by-step reasoning

1. `1. Na3` develops — but onto a square Black can attack with one pawn step.
2. `1...b4` is a threat (pawn attacks knight: the engine's biggest positional term) *and* an improving move (space, control of a3 and c3).
3. `2. Nb1` answers the threat but improves nothing. Net result: Black gained a full move. **Engine judgement:** this is the engine's own line at depth 12; `1. Na3` scores −72 at depth 12 and −60 at depth 14.

Common mistakes

- Making a threat that the opponent answers with a move they wanted to make anyway.
- Retreating a piece twice in the opening. Each retreat is a tempo given away.
- Leaving pieces where they do not protect one another. The engine's "hanging piece" term (+25 for the attacker) is small, but the tactical cost is not.

Advanced insight

On a board with 12.9 moves per position, the defender's resources are fewer than in the older game. A single tempo can turn "I can defend" into "I am one move short". This is why the engine's tempo term is so large compared with the material of a pawn (65 versus 89).

Worked example

Coordination and a discovered attack, from engine play:

Black plays `8...Nf4+!`. The knight checks the king on e2 from f4 and, by leaving d5, opens the queen's diagonal c6–d5–e4–f3. After `9. Ke1` the queen takes the knight: `9...Qxf3`. A check and a discovered attack in one move. **Engine judgement (depth 10):** +459 for Black; the quiet `8...Re6` only −132. Knight, queen and the advanced e3-pawn (which keeps the king boxed in) worked together around the exposed king.

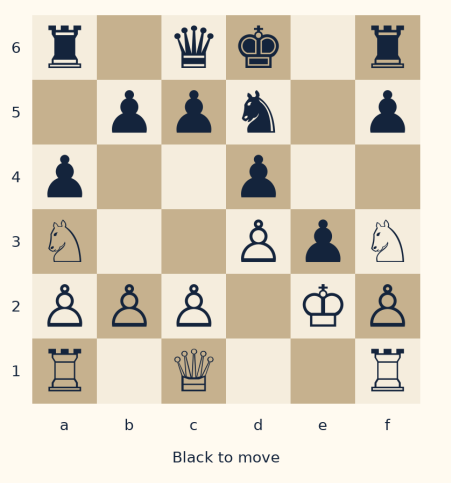


Figure 83: Academy puzzle pz-p05pjm. Black to move. White's king stands on e2; Black's knight on d5 and pawn on e3 are close by.

Summary

- Tempo: +65 cp to the engine; White scores 55.9% in balanced engine games.
- Threats gain time only if the threatening move also improves your position.
- Few legal moves per position (12.9 on average) make every tempo count.
- Queen + knight and rook + rook are the natural teams of a bishopless army.

Open Roads and Safe Harbours: Files, Weak Squares and Outposts

Explanation

Open files. A rook on a file with no pawns earns **+68 cp** in the middlegame; on a half-open file (only enemy pawns) **+48** (method D). These are among the largest positional terms the engine has. With only six files, a single open file often decides who controls the game.

Facing rooks. Remember the opening fact: at the start, nothing defends a1 or f1 (or a6 and f6). When the a-pawns or f-pawns are exchanged early, the two rooks on that file stare at each other with nothing between them and **neither rook defended**. Whoever is on move captures a whole rook. Many Academy puzzles from engine play (for example pz-1alc6nh, pz-1kvvn1q, pz-146whg9, pz-54922z, pz-1cm516w, pz-sawrgf) are exactly this, with the solution "take the rook first".

Weak squares and outposts. A square is *weak* for a side when none of its pawns can ever attack it again (because the pawns on the neighbouring files have advanced past it or disappeared). An enemy knight on such a square, defended by a pawn, is called an outpost. The engine's code detects outposts (a knight on its 3rd to 5th rank, defended by a pawn, not attackable by enemy pawns) — but in the shipped weights the outpost term is **zero**. The engine values those knights through its square table instead: in the middlegame a White knight earns +22 on c4, +28 on e5 and +12 on d5, but −38 on c3 and −31 on d3 (method D). So: the engine likes advanced central knights; it does not pay extra merely because a pawn defends them.

Illustrated example

Step-by-step reasoning

1. **See the file.** No pawn stands on f2, f3, f4 or f5. The rooks on f1 and f6 attack each other.
2. **Check the defenders.** White's f1-rook: the e1-knight covers c2, d3, f3 — not f1. The king on d1 does not reach f1. Undefended. Black's f6-rook: the e6-knight covers c5, d4, f4 — not f6. Undefended.

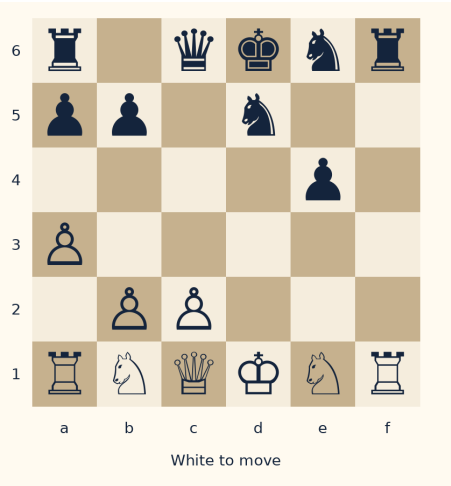


Figure 84: Academy puzzle pz-1cm516w. White to move. The f-file is open; Rf1 and Rf6 face each other, and neither is defended.

3. **Conclusion:** the side to move wins a rook. White plays `7. Rxf6`, and after `7...Nxf6` the material is level again — whereas any other move allows `...Rxf1`, losing a rook. **Engine judgement (depth 10):** `7. Rxf6` −37; every alternative −489 or worse.

Common mistakes

- Opening the a- or f-file in the opening without checking whether your rook is defended.
- Putting a rook on an open file that the opponent can contest at once with an undefended rook of their own — the same pattern in the middlegame.
- Paying a pawn to plant a knight on an "outpost" that the opponent can simply exchange off with a knight. Without bishops, only knights (and the queen) can challenge a knight on a pawn-proof square.

Advanced insight

On six files, a half-open file is only one pawn exchange away from an open one, and a rook's whole file is just six squares long. That is why the engine rates the open file at more than half a pawn (68 cp) while rating an outpost at nothing extra: rooks gain far more from a single open line than knights gain from a protected square.

Worked example

A real outpost, as the engine's code defines it:

What *can* challenge the knight? Only Black's knight, by reaching a square from which it attacks d4 (b3, c2, c6, e2, e6 or f3) — from b6 that takes three moves, for example `...Nc4`, `...Ne5`, `...Nc6`. **Guidance:** a pawn-proof square is valuable when the opponent has no knight left to contest it. Without bishops, that question is asked of knights alone.

Summary

- Open file +68, half-open +48: rooks love the few open lines of a six-file board.
- Facing rooks on the a- and f-files: both start undefended; the side to move takes first.
- The engine detects outposts but weights them zero; its knight square table (best: e5, c4, d5) carries the knowledge. Without bishops, only a knight can challenge a pawn-proof knight.

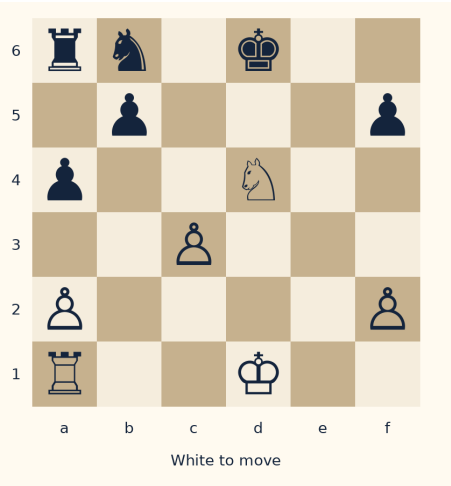


Figure 85: White's knight on d4 is defended by the c3-pawn, and no Black pawn can ever attack d4 (there are no Black pawns on the c- or e-files). The engine's trace flags it as an outpost — and adds nothing for it, because the outpost weight is zero.

The Long Spell: Planning and Prophylaxis

Explanation

A plan answers four questions, in this order:

1. **What is threatened?** Your first duty is to see the opponent's threat. With 12.9 moves per position and many loose pieces, threats are everywhere.
2. **What does the structure ask for?** Open files for rooks, passed pawns to push or stop, a weak square to occupy.
3. **Which is my worst piece?** Improve it.
4. **Where is the target?** A weak pawn, the exposed king, a loose piece.

Prophylaxis is question 1 turned into a habit: before every move, ask what the opponent would play if it were their move again. A prophylactic move stops that idea *and* improves your position.

Engine judgement. The engine is not a planner in words; it searches. But when its top choices are equal, each corresponds to a different plan, and you may choose by understanding. When one quiet move is far ahead of all others, that move is usually prophylaxis: it meets a threat you might not have noticed.

Illustrated example

Step-by-step reasoning

1. **The threat.** Black's knight on c4 attacks the b2-pawn, and from b2 a knight attacks d1: `...Nxb2+` would win a pawn with check. That must be met.
2. **The structure.** The d-file is fully open. Black's e5-pawn controls d4; if it advances to e4, d4 becomes a fine square for a White knight.
3. **Worst piece.** White's knight on c1 and king on d1 (blocking the rooks).
4. **Candidate plans (Engine judgement, method B, depth 12, every move scored):**

Move	Score	Plan
`Nd3`	+31	defend b2 with the knight, which also eyes c5, e5, b4, f4
`Kc2`	+31	defend b2 with the king, connect the rooks
`Rb1`	−14	defend b2 with a rook (passive)
`Ke2`	−73	allows `...Nxb2`

Move	Score	Plan
`b3`	−93	"kick the knight": after `...axb3 Nxb3` White's a2-, c3- and f2-pawns are all isolated

Two plans are equal, and both do two jobs at once: they meet the threat **and** improve a piece. The tempting `b3`, which "gains a tempo" on the knight, is the worst of the natural moves.

Verified line: Kc2 e4 Ne2 Kc5 Rfe1

At depth 15 the engine switches to `Nd3`; both plans are within a few centipawns. That is a lesson in itself: in balanced positions, good plans are plural.

Common mistakes

- Choosing a plan before checking the opponent's threat.
- "Gaining a tempo" by attacking a piece that simply steps to a better square (`b3?` above).
- Making a plan with your best piece instead of your worst one.

Advanced insight

Every pawn move is a plan you cannot take back. In the diagram, Black's `...e4` gains space but hands White the d4-square forever. Before you push a pawn, name the square it stops guarding.

Worked example

Prophylaxis in its purest form: a quiet rook move that the engine prefers by more than four pawns.

What does Black want? `...Qxe2+` (the queen reaches e2 along the b5–e2 diagonal, supported by the f4-knight). White's answer is `8. Re1!`, defending e2 with the rook. **Engine judgement (depth 10):** −24 for White after `8. Re1`; the active-looking `8. Ne4+` is −445 and `8. c3` is −822. One quiet move, found by asking "what does my opponent want?"

Summary

- Plans start from threats (prophylaxis), then structure, worst piece and target.
- When the engine shows several equal moves, they are several equal plans; choose the one you understand.
- When one quiet move is far ahead, it is usually prophylaxis.

Turning Gold into Crowns: Conversion, Tactics and Patience

Explanation

Evidence (method A, balanced games). At the first quiet moment from ply 40 onward (no capture or pawn move in the previous two plies, side to move not in check), material was counted (pawn 1, knight 3, rook 5, queen 9). How did the side ahead fare?

Material lead	Games	Win	Draw	Loss
Level (White's result)	1,323	24.9%	54.5%	20.6%
+1	1,501	48.0%	39.6%	12.3%
+2	1,003	68.3%	23.9%	7.8%
+3 or +4	852	83.1%	11.6%	5.3%
+5 or more	615	93.0%	5.9%	1.1%

A single extra pawn wins less than half the time even between engines. Conversion is a skill, not a formality.

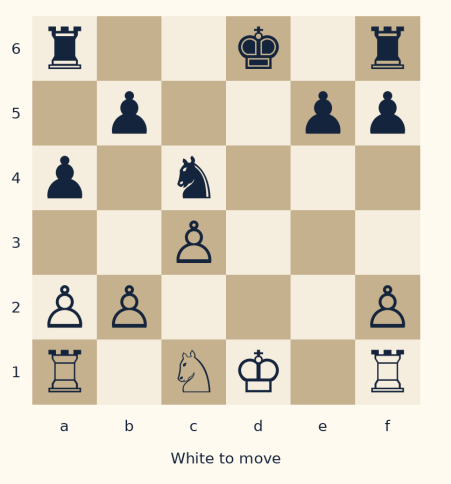


Figure 86: A planning position. Rooks and a knight each, four pawns each. Black’s knight on c4 attacks b2. White to move.

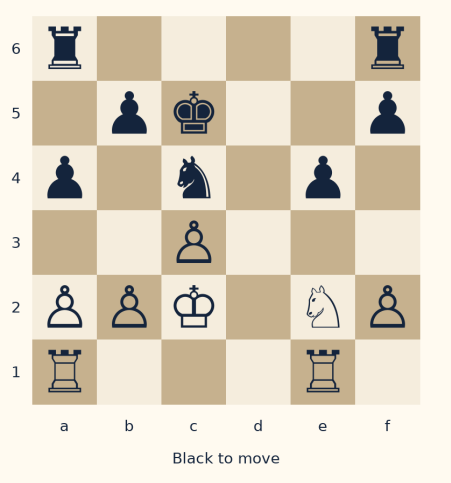


Figure 87: The king plan (engine line at depth 12): 1. Kc2 e4 2. Ne2 Kc5 3. Rfe1. White’s king guards b2 and connects the rooks; after …e4 the d4-square is free of Black pawn control and the knight heads there.

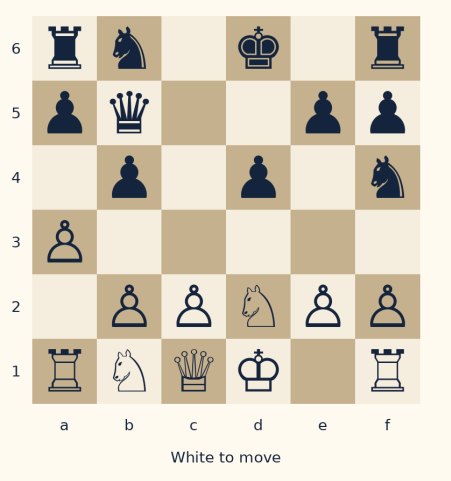


Figure 88: Academy puzzle pz-qcmiyc. White to move. Black’s queen on b5 and knight on f4 look at e2.

Three rules of conversion (Guidance, built on the evidence above and in III.7): 1. **Trade pieces, not pawns.** Pawns are what you will crown; and with no pawns the engine's own rule multiplies a small material lead by 0.25. 2. **Aim for the ending that wins most.** With an extra pawn, pawn endings won 77.4% of the time, knight endings 49.0%. 3. **Prefer certainty to brilliance.** If a simplification leads to a position you *know* is won, take it, even if a sharper line wins faster.

Tactics or patience? Look at the position's forcing moves: checks, captures, threats. If they exist and lead somewhere, calculate (Book IV). If they do not, improve your worst piece. On this board the share of forcing moves is high — 21.4% of moves played by engines were captures and 14.1% gave check (method A) — so the tactical check is never optional.

Illustrated example

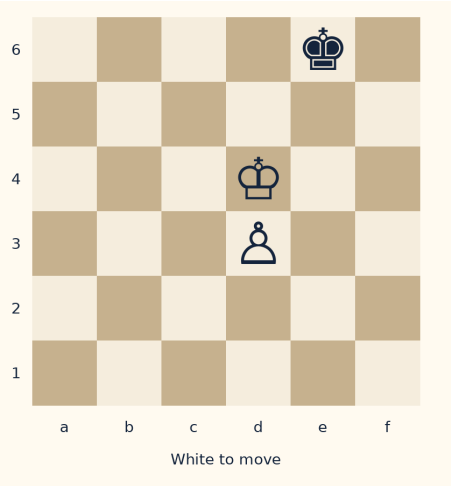


Figure 89: A conversion choice. White traded rooks with 1. Rxe6+ Kxe6, reaching a king-and-pawn ending that is won with certainty: proven by exhaustive enumeration (mate in 9 more moves with best play).

Verified line: Rxe6+ Kxe6

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

Step-by-step reasoning

1. **Before the trade.** White has rook and pawn against rook. **Engine judgement (depth 12):** both `1. Ra1` (+1389, threatening `Ra6+` and a skewer) and `1. Rxe6+` (+1060) win.
2. **After the trade.** White's king on d4 stands directly in front of its pawn on d3, on one of the pawn's *key squares* (Book IV). **Evidence (exhaustive enumeration):** every legal king-and-pawn position was solved by retrograde analysis over `src/rules` for this book, and this one is a forced win: White mates in 9 moves.
3. **The choice.** `Ra1` is stronger in the engine's eyes, but it requires seeing a skewer; `Rxe6+` turns the game into an ending whose result is known. For a human converting under a clock, the second is the safer road.

Common mistakes

- Trading pawns when ahead: every pawn trade brings you closer to a pawnless ending, which rarely wins.
- Rushing a winning position into complications.
- Playing "slow" moves in a position where the opponent has forcing moves.

Advanced insight

The engine's evaluation does not "know" which endings are won; it estimates. A strategist who knows the exact facts of simple endings (Book IV) can convert more reliably than an engine's estimate suggests. This is one of the few places where knowledge beats calculation.

Worked example

Tactics first, even in an ending.

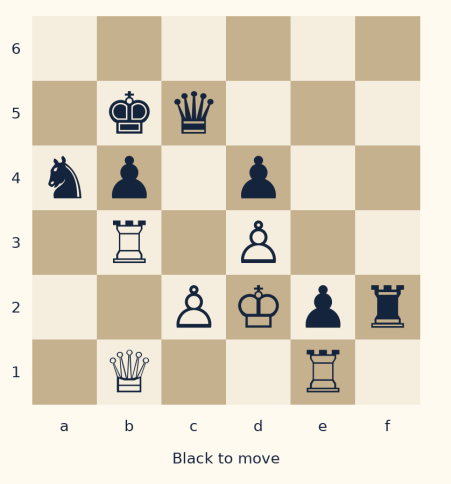


Figure 90: Academy puzzle pz-15sshpo. Black to move. Black is under pressure, but White's pieces on b3, b1 and e1 are loosely placed.

Look first at the second rank: Black's pawn on e2 has almost crowned, defended by the rook on f2, and White's king (d2) and rook (e1) blockade it. Black plays `22...Nc3!`. The knight, protected by the b4- and d4-pawns, attacks White's queen on b1 and also covers d1 and e2's neighbourhood. White cannot save the queen and keep the blockade together. **Engine judgement (depth 10):** +467 for Black; the "patient" `22...Kc6` only +76.

Summary

- One extra pawn wins about half the time; three points about 83% of the time (balanced engine games).
- Trade pieces, not pawns; head for pawn endings when a pawn up; take certainty over brilliance.
- Always check the forcing moves first: on this board they are frequent.

What "best-supported" means

No first move is claimed solved. The initial position has ten legal moves. The published crown-repertoire-1 artifact stores a depth-7 shortlist with d3 highest in that particular finite search. That is a historical engine preference, not minimax proof. A move missing from the shortlist is not thereby illegal or refuted. Study a central-pawn plan, a knight-first plan, and a response to an early capture; those are descriptive categories, not invented official opening names.

A gambit needs concrete compensation: time, a line, a forcing threat or a recoverable pawn. Calculate refusal as well as acceptance. A trap is conditional: identify the opponent's mistake and the safe deviation. Do not sell a tactical pattern as a forced win from the starting position.

PART V · Endgame Mastery

The Crown's March: King Activity and King-and-Pawn Endings

Explanation

King activity. In the endgame the engine's king table rewards the centre (d4 +21, c4/d5 +18) and its shield term turns negative (Book III). The king is a fighting piece: walk it forward.

Evidence (exhaustive enumeration): king and pawn against king. All 22,484 legal positions with White to move were solved. White wins **76.0%** of them; by file of the pawn: a- or f-pawn 71.6%, b- or e-pawn 76.7%, c- or d-pawn 79.8%. The longest win takes 20 moves (to mate, including the promotion), from `6/k5/6/6/K3P1/6 w`. The rook pawn is harder to win, but far from the hopeless case of the older game.

Key squares (Evidence, exhaustive enumeration). A *key square* is a square such that, if the stronger king stands there, it wins whatever the defender does (with the defender to move; positions where the pawn simply hangs are excluded). Computed for a d-pawn:

```
Pawn on d2          Pawn on d3          Pawn on d4
6 . . . . . .       6 . . . . . .       6 . . * * * .
5 . . . . . .       5 . . * * * .       5 . . * * * .
4 . . * * * .       4 . . * * * .       4 . . . P . .
3 . . . . . .       3 . . . P . .       3 . . . . . .
2 . . . P . .       2 . . . . . .       2 . . . . . .
1 . . . . . .       1 . . . . . .       1 . . . . . .
  a b c d e f         a b c d e f         a b c d e f
```

Rule: **for a pawn on its 2nd rank the key squares are two ranks ahead of it; from the 3rd rank on, the key squares are the squares one and two ranks ahead** (on the pawn's file and the two neighbouring files). For a b-pawn on b3 the key squares are a4–c4 and a5–c5; for a rook pawn on a3 the only key square is b5.

Opposition. When the kings stand on the same file with one square between them, the side *not* to move "has the opposition": the other king must give way. In king-and-pawn endings this often decides everything (see the illustrated example).

The rule of the square. Draw a square from the pawn to its promotion rank (for a pawn on f3: f3–f6 and across to c3–c6). If the defending king is inside it — or, with the defender to move, can step into it — the king catches the pawn. **There is no exception for the starting rank here**, because pawns never move two squares. **Evidence (exhaustive enumeration):** pawn on f3, White king far away on a1, Black king on a5 with Black to move: lost (a5 cannot step into the c-to-f square); Black king on b5: drawn. Pawn on f2, Black king on a5: drawn with Black to move (b5 is inside the larger square), lost with White to move.

Illustrated example

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

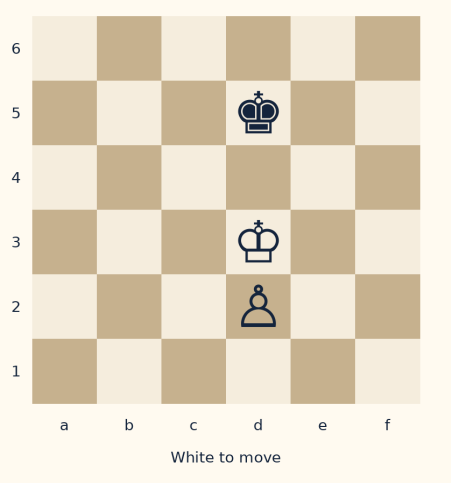


Figure 91: Opposition, White to move. Kings on d3 and d5, pawn on d2. Evidence (exhaustive enumeration): with White to move this is a draw — White must give way.

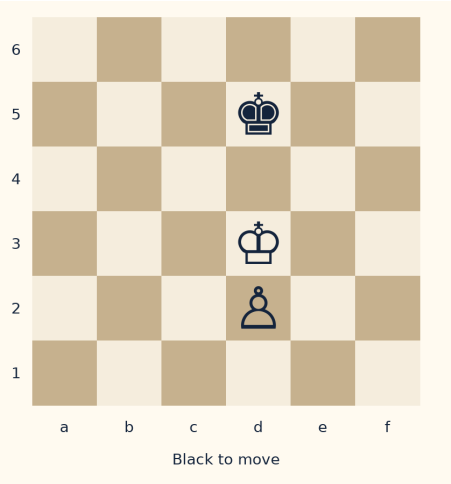


Figure 92: The same position with Black to move. Evidence (exhaustive enumeration): White wins (mate in 12 with best play), because Black’s king must step aside and White’s king reaches a key square (c4, d4 or e4).

Step-by-step reasoning

Why does the move matter?

1. **Black to move.** Black's king on d5 must leave. If it goes to c5 or c6, White's king steps to e4; if to e5 or e6, White goes c4. White's king reaches a key square of the d2-pawn (c4, d4, e4) and wins.
2. **White to move.** Every White king move gives up the opposition: after `Kc3` Black answers `...Kc5`, after `Ke3` `...Ke5`, and Black's king keeps White out of the key squares. Draw.
3. **The lesson.** With the kings face to face, the side that does *not* have to move is the one in control.

Common mistakes

- Pushing the pawn first. The pawn should usually follow the king, not lead it: the key squares are in front.
- Stepping back with the king. **Evidence (exhaustive enumeration):** in the Book III conversion example (`4k1/6/3K2/3P2/6/6 w`), `Kc5`, `Kc4` and `Ke4` win but `Kc3` and `Ke3` only draw.
- Applying the eight-by-eight "starting-rank exception" to the rule of the square. There is none here.

Advanced insight

The key-square rule on six ranks is the older game's rule compressed: the board's two empty middle ranks mean a pawn on its 3rd rank is already "half-way", and its key squares reach the far side of the board. That is why, in a pawn race, the king that arrives first in front of its pawn decides the game.

Worked example

Taking the opposition at a distance.

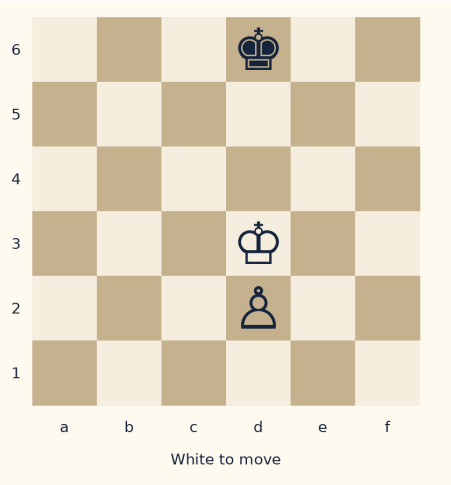


Figure 93: White to move. Black's king on d6, White's on d3, pawn on d2. Evidence (exhaustive enumeration): White wins with 1. Kc4, 1. Kd4 or 1. Ke4 — each reaches a key square. With Black to move, only 1…Kd5! (taking the opposition) draws.

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

With White to move, `1. Kd4` puts the kings on d4 and d6 with Black to move: Black must give way and White's king advances to c5 or e5 next. With Black to move, `1...Kd5` stops the king at once: White's king cannot reach c4, d4 or e4 without Black answering in opposition.

Summary

- Key squares: two ranks ahead for a 2nd-rank pawn; one and two ranks ahead from the 3rd rank on. King on a key square = win.
- Opposition: with the kings face to face, the side not to move holds the opposition.
- The square rule has no starting-rank exception. Rook pawns win less often (71.6% of positions with White to move) but often still win.

The Tower's Patience: Rook Endings

Explanation

Rook endings are the commonest endings. Evidence (archive): of the "pure" endings first reached in the balanced games, rook endings were by far the most frequent (1,511, against 428 knight, 286 pawn and 125 queen endings). With equal pawns they were drawn 53.5% of the time; the side with more pawns won 62.2% and drew 28.7%.

King and rook against king (Evidence, exhaustive enumeration). Every one of the 25,440 legal positions with White to move is a win, and the longest takes **12 moves** (from `6/6/6/6/2k3/KR4 w`). The technique: use the rook to cut the enemy king off on a rank or file, bring your own king up in opposition, and check on the edge.

Rook endings with pawns (Guidance). 1. **Active rook.** A rook attacking pawns or cutting off a king is worth more than a rook passively defending. 2. **Rook behind passed pawns** — yours (it pushes) or the opponent's (it stops). 3. **Bring the king.** In rook-against-pawn endings, the defending king next to its pawn often draws.

Illustrated example

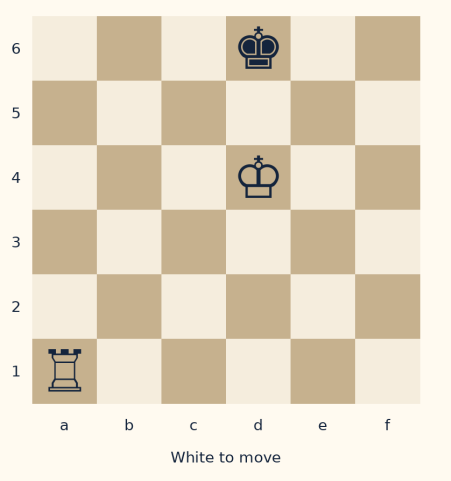


Figure 94: The basic rook mate. Kings on d4 and d6; the rook mates on the sixth rank because White's king covers c5, d5 and e5.

Verified line: Ra6#

Step-by-step reasoning

The technique in three steps (illustrated by a real engine game):

Verified line: Kd4 Kf6 Ke4 Ke6 Ra6#

1. **Cut off.** The rook on a5 confines Black's king to the sixth rank.
2. **Approach in opposition.** White's king walks up: `Kd4`, `Ke4`. When Black's king stands opposite (e6 against e4), the rook checks.
3. **Mate on the edge.** `Ra6#`.

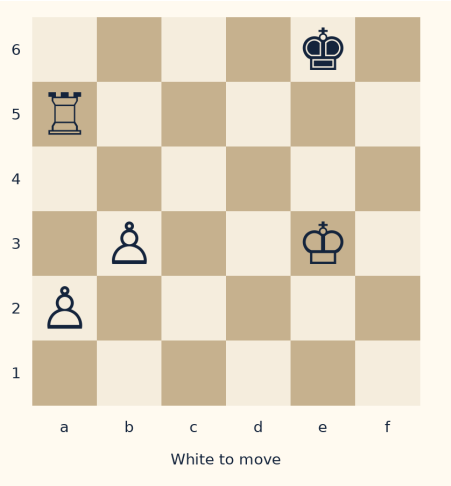


Figure 95: From an engine game. The rook already cuts Black’s king off on the sixth rank (it stands on a5, guarding the fifth). White mates in three with two quiet king moves: 1. Kd4 Kf6 2. Ke4 Ke6 3. Ra6#.

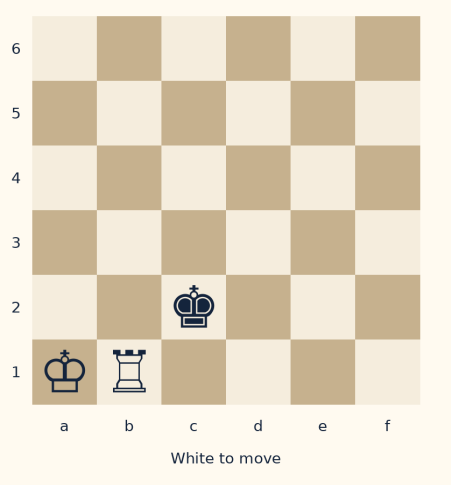


Figure 96: Evidence (exhaustive enumeration): the longest king-and-rook win on 6×6 — White mates in 12 moves with best play. Every legal position of this ending is won in 12 moves or fewer, well inside the 100-halfmove limit.

For the longest case, here is the starting position of the 12-move win:

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

Common mistakes

- Checking aimlessly with the rook. Checks are for the last move; cutting off comes first.
- Letting the enemy king attack the rook: keep the rook far from the enemy king.
- In rook-and-pawn endings, keeping the rook passive in front of your own pawn.

Advanced insight

The 6×6 edge is only two moves from the centre, so the defending king is driven to the edge almost at once. That is why king-and-rook wins never take more than 12 moves here. In practical games, use the rook to restrict before you check.

Worked example

Black mates with king and rook in three moves, from an engine game.

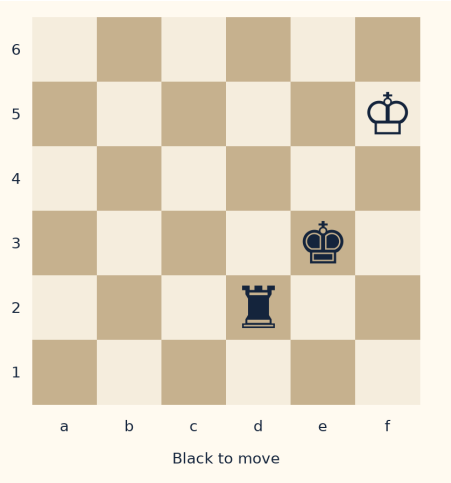


Figure 97: Black to move mates in three: 1…Kd4 2. Kf6 Kd5 3. Kf5 Rf2#. White's king is already on the edge file; Black's king approaches in opposition and the rook mates along the f-file.

Verified line: Kd4 Kf6 Kd5 Kf5 Rf2#

After `1...Kd4` and `2...Kd5` Black's king stands opposite White's on the fifth rank (d5 against f5, with e5 between them covered). The rook swings to f2 and checks along the f-file; the king on d5 covers e4, e5 and e6.

Summary

- King and rook beat king in at most 12 moves: cut off, approach in opposition, mate on the edge.
- Rook endings are the commonest pure endings; an extra pawn wins about 62% of them.
- Against a rook, a pawn one step from crowning, defended by its king, often draws.

Knights in Moonlight, Queens at Dawn: Knight and Queen Endings

Explanation

Knights (Evidence, exhaustive enumeration). - King and knight against king: **no position is a win**; mate is impossible. - King and **two** knights against king: no forced win. In 447,832

positions with White to move, 0.1% are wins, and every one of those is a mate in one that already stands on the board. With the defender to move, the only lost positions (88 of 583,440) are those where the defender is already checkmated on the board. - **Evidence (archive):** knight endings were drawn 61.0% of the time, and even with an extra pawn the stronger side won only 49.0%.

Rook against knight (Evidence, exhaustive enumeration). 852,400 positions with White (the rook) to move; we report the 592,704 *quiet* positions with the knight's side to move (not in check, no capture available). White wins only **11.7%**; the rest are draws. The win rate depends on two things:

Defender's king–knight distance	1	2	3	4 or more
White wins	2.5%	3.4%	7.3%	27.1%

Defender's king on	the edge (corners included)	second ring	the four centre squares	a corner
White wins	17.5%	4.0%	0.8%	33.0%

Rule (proven by these counts): keep the knight next to the king and the king out of the corner. The longest win in this ending takes 29 moves.

Queens (Evidence, exhaustive enumeration). - King and queen against king: every position is won, in at most **7 moves** (longest from `6/6/4k1/6/1Q4/K5 w`). - Queen against rook: in the 326,672 quiet positions with the rook's side to move, White wins **87.8%**, draws 10.6% (and loses 1.6%, to tactical rook tricks). The draws do not depend on geometry the way rook-against-knight does: the defender's chances are tactical — checks, skewers and stalemates. The longest win takes 23 moves. - **Evidence (archive):** queen endings with an extra pawn were won 63.8% of the time.

Illustrated example

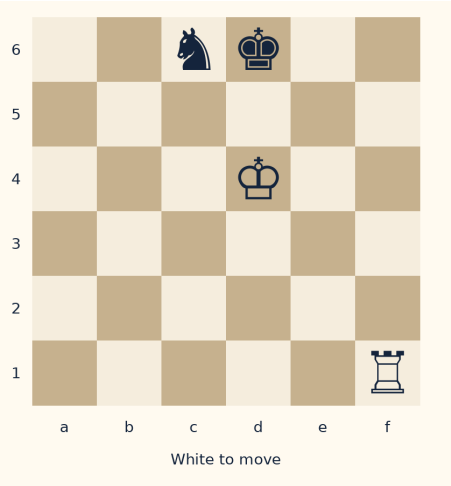


Figure 98: Rook against knight, knight beside its king. Evidence (exhaustive enumeration): with White to move, a draw.

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

Model note: DTM tables ignore repetition and 100/150-halfmove clocks; check the actual game history.

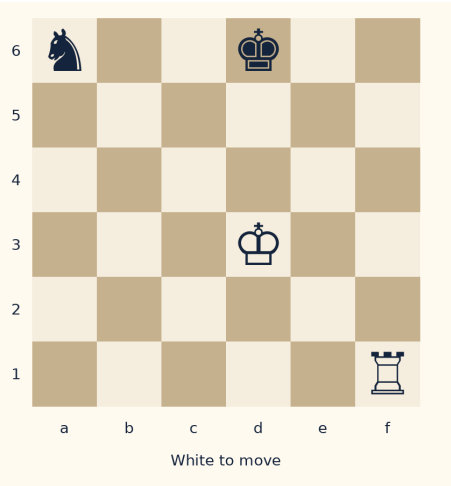


Figure 99: The same material with the knight stranded in the corner, away from its king. Evidence (exhaustive enumeration): White wins (mate in 10).

Step-by-step reasoning

1. **Why does distance matter?** A rook wins against a knight mostly by trapping or forking the knight, or by mating the king while the knight is too far to block. A knight next to its king is protected and can block checks.
2. **Why does the corner matter?** In the corner the defending king has only three squares; the rook and king can build a mating net, and the knight cannot cover everything. In the centre the king has eight squares.
3. **What the defender does.** Keep the king central and the knight within one or two squares of it. In the quiet positions satisfying both, the attacker almost never wins (under 1% with the king on the centre squares).

Common mistakes

- Running with the king to the corner "for safety". That is where the rook wins most often (33.0%).
- Sending the knight on a long trip. At distance four or more, the rook wins over a quarter of the time.
- Playing on for a win with king and two knights against a careful defender: there is no forced mate.

Advanced insight

Time and the rules matter in these endings (Chapter IV.7): with king and knight against a lone king you cannot mate, **but if the lone king's side runs out of time, it loses** — the crown-adj-1 rule awards the game to the side that still has any piece. The defender of a "dead draw" must still watch the clock.

Worked example

A knight that saves the game by blockading a pawn.

`24. Nb1!` solves both problems at once: the knight leaves the attacked square and blockades the queening square b1. **Engine judgement (depth 10):** −36 (holding); `24. Kd1` is −1,142 and `24. Kf1` allows mate. A knight in front of a pawn is a perfect blockader: it cannot be driven away by the pawn.

Summary

- K+N vs K and K+N+N vs K cannot be won by force. Knight endings are drawish (61% draws in the archive).

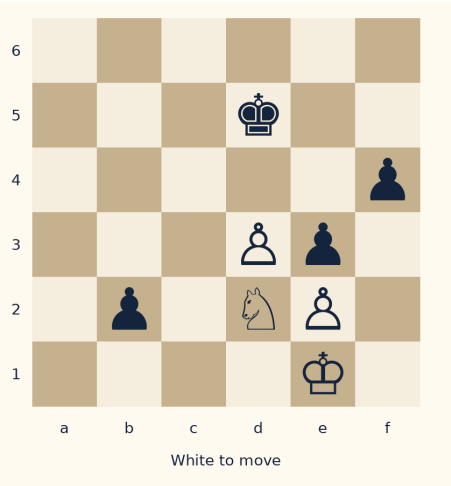


Figure 100: Academy puzzle pz-1ce067n. White to move. White's knight on d2 is attacked by the e3-pawn, and Black's b2-pawn wants to crown on b1.

- Rook vs knight is a draw in 88% of quiet positions: keep the knight close and the king central.
- K+Q vs K: at most 7 moves. Q vs R: a win in about 88% of quiet positions, at most 23 moves.

The Law of the Realm: Draws, Fortresses and the Rules

Explanation

The rules (verified in `src/rules/adjudication.ts`, policy `crown-adj-1`):

Event	Result	Automatic or claimed?
Checkmate	win	automatic (takes precedence over every draw)
Stalemate	draw	automatic
Only the two kings remain	draw	automatic ("bare kings")
The same position for the 3rd time	draw	**claimable**
The same position for the 5th time	draw	automatic
100 halfmoves without a pawn move or capture	draw	**claimable**
150 halfmoves without a pawn move or capture	draw	automatic
Agreement	draw	only after each side has made at least 10 moves
A player's time runs out	loss — **if the opponent has any piece other than the king**; otherwise a draw	automatic

Two differences from the older game's rules matter in practice: 1. **Only bare kings is an automatic "dead position".** King and knight against king is *not* automatically drawn; it continues until repetition, the halfmove rules or agreement. 2. **The timeout rule is strict.** If you run out of time and your opponent has even a lone knight, you lose — even though that knight could never mate.

Evidence (archive, balanced games): checkmate ended 66.2% of games; threefold-repetition claims 22.9%; bare kings 4.9%; 100-halfmove claims 4.9%; stalemate 1.0%.

The halfmove limits are never in the way of correct technique in the small endings.

Evidence (exhaustive enumeration): the longest forced wins are KQK 7 moves, KRK 12, KPK 20, KQ–KR 23 and KR–KN 29 — all inside 50 moves (100 halfmoves).

Illustrated example

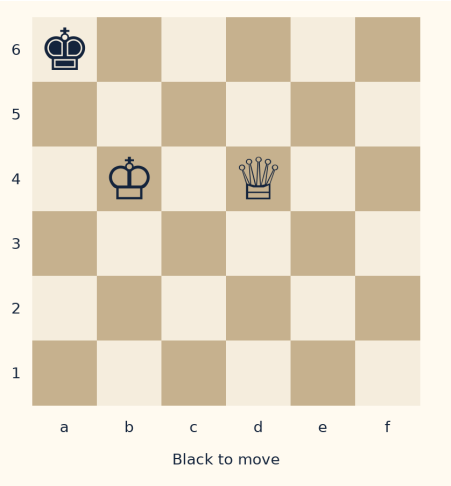


Figure 101: The trap. 1. Qd4?? looks like it tightens the net, but Black's king on a6 has no legal move and is not in check: stalemate, a draw.

Verified line: Qd4

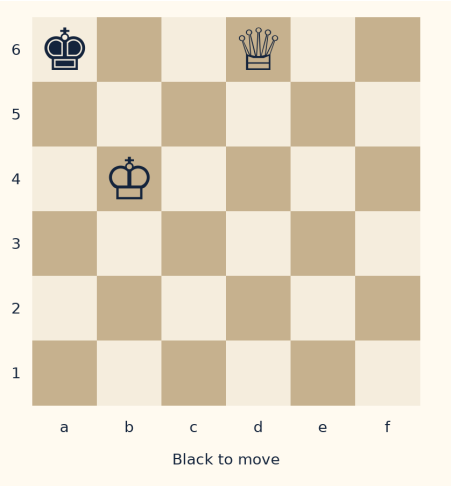


Figure 102: The right move from the same position: 1. Qd6#. The queen checks along the sixth rank; the king on b4 covers a5 and b5.

Verified line: Qd6#

Step-by-step reasoning

How to avoid stalemate with a queen: 1. **Before every queen move, count the enemy king's legal moves after it.** If the answer is zero and it is not check, stop. 2. **Give the king a square until you are ready to mate.** A king with one square cannot be stalemated. 3. **Mate with check.** The final move must attack the king.

Common mistakes

- Playing "safe" queen moves near a cornered king without counting its squares.
- Forgetting to *claim* a threefold repetition or the 100-halfmove draw: they are not automatic until the fifth repetition or 150 halfmoves.
- Relaxing on the clock with a "drawn" lone king against a knight: if you flag, you lose.

Advanced insight

Stalemate as a defence: a king with no pawns and no pieces left, boxed into a corner, may be stalemated by an over-eager attacker. In the rook-pawn ending of Chapter IV.4 (`pz-1og62nt`), the defender's king sits in the corner precisely because the attacker must avoid stalemate and can never drive it out.

Worked example

A draw by repetition, found by the defender.

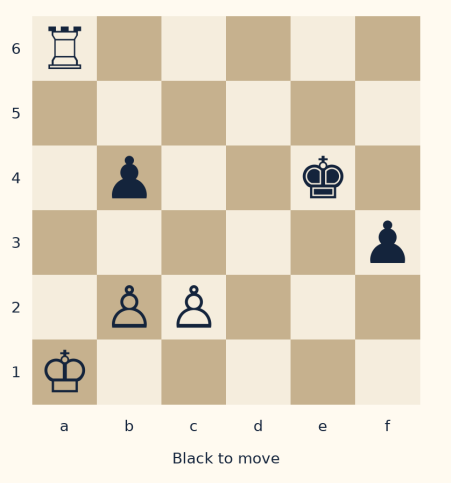


Figure 103: Academy puzzle pz-rmet01. Black to move. Black has two pawns for nothing against a rook, but White's king is far away.

`32...Ke5!` keeps Black's king between both pawns: it guards the f3-pawn's path and stays near the b4-pawn. White's rook can only check: `33. Ra5+ Ke4 34. Ra6` and the position repeats. **Engine judgement (depth 10):** 0.00 (draw); `32...Kf5` −135 and `32...f2` −323. A draw by repetition must be claimed on the third occurrence, or it becomes automatic on the fifth.

Summary

- Automatic: checkmate, stalemate, bare kings, fivefold repetition, 150 halfmoves. Claimable: threefold, 100 halfmoves. Agreement after ten moves each.
- Timeout loses if the opponent has any non-king piece; otherwise it is a draw.
- Count the lone king's moves before every queen move: stalemate is one careless move away.

The Sovereign's Hourglass: Conversion, Time and Practical Decisions

Explanation

Conversion. Evidence (archive): at the first quiet moment after move 20, a one-pawn lead won 48.0% of games, two pawns 68.3%, three or four points 83.1%, five or more 93.0% (Book III, Chapter III.9). The methods: 1. **Take what is offered** — but only after checking the opponent's checks. 2. **Simplify toward the endings you know are won**: king and pawn with the king on a key square; queen against rook; rook against pawns without their king. 3. **Avoid the drawing materials**: knight endings, rook against knight, two knights. 4. **Promote to a queen** unless a knight check or fork wins more (3.1% of promotions in the archive were knights, 1.3% rooks).

Time. The Academy's time controls (`src/games/types.ts`) are **Blitz 3+2**, **Rapid 5+3** and **Classical 15+10** (minutes + seconds added per move). **Evidence (archive):** balanced engine games lasted a median of **75 plies** (38 moves each), with 10% shorter than 49 plies and 10% longer than 124 (62 moves). In a Rapid game, a 38-move game gives you 300 + 38 × 3 = 414

seconds, about 11 seconds a move; a 62-move game, about 8 seconds a move. The engine itself spends roughly (remaining time − a small reserve) / 22 + ¾ of the increment on each move (`src/engine/levels.ts`).

Guidance for the clock: - Spend time where the position is *critical*: when forcing moves exist, or when you are choosing a plan or a pawn structure. Spend little on recaptures and forced replies. - Keep a reserve for the endgame: half of all games here see a promotion (median first promotion on ply 64), and pawn races are decided by a single tempo. - Remember the timeout rule: if your opponent has any piece, flagging loses.

Practical decisions. - **Claims.** You may claim a draw at the third repetition or after 100 halfmoves. If you are worse, claim. If you are better, avoid repeating. - **Playing on.** With a material lead of two points or more, the archive says play on (68% wins). - **Agreement.** Only after ten moves each.

Illustrated example

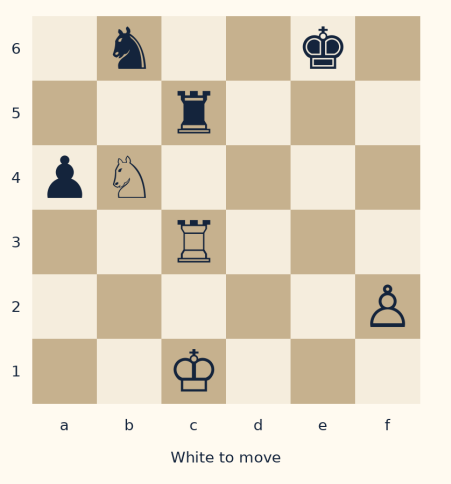


Figure 104: Academy puzzle pz-b8tld2. White to move. Black's rook on c5 stands undefended in front of White's rook.

Step-by-step reasoning

1. **Checks for the opponent?** After `29. Rxc5`, Black has no check that wins anything.
2. **Is the rook really free?** Black's king on e6 does not reach c5, and the b6-knight covers a4, c4, d5 — not c5. Yes.
3. **Take it.** `29. Rxc5`. **Engine judgement (depth 10):** +843; any other move (`29. Kb2`, `29. Kc2`) is about +30. When something is genuinely free, conversion begins by taking it.

Common mistakes

- Spending your clock early on routine opening moves and then losing a won pawn race on time.
- Avoiding a simplification into a won pawn ending because "the rook ending is safer". The archive says otherwise (77.4% against 62.2% wins with an extra pawn), and the key squares tell you exactly.
- Under-promoting without a concrete reason.

Advanced insight

In converting, the opponent's best chances are tactical: checks, discovered attacks and stalemate tricks. Your job is to remove them one at a time — trade the active piece, centralise your king, then push. On a six-rank board a passed pawn needs at most four steps, so every move spent on safety is cheap compared with the cost of a counter-chance.

Worked example

Converting with a discovered attack.

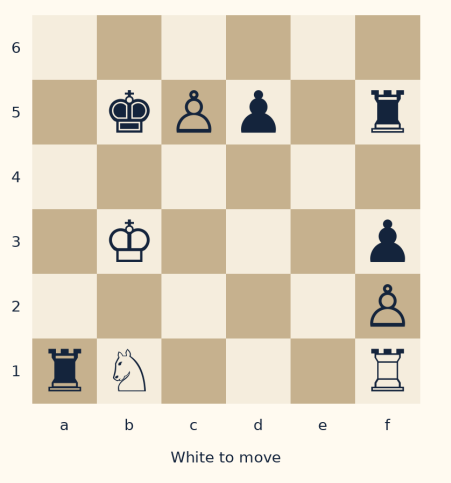


Figure 105: Academy puzzle pz-ephpsb. White to move. Black’s rook on a1 and White’s rook on f1 stand on the same rank, with only White’s knight on b1 between them.

`20. Nc3+!` The knight jumps with check (from c3 it attacks b5) and uncovers the f1-rook’s attack along the first rank on the a1-rook. After `20...Kxc5 21. Rxa1` White has won a rook. **Engine judgement (depth 10):** +271; `20. Na3+` only −87.

Summary

- Convert by taking what is free, simplifying into known wins, and avoiding drawish materials.
- Budget about 11 seconds a move in Rapid for a median-length game; save time for the endgame.
- Claim draws when worse; play on with a two-point lead.

PART VI · The Competitive Federation

Reading the Opponent: profiling and style adaptation

Explanation

Opponent profiling means predicting how an opponent will choose moves, and steering towards positions where those choices go wrong. In Crown & Calculation every opponent is a program, and its method is documented. That is an unusual gift. You can profile with facts instead of rumours.

The engine levels (profile set crown-levels-6). [Demonstrated] src/engine/levels.ts and the measured calibration src/engine/calibration.json (500 engine-vs-engine games, L1 anchored at 800; this is the Crown scale, not FIDE):

Level	Name	How it chooses	Crown rating (± RD)
L1	Novice	1-ply search; random choice among moves within 3.2 pawns of its best	800 (anchor)
L2	Apprentice	1-ply plus quiescence; within 2 pawns	1122 ± 86
L3	Club	2-ply; within 1.3 pawns	1582 ± 116
L4	District	2-ply; within 0.7 pawns	1710 ± 121
L5	Regional	full search, 600 nodes per move, always its best	1976 ± 129
L6	National	full search, 4,000 nodes	2375 ± 148
L7	Master	full search, 30,000 nodes	2641 ± 157
L8	Crown	full search limited by time (up to 6 s per move)	2926 ± 169
L9	Sovereign Unbound	canonical engine, larger hash table, your chosen budget (rated: 10 s)	3038 ± 171 (derived)

What this means in practice:

- **L1–L4 never play a move worse than their margin below their own best,** and they choose among the near-best moves with a softmax “temperature”. They are *loose*, not suicidal. They do not throw games. Their weakness is **shallow sight**: a 1- or 2-ply search cannot see a quiet move whose point arrives three plies later. So set slow traps: quiet moves that prepare a double attack ([Counsel], based on the documented depth).

- **When L1–L4 find a mate, they play it.** Bounded selection is skipped when the best score is a mate.
- **L5–L7 always play their best move,** but within a fixed node budget. They are consistent, and their horizon is short in long, quiet manoeuvring. Positions with a slow plan (a pawn majority that matters only in the ending) test them more than sharp tactics do ([Counsel]).
- **L8 and L9 are limited by time, not nodes.** A faster computer makes them stronger, and in timed games their clock lowers their thinking time (see Chapter V.8).

Federation competitors. [Demonstrated] src/federation/. *Each of the 2,048 competitors has a* public* Glicko-2 rating and RD, a division (1 Crown … 8 Open), a region, and a style label. Their *hidden* playing strength ("latent") chooses their engine profile: below the L5 rating the nearest calibrated level L1–L5 is used. From L5 upward it is a full search whose node budget interpolates the measured ladder (crown-profile-map-1, about 118 Elo per doubling of nodes). **The hidden strength is never shown and is never used by forecasts or NPC traders.** The public rating is only an estimate updated from results. So a high-RD competitor's rating is a weak signal, and a low-RD rating is a better one.

Styles. [Demonstrated] src/engine/styles.ts, styles-validation.json and docs/STYLES_REPORT.md. Six labels exist: Universal, Tactical attacker, Positional strategist, Defensive specialist, Endgame technician, Risk-taking aggressor. A style is **played only if it passed validation** (validation id styles-5-2026-11-04). Four passed: **Tactical, Positional, Defensive and Endgame**. Each one changes the engine's evaluation **for the whole game**, for its own side only, and stays within ±50 Elo of Universal (200 colour-swapped games per style):

- **Tactical** values threats and loose enemy pieces more and avoids draws a little (contempt +15). Measured: 21 % draws against Universal's 31 % in self-play.
- **Positional** values supported pawns, pawn chains, outposts and mobility, and threats slightly less.
- **Defensive** values its king's pawn shield and fears the opponent's attackers, loose pieces and queen more, so it welcomes queen trades and accepts draws (contempt −20). Measured: about 10 % fewer attacks on its king zone, 35.5 % draws.
- **Endgame** values passed pawns and an active king in the endgame and likes to simplify. Measured: queens leave the board earlier (median ply 28 against 31).

In addition, Tactical and Positional keep their **opening repertoire for the first six plies**: a choice among book moves the engine judges within **30 cp** of the best at depth 7. Tactical prefers moves that attack pieces, capture, check, or develop knights and the queen; Positional prefers knight moves, piece development and central files. **Risk-taking aggressor failed validation** (it did not give more checks than its opponent) and plays as Universal; its profile says so.

That is the honest profile: a style shapes *tendencies over a whole game* and never the strength. Single moves barely differ (10–21 % of positions get a different move); the style shows in what kind of game you get.

Illustrated example

Verified line: Nf3 Nf4

Suppose your Federation opponent is a *Positional strategist*. Its repertoire weights favour knight moves (+2), any piece move (+1) and central c/d files (+1). So from the start position a move such as Nf3 or Nc3 is much more likely than a3, *provided the move is in its 30 cp book*. You can prepare a reply to its most likely first moves. Its choices are random, though (seeded per game), so prepare for two or three, not one.

Step-by-step reasoning

1. **Read the profile:** public rating, RD (is the rating trustworthy?), division, style label.
2. **Translate the label into behaviour:** Tactical, Positional, Defensive and Endgame change the whole game (and Tactical and Positional also the first six plies). Aggressive and Universal mean the engine's main moves.
3. **Translate the rating into an engine profile:** under about 1976, a sampling level (loose choices, shallow sight); above it, a full search with a node budget (precise but short-sighted in quiet positions).

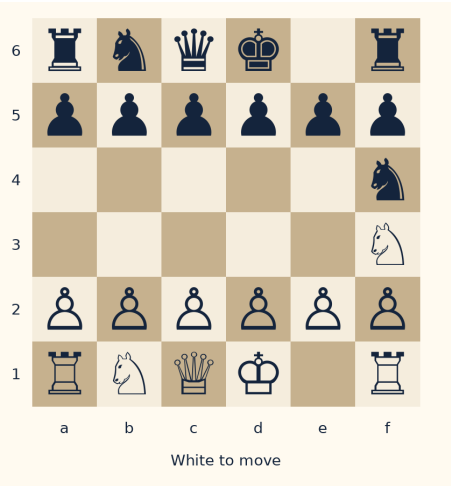


Figure 106: After 1.Nf3 Nf4: knight moves score well with both opening repertoires (Tactical and Positional).

4. **Choose your battlefield:** against shallow searchers, slow multi-move threats; against node-limited full searchers, long quiet plans and endgames where the horizon matters ([Counsel]).
5. **Keep it legal and fair:** you never need hidden information. Everything above comes from what the game shows you or from its documentation.

Common mistakes

- **Expecting a style to change strength or every move.** It changes tendencies (draw rate, king safety, when the queens come off), not the level of play, and most single moves are the same as Universal's.
- **Trusting a provisional rating.** A rating with a high RD can be hundreds of points away from the real strength.
- **Treating L1–L4 as random.** They choose among *near-best* moves. A 0.7-pawn margin (L4) is quite strict.
- **Over-adapting.** If you play badly to "exploit" an opponent, you also lose to the moves it plays when it plays its best ([Counsel]).

Advanced insight

Style adaptation is really self-adaptation. The opponent's method is fixed. The variable you control is *which positions you reach*. Keep a short list of positions you play well (from the Lab dimensions with the narrowest intervals and highest rates) and steer towards them. That is profiling yourself, and it is measured ([Demonstrated], Lab methodology crown-lab-1: every rate has a denominator and a 95 % interval).

Worked example

You are rated 1650 (established). Your next league opponent is shown at 1580 ± 60, Tactical attacker, Club Division.

- The RD of 60 means the rating is reasonably established.
- The hidden strength picks the engine profile. If it is near 1580, that is close to L3 (1582): a 2-ply search choosing within 1.3 pawns of its best.
- "Tactical" means its first three moves favour captures, checks and attacks on pieces.
- **Plan:** in the opening, avoid leaving pieces where a 2-ply search sees a cheap capture. In the middlegame, set multi-move traps (a quiet move that prepares a fork), which a 2-ply search often walks into ([Counsel]).
- **Expected score:** the Federation's result model (crown-result-model-1, Davidson model with $\nu = 0.9$) gives a 70-point favourite about 0.57. A 100-point favourite gets 0.598 (W 0.447, D 0.302, L 0.251). You are a modest favourite, not a certainty.

Summary

Profile with facts. Levels differ in search depth and selection margin, competitors differ in hidden strength (mapped to a profile) and in a six-ply opening preference if their style is validated, and ratings differ in reliability (RD). Adapt by choosing your battlefield, not by playing worse moves.

The Tournament Hall: Swiss, leagues and knockouts

Explanation

The season. [Demonstrated] src/federation/model.ts: 19 rounds per season. League rounds run 1–15, Swiss events fill rounds 4–8, the Club Weekend rounds 10–11, and the Crown Championship knockout rounds 16–19. Federation games for the player use **Rapid 5+3** (src/app/federation.ts). There are 8 divisions × 16 regional leagues × 16 competitors (2,048 competitors).

Leagues (round robin). [Demonstrated] src/federation/formats.ts: a circle-method round robin with colour balancing. **Order:** points, then head-to-head among the tied group, then Sonneborn–Berger, then pre-season seed. At season end the **top two are promoted and the bottom two are relegated** between adjacent divisions of the same region.

Swiss (crown-swiss-1, a simplified score-group system, not FIDE Dutch). [Demonstrated] Players are sorted by points then seed. With an odd field, the bye goes to the lowest-ranked player who has not had one. Then each player meets the nearest-ranked opponent not met before, preferring the same score group. The player with fewer Whites gets White. **Order:** points, Buchholz Cut-1, Buchholz, Sonneborn–Berger, seed.

Knockout (Crown Championship). [Demonstrated] Each match is two classical games with colours swapped. If level, one rapid pair; if still level, one blitz pair. If still level, the **higher pre-event seed advances**, except in the championship **final**, where a level match gives **joint champions**.

Performance and norms. [Demonstrated] src/heraldry/titles.ts: the performance rating is the average opponent rating + dp, with $dp = -400 \cdot \log_{10}(1/p - 1)$ for score fraction p (capped at ±800). An event with **at least 5 games** counts as a *norm* for a title tier when both the performance and the average opposition reach the tier's thresholds:

Federation title	Performance ≥	Avg. opponents ≥	Norms needed	Norm games ≥	Peak rating ≥	Other
Squire	1700	1100	1	5	1600	—
Knight	2000	1500	2	15	1900 (established)	—
Paladin	2300	1800	3	27	2200 (established)	an event title
Sovereign	2600	2100	3	27	2500 (established)	a Championship final

Federation titles are **permanent**. Rating titles (31 bands of 100 points, from *Wandless Wanderer* to *Eternal Sovereign*) describe your current rating and are a separate system (src/ratings/titles.ts).

Circuit Grades (v0.11). [Demonstrated] src/federation/ranks.ts: a rank ladder of 16 feudal grades, each with three sublevels III → II → I: 48 steps from **Commoner III** to **High Sovereign I** (the names are a proposed list, docs/FEUDAL_RANKS.md). Each division owns two grades: the Open Division Commoner and Page, the Club Division Squire and Man-at-Arms, and so on up to King and High Sovereign in the Crown Division. Your **division** decides the pair; promotion starts you at the new division's lower grade III, relegation puts you at the lower division's upper grade III. Inside the division a **standing score** (0–100) moves you at two **checkpoints** per season, after round 8 and at season end: 45 % league position (1st = 100, 16th = 0), 25 % your score in sanctioned rated games (league, the Opens, the Zonal Championships, the Regional Cups, the

Crown Championship) and 30 % your rating's percentile within the division. The six steps of a division are reached at 0 / 30 / 42 / 54 / 64 / 76; you drop a step only below the step's mark minus **12**; at most one step per checkpoint; never down at the checkpoint right after a step up; never up from the bottom two of your league. Exhibitions, unrated, practice and sandbox games do not count, and nothing can be bought. Your **highest rank ever** is kept for good. A Circuit Grade is not your Career Elo (the Glicko-2 rating) and not Played Like.

Illustrated example

A Swiss, last round. You are on 3.5/4, half a point clear of three players on 3. You are White against one of them.

- A **draw** keeps you on 4 and the opponent on 3.5. The other two on 3 can reach 4 by winning, and then **Buchholz Cut-1** decides between you. Your Cut-1 depends on your opponents' scores, which you cannot control.
- A **win** makes you sole winner.
- A **loss** drops you into a tie at 3.5, possibly behind several players.

The choice between "play solidly for a draw" and "press for the win" depends on your Cut-1 relative to the chasers. Look it up in the standings table before the round ([Counsel] on the decision; the tiebreak order is [Demonstrated]).

Step-by-step reasoning

1. **Know the format's tiebreak** before the event (table above).
2. **Before each round, list the results you need** and which ones you control.
3. **Set your risk level:** in a league, the target is top two (promotion) or avoiding the bottom two. In a knockout, a drawn classical pair leads to rapid and blitz, so your speed-chess level matters.
4. **After the event, compute your performance** and check it against the norm table.

Common mistakes

- **Playing every round the same way.** Round 1 of a 15-round league and the last round of a Swiss are different decisions.
- **Forgetting that knockout tiebreaks are faster games.** If your blitz is much weaker than your classical play, a drawn classical pair is bad news.
- **Assuming a high score is a norm.** It needs ≥ 5 games *and* a high enough average opposition.
- **Reading your Circuit Grade as a rating.** "Knight II" says where you stand in your division's ladder, not how strong you are: it is tied to your division and your checkpoint standing, while your Career Elo moves with every rated game.
- **Entering an event you have traded on.** If you enter an event, its open winner market is **cancelled at the published 1/n vector before your entry takes effect** ([Demonstrated], src/exchange/families.ts rules). See Book VI.

Advanced insight

Sonneborn–Berger rewards beating strong finishers. In a league tie, a win against the eventual runner-up is worth more on SB than a win against the bottom seed. Head-to-head comes first, though. So in a race for second place, the direct game against your rival counts twice: once in points and once in the head-to-head tiebreak ([Demonstrated] order; the conclusion follows directly).

Worked example

Norm check. In a league season you play 15 games against an average opposition of 1,820 and score 10. $p = 10/15 = 0.667$, so $dp = -400 \cdot \log10(1/0.667 - 1) = -400 \cdot \log10(0.5) =$ **+120**. Performance = 1820 + 120 = **1940**. Squire tier: 1940 ≥ 1700 and 1820 ≥ 1100, so it **qualifies**. Knight tier: 1940 < 2000, so it **does not**. This season gives one Squire-level norm. (Computed with the formula in src/heraldry/titles.ts.)

Summary

Know the pairing rules, tiebreak order and advancement rules of each event, and choose your risk round by round. The performance rating turns your score into a number, and norms turn sustained performances into permanent titles.

Critical Moments, Stronger Opponents and Predictable Mistakes

Explanation

The game's definition. [Demonstrated] src/analysis/analyzer.ts (crown-critical-1): a position is **critical** when the best move is at least **0.10 expected-score points** better than the second-best (or when a mate is available and the second move is not a mate). The game summary reports how many critical moments you **handled** (expected-score loss ≤ 0.05). At the board you have no engine, so learn the signs ([Counsel]):

- **Your opponent has just changed the position's character** with a capture, a pawn break or a sacrifice.
- **You have only one or two moves that do not lose material** (an "only move").
- **A pawn is two steps from promotion,** which on this board happens early.
- **Checks are available for either side.**

When you notice a sign, **stop and spend time.** The Lab's *Time management* dimension measures whether you do.

Against stronger opponents the honest goal is often the draw. Crown has two automatic draws you can aim for: **bare kings** and **fivefold repetition**. Threefold repetition and the 100-halfmove rule can be **claimed**. Perpetual check is the classic defensive weapon: if you can check forever, the position repeats ([Demonstrated], src/rules/adjudication.ts).

Predictable mistakes. [Demonstrated] src/academy/coach.ts lists the coach's topics: hanging pieces, knight forks, double attacks, mating nets, the opponent's threats before your move, promotion races, converting winning positions, and calculation. Each note is derived from an actual engine line replayed under the rules. The Lab marks a pattern as **recurring** only after at least **5 occurrences in at least 3 games**.

Illustrated example

Academy puzzle pz-1w5drbc (goal: "Find the only move that holds", Defensive Survival pool).

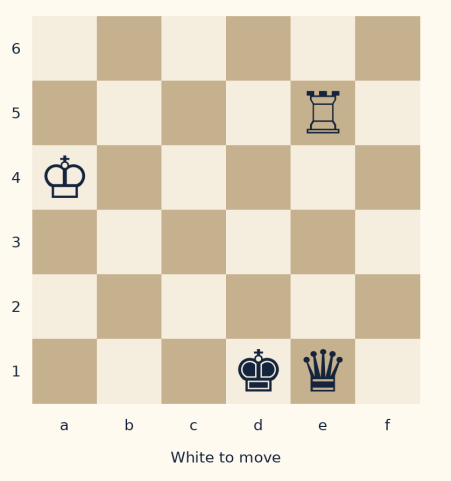


Figure 107: White to move (Academy puzzle pz-1w5drbc). A rook against a queen. One move holds.

Verified line: Rxe1+ Kxe1

[Demonstrated] 1.Rxe1+! Kxe1 and only the two kings remain: an **automatic draw by bare kings** (the checker confirms the final position). Black could decline the rook with a king move, but then K+R against K is lost for Black. Every other White move loses to the queen. This is a critical moment in its purest form: one move is far better than every other, and the weaker side saves the game by trading everything off.

Step-by-step reasoning

1. **Notice the sign:** you are a queen against a rook down in material. That is a critical moment.
2. **List forcing moves first.** Here a capture with check exists, and the board's draw rules (Rulebook: bare kings) make the trade a full save.
3. **Ask "what does my opponent threaten if I do nothing?"** This is the coach's *threat-awareness* topic. It would have shown you that a quiet move loses.

Common mistakes

- **Playing fast in critical moments and slowly in quiet ones.** This is the most common time-management fault in competitive play ([Counsel]).
- **Going for a perpetual too early.** If the position is objectively fine, repeating throws away chances. Use the evaluation graph in later study to judge whether the draw was needed.
- **Forgetting that threefold repetition must be claimed.** It is not automatic until the fifth occurrence.

Advanced insight

Against a much stronger opponent, **reduce the number of critical moments you face.** Every critical moment is a chance to lose. Simplifying lines (Chapter V.2) and forcing repetitions reduce them ([Counsel]). The reverse holds too: against a weaker opponent, *create* critical moments, because each one is a chance for them to err.

Worked example

Academy puzzle pz-1qogh9k ("Find the only move that holds", difficulty 3).

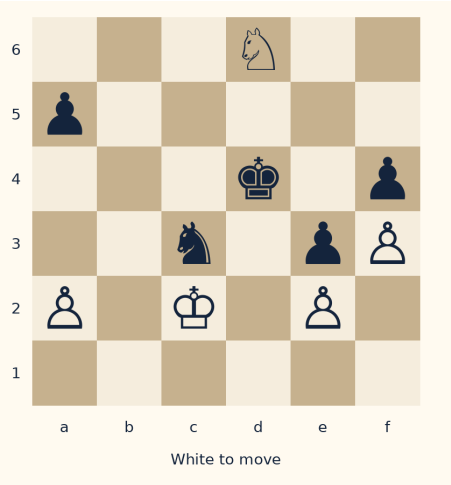


Figure 108: White to move. Black's king and knight are centralised. Can White's knight hold the game?

Verified line: Nf5+ Kc4 Nd6+ Kd4 Nf5+

1.Nf5+ Kc4 2.Nd6+ Kd4 3.Nf5+. The knight checks from f5 and d6 in turn, and after move 3 the position after 1.Nf5+ has occurred a second time. **[Demonstrated]** this is the verified line of the puzzle. Continuing to repeat leads to a threefold-repetition claim (or the automatic fivefold draw). The weaker side saved half a point by spotting a defensive resource at the critical moment.

Summary

A critical moment is one where the best move is clearly better than the second, and at the board you feel it as forcing play, only moves, near-promotions and available checks. Spend your time there. Against stronger opponents, reduce critical moments and look for perpetual checks and repetitions. Learn your predictable mistakes from the coach's topics and fix them in the Academy.

The Engine's Mirror: postgame study, evaluations and losses

Explanation

The tools. [Demonstrated] src/analysis/analyzer.ts, src/ui/AnalysisScreen.tsx, src/academy/*:

Tool	What it does
Presets	Quick (depth 5), Standard (depth 7, the default), Deep (depth 9; 4 candidate lines instead of 3). Deeper is slower and more reliable.
Candidate lines	for each move: the best line and alternatives, each with its expected score.
Explore a variation	step through a candidate line on the board; *Return to Main Line* goes back.
Bookmarks	*Bookmark Position* and *Mark Memorable* save plies of the game for later study.
Retry from This Position (Unrated)	shown on your moves that lost more than 0.05; starts an untimed **training** game from the position before your move, against the same opponent. Training games are never rated and are excluded from the Lab's game evidence.
Replay Your Mistakes (Academy)	turns every move of yours that lost **≥ 0.12** expected score in an analysed game into a puzzle (accepting any move within 0.02 of the best).
The Lab	chess skill dimensions (Tactical vision, Calculation, Positional understanding, Defensive awareness, Endgame technique, Threat recognition, Opening understanding, Time management, Decision consistency, Adaptability), each a success rate with its denominator and a 95 % interval. It says "insufficient" when there is too little evidence.

The expected-score scale. [Demonstrated] crown-outcome-map-v2 is an isotonic map fitted on 43,305 positions from engine games and validated on held-out opening families. Values for the side to move:

Engine score	−500	−300	−200	−100	−50	0	+50	+100	+150	+200	+300	+500
Expected score	0.009	0.073	0.160	0.329	0.406	0.502	0.575	0.676	0.769	0.820	0.932	0.989

(Computed by calling `mapV2` from src/analysis/outcomeMap.ts.) A one-pawn edge (+100) is worth about two-thirds of a point *in engine games*. A three-pawn edge is close to decisive. The map comes from engine games, and humans convert advantages less reliably, so read it as "how winning the position is", not "how likely *you* are to win".

Grades. [Demonstrated] The grading methodology is **crown-grade-v3-experimental**. A move's *loss* is the drop in expected score compared with the best move. The bins are Best ≤ 0.002, Excellent ≤ 0.02, Good ≤ 0.05, Inaccuracy ≤ 0.12, Mistake ≤ 0.25, Blunder above. The **bins and the accuracy constant are uncalibrated placeholders**: Accuracy = 100·exp(−4 × mean loss). **Great** and **Brilliant** labels are **not validated**, and Brilliant-move achievements wait for validated grading. **Played Like** is shown as **"Calibrating"** because the model failed its release gate: it did no better than a naive accuracy baseline. Treat grades as *pointers to look at*, not verdicts. Where a grade is uncertain between depths, the game says so.

Illustrated example

You lost a Rapid game. The summary shows two turning points (plies 23 and 31) and one Blunder at ply 31 with a loss of 0.41. The coach note on ply 31 is under *Threat awareness* and shows the engine line in which your opponent wins a rook.

Routine:

1. Open ply 23 first. The *first* turning point is usually the real cause; the blunder is often just the consequence.
2. **Explore** the best candidate at ply 23 and step through it. Can you explain in words why it is better?
3. **Bookmark** ply 23, and use *Retry from This Position (Unrated)* to play the position again against the same opponent.
4. After the game, open **Replay Your Mistakes**. Ply 31 (loss 0.41 ≥ 0.12) will be there as a puzzle.
5. A week later, check whether the Lab's *Threat recognition* rate has moved, and look at its interval, not just its value.

Step-by-step reasoning: the loss routine

1. **Before the engine:** write down the move where you think you went wrong, and why. (Your own view first: the comparison is the lesson.) ([Counsel])
2. **Run Standard analysis.** If a critical moment is graded with an uncertainty warning, rerun with **Deep**.
3. **Find the first turning point,** not the biggest blunder.
4. **Explain the better move in words.** If you cannot, the engine line is not yet knowledge.
5. **Retry** the position against the same opponent until you hold it.
6. **Classify the error** by coach topic, and count it. Five occurrences in three games is a pattern.
7. **Assign training:** puzzles in that theme (Tactical Academy), Defensive Survival, or the Endgame Workshop.

Common mistakes

- **Analysing only the blunder.** The position was often already difficult.
- **Trusting a single grade.** The bins are experimental, and the label can depend on depth.
- **Reading "+1.0" as "I should have won".** +100 cp ≈ 0.68 expected score: likely, not certain.
- **Counting Brilliant labels as skill.** They are not validated.
- **Using the engine during rated play.** Rated play hides engine information by design, and assisted games are excluded from evidence and markets.

Advanced insight

The Lab aggregates only reports from **one methodology** (the most common grading id + engine build + preset) and counts the rest as excluded. If you mix Quick and Deep analyses, your Lab evidence splits. **Pick one preset for your record** (Standard is the default) and use Deep only for extra study ([Demonstrated], src/academy/lab.ts `eligibleItems`).

Worked example

A move goes from +150 (expected 0.769) to +50 (0.575) after your reply, while the best move would have kept +150. Loss = 0.769 − 0.575 = **0.194** → **Mistake** (0.12 < 0.194 ≤ 0.25), and it qualifies for Replay Your Mistakes (≥ 0.12). In words: you turned a large edge (about three-quarters of a point) into a small one (just over half).

Summary

Use presets, candidate lines, variations, bookmarks and retries to turn a lost game into training. Read centipawns through the expected-score map. Treat grades as experimental pointers. Brilliant is unvalidated and Played Like is Calibrating. Start with the first turning point, explain the better move in words, and train the pattern until the Lab's interval moves.

The Sixty-Four Gates: building a repertoire

Explanation

[Demonstrated] The engine's benchmark book has **64 two-ply openings** (one White move, one Black reply), chosen because Fairy-Stockfish at depth 10 judged each within **±80 cp** from White's view. They range from −76 to +76 (mean +16). The game's opening classifier (crown-open-0) names them descriptively. **No established 6×6 opening theory exists**, and the Lab's *Opening understanding* dimension measures soundness (loss ≤ 0.05 in the opening phase), not memorised lines.

White's ten first moves and the replies in the book, with the evaluation (cp, White's view):

White's first move	Book replies (eval)
Nc3	a4 (−53), e4 (−74), Nc4 (0), Na4 (+15), Nf4 (−23), Nd4 (0)
Na3	a4 (+45), d4 (+7), e4 (−14), Nc4 (+7), Nf4 (+22)
Nf3	a4 (+14), b4 (+14), c4 (−30), d4 (−30), f4 (+30), Nf4 (+76), Nd4 (+22)
Nd3	a4 (+61), b4 (−46), f4 (−76), Nc4 (+71), Na4 (+38), Nd4 (+7)
a3	a4 (+38), c4 (−59), d4 (−23), e4 (+75), f4 (0), Nc4 (+75), Na4 (−17), Nf4 (+30), Nd4 (+69)
b3	a4 (−51), b4 (+75), c4 (−22), d4 (+23), e4 (+71), f4 (+7), Nf4 (+14)
c3	b4 (0), c4 (−7), d4 (+45), f4 (+67), Na4 (+72), Nf4 (+53)
d3	b4 (+15), c4 (0), d4 (+46), e4 (+22), f4 (+67), Nf4 (+76)
e3	a4 (−46), b4 (−23), e4 (+67), Na4 (+36)
f3	a4 (+60), b4 (−7), c4 (+52), d4 (−38), e4 (+75), f4 (−23), Na4 (+56), Nf4 (0)

(Converted from the book's move codes to SAN by replaying each line under the rules.) Two things matter. First, **every pawn move is a single step**: there is no double step on this board. Second, after one move each, the pawns are already one or two steps from contact.

Illustrated example

Verified line: Nc3 e4

After 1.Nc3 e4 Black's e-pawn already attacks d3 and f3, the squares White's d- and f-pawns would step to. It is three steps from promotion. For Black this is a good choice against Nc3. For White it is a reason to choose another first move.

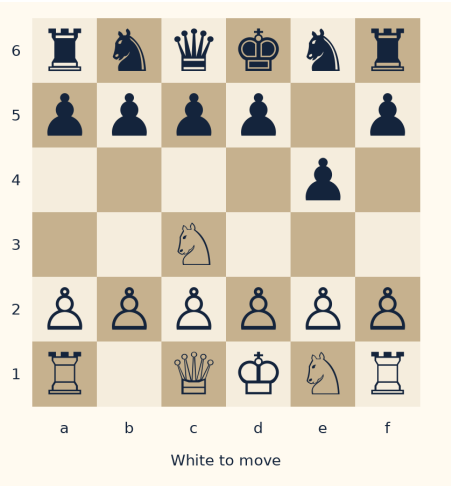


Figure 109: Nc3 e4 — the most Black-favoured book line (−74 for White at Fairy-Stockfish depth 10).

Step-by-step reasoning: building a repertoire

1. **White: choose one or two first moves.** Pick ones where *most* book replies are ≥ 0, so that Black's best reply still leaves you a playable game. For example, **d3** (lowest reply 0) or **c3** (lowest −7).
2. **Black: prepare one reply to each of White's ten first moves.** Choose the most negative eval from White's view, if you understand the resulting structure. Example: against Nc3, …e4 (−74); against Nd3, …f4 (−76); against a3, …c4 (−59); against b3, …a4 (−51); against e3, …a4 (−46); against f3, …d4 (−38).
3. **Attach a plan, not a move list:** for each line write one sentence. *Which pawn will advance, where my knights go, where my king is safe.*
4. **Test it:** play training games from each line against L5–L7 and check the Lab's *Opening understanding* dimension.

Common mistakes

- **Treating ±76 cp as a big difference.** Read through the game's expected-score map (fitted for the Crown engine, so only a rough guide for another engine's centipawns), it is about 0.37–0.63 at the extremes, and these are depth-10 judgements. Either side can play all 64 lines.
- **Memorising moves beyond move 3.** There is no theory to memorise. Your plan matters more.
- **Ignoring the styled opponents' preferences.** A Tactical or Positional competitor picks among its own 30 cp book for six plies (Chapter V.3).

Advanced insight

Long-term plans on 6×6. With only four steps to promotion, a pawn majority on one wing is a much more immediate asset than in orthodox chess. Middlegame plans often come down to "which majority promotes first, and whose king is closer" ([Counsel], consistent with the coach's promotion-race topic). Choose openings whose pawn structure gives *you* the faster majority.

Worked example

A Black repertoire against 1.Nf3. The book's replies range from −30 (…c4, …d4) to +76 (…Nf4). Choose **1…c4** or **1…d4** (−30 each). Plan for 1…d4: the pawn gains space and attacks c3 and e3 at once. Develop the knights towards the queenside, keep the king on d6 until the centre clears, and aim the c-pawn at White's queenside. Avoid 1…Nf4 (+76), the worst book reply for Black.

Summary

The 64 book openings are all balanced at depth 10 (±80 cp). Choose a small repertoire by the worst reply you might face, attach a plan to each line, and test it in training games. There is no 6×6 theory, so understanding beats memory.

Versioned competition and honest progress

The detailed older competition examples above describe their recorded policy. Newer worlds may use crown-competition-3; existing worlds finish their season under its saved policy. Read Qualification · What You Need before planning a route. A current Open invitation, Crown Path place or zonal field must be assessed under that world’s policy, not assumed from an older diagram.

Career Elo is Glicko-2 on the Crown scale, not FIDE strength. New v0.11+ careers begin at zero; legacy careers retain their stored origin. The displayed rating is floored at zero while the underlying value can fall below it. Established status requires twenty rated games **and** RD at most 100. Circuit Grades, permanent Federation Titles, cosmetic rating titles and Played Like analysis estimates are separate concepts. None is bought with Exchange profits.

Record your candidate and predicted defence before opening engine analysis. Compare the same position under documented budgets. A changing score is a reason to investigate, not to quote whichever number flatters your play. Keep private search out of rated assistance and official market pricing.

PART VII · The Crown Exchange

The Language of Chance: probability, prices and expected value

Explanation

A **probability** is a number from 0 to 1 that says how likely an outcome is. For a single game, the three outcomes *White wins, Draw, Black wins* have probabilities that add up to 1.

Because a share pays exactly $1 if its outcome happens, a **price of $0.40 says "40 % likely"**. That is the **implied probability**. In the LMSR the prices of all states always add up to exactly $1 ([Demonstrated], `prices()` normalises them). The price of a claim covering several states is the sum of their prices.

The **expected value** of a trade is what it earns on average, if your belief is right:

> EV = (your probability) × $1 × shares − (total paid, including the fee)

For a small trade at price p, the total paid per share is about $p \times 1.01$, so the trade has positive EV only if your belief q is above **1.01 × p**. That is the **break-even belief**. For larger trades the price moves against you while you buy (Chapter VI.2), and the break-even rises.

Illustrated example

A featured match is priced **White $0.40 · Draw $0.30 · Black $0.30**. You believe White wins 50 % of the time.

You buy **10 White shares**. The game's quote is $4.14 in total, fee included (computed with `buyClaimMicro` and `feeMicro`; the price after your trade is $0.420).

EV = 0.50 × $10 − $4.14 = **+$0.86**. If White wins you receive $10.00 (profit $5.86). Otherwise you receive $0 (loss $4.14). Neither result tells you whether the trade was good. The EV does, and only if your 50 % is right.

Step-by-step reasoning

1. **Write your probability first,** before you look at the price ([Counsel]: this guards against anchoring).
2. **Read the price** as the market's implied probability.
3. **Compare:** is your belief above 1.01 × price?
4. **Get the quote** for the size you want. It includes the price movement and the fee.
5. **Compute EV** = belief × shares − quoted total.
6. **Decide size** separately (Chapter VI.4). A positive EV does not mean "buy as much as you can".

Common mistakes

- **Treating the price as the probability of winning *money*.** The price is the probability of the *outcome*. Your profit also depends on what you paid.

- **Forgetting the draw.** In chess-like games draws are common. The Federation's result model gives about 31 % draws between equals ([Demonstrated], crown-result-model-1, ν = 0.9).
- **Forgetting the fee on the way out.** Selling also costs 1 %.

Advanced insight

The Federation's **public forecast** for a match uses the Davidson model on public ratings ([Demonstrated], src/federation/resultModel.ts): with π = 10^(rating/400) for each side, P(W) = π_w / Z, P(D) = ν·√(π_w·π_b) / Z, P(B) = π_b / Z, where Z is the sum of the three numerators and ν = 0.9. Computed values:

White − Black (public rating)	P(White)	P(Draw)	P(Black)	White's expected score
0	0.345	0.310	0.345	0.500
+100	0.447	0.302	0.251	0.598
+200	0.549	0.278	0.174	0.688
+400	0.722	0.206	0.072	0.825

Prices that stray from this public model are either noise or information. Chapter VI.7 shows which.

Worked example

A promotion market (*Yes / No*) shows **Yes at $0.30**. You think the competitor's chance of finishing in the top two is 40 %.

- Break-even for a small trade: 1.01 × 0.30 = **0.303**. Your 0.40 is above it.
- Quote for 25 Yes shares (liquidity b = 80 for promotion markets): **$8.44** total, average price $0.334, price after $0.369 (computed).
- EV = 0.40 × $25 − $8.44 = **+$1.56**.

Summary

Price = implied probability. EV = belief × payout − total cost. Break-even belief ≈ 1.01 × price for small trades, higher for large ones. Write your belief before you look at the price, and judge trades by EV, not by results.

The Engine of Prices: the LMSR market maker, liquidity and slippage

Explanation

[Demonstrated] src/exchange/lmsr.ts: the Crown Exchange uses Hanson's **logarithmic market scoring rule** with a fixed liquidity parameter *b*. For outstanding shares q = (q_1, …, q_n) across the n states:

> **Cost function:** C(q) = b · ln(Σ_i e^(q_i / b))
>
> **Price of state i:** p_i = e^(q_i / b) / Σ_j e^(q_j / b)
>
> **Cost of a trade** that changes q to q′: **C(q′) − C(q)**

Consequences:

- **Buying a state raises its price and lowers the others.** Prices always add up to 1.
- **Path independence:** buying 30 shares at once costs the same as buying 10 three times (apart from rounding and the fee on each trade).
- **Rounding favours the market:** buy costs are rounded *up* to the next micro-dollar and sale proceeds *down*.

- **The maker's worst-case loss is b · ln n,** and that **subsidy is funded before the market opens**. For a match market (b = 120, n = 3) it is **$131.83**.
- **Liquidity by market family:** match 120, tournament winner 150, promotion 80, qualification 80, season title 100, bracket 100; Sandbox 60 (src/exchange/families.ts, src/app/federation.ts).

A larger *b* means **more liquidity**: prices move less per share. **Slippage** is the difference between the price you see and the average price you pay. Near a price *p*, each share moves the price by about **p(1 − p)/b**.

Illustrated example

A fresh match market: q = (0, 0, 0), b = 120, all prices $0.3333. You buy **30 White shares**.

- C(0,0,0) = 120 · ln 3 = **131.8335**
- C(30,0,0) = 120 · ln(e^0.25 + 2) = 120 · ln(3.284025) = **142.6884**
- Cost = 142.6884 − 131.8335 = **$10.8549** (the game's exact figure: $10.854918)
- Fee: 1 % rounded up = **$0.108550**, so the total is **$10.963468**
- New White price = e^0.25 / (e^0.25 + 2) = 1.284025 / 3.284025 = **$0.3910**; Draw and Black fall to **$0.3045**
- Average price paid: $10.8549 / 30 = **$0.3618**, about 2.9 cents of slippage per share

Step-by-step reasoning: estimating impact before trading

1. Read the current price *p* and the market's *b*.
2. Estimate the price after *n* shares: about p + n·p(1 − p)/b for small n. Near $0.50 with b = 120, that is about 0.2 cents per share.
3. The average price you pay is roughly halfway between the start and the end price.
4. Ask the game for the exact quote. The estimate only tells you whether the size is sensible.

Measured slippage when **White is at $0.40** (Draw and Black $0.30), b = 120 (computed):

Shares bought	Total incl. fee	Average price	White price after
50	$22.78	$0.451	$0.503
100	$50.78	$0.503	$0.605
150	$83.78	$0.553	$0.699
200	$121.18	$0.600	$0.779
250	$162.20	$0.642	$0.843

Common mistakes

- **Judging a trade by the displayed price.** You pay the *average* price over the trade. Use the quote.
- **Buying and selling back quickly.** A round trip costs two fees. Buying 10 shares at $0.50 costs $5.1552, and selling them straight back nets $5.0531: **$0.10 lost** for nothing.
- **Forgetting quotes expire.** A quote is tied to the market's *revision*, which changes with every trade and every published live move. A stale quote is rejected, not filled at a worse price ([Demonstrated], `revisionOf`).

Advanced insight

Because every claim in a market shares **one inventory**, prices stay coherent across claims. In a bracket market (8 states: the possible final-four paths), "A is champion" covers 2 states and "A reaches the final" covers 4, so the champion claim can never cost more than the finalist claim ([Demonstrated], src/exchange/market.ts). In a fresh bracket market every path costs 1/8, so the champion claim starts at **$0.25**.

Worked example

How large can a trade be before it stops making sense? White is at \$0.40 and you believe 0.50. As you buy, the marginal price rises towards your belief. Buying beyond the point where the marginal price passes your belief (minus the fee) *loses* EV. Computed EV, belief 0.50:

Shares	Total paid	EV	Price after
10	\$4.14	+\$0.86	\$0.420
25	\$10.74	+\$1.76	\$0.451
40	\$17.80	+\$2.20	\$0.482
50	\$22.78	+\$2.22	\$0.503
60	\$27.96	+\$2.04	\$0.524
75	\$36.13	+\$1.37	\$0.555
100	\$50.78	−\$0.78	\$0.605

EV peaks at **about 45–50 shares** (the price needed to reach \$0.495 is 46 shares). Beyond that, every extra share costs more than it is worth to you. **The market's liquidity, not your enthusiasm, sets the useful size.**

Summary

The LMSR prices states by $p_i = e^{(q_i/b)/\Sigma e}(q_j/b)$ and charges $C(q') - C(q)$ for a trade. Liquidity b sets the slippage, about $p(1 - p)/b$ per share. The maker's loss is capped at $b \cdot \ln n$, funded in advance. Size your trade by where the marginal price meets your belief, not by your budget.

Measuring the Oracle: calibration and the Brier score

Explanation

Paper forecasts. [Demonstrated] src/app/forecasting.ts (crown-forecast-1): for any market you may record a **free probability vector** over its states. It costs nothing and is **locked when the market closes**. On resolution it is scored with two proper scoring rules:

> **Brier score:** $B = \Sigma_s (p_s - y_s)^2$, summed over **all states**, from 0 (perfect) to 2 (worst). Lower is better.
>
> **Log score:** $L = -\Sigma_s y_s \cdot \ln \max(p_s, 0.001)$. Lower is better.

Here y is the resolution vector: 1 for the state that happened and 0 for the others (½/½ for joint champions). **Cancelled markets are not scored.** Because the game sums over every state, a two-state market's Brier score is **twice** the "binary Brier" you may know from elsewhere.

Your forecasts are compared with two baselines: the **market price** at the time of your forecast, and the **public forecast** (the model). Over your scored forecasts the game shows a tier ([Demonstrated], src/ui/ForecastScreen.tsx): **Apprentice** (fewer than 10 scored), **Sharp** (your mean Brier more than 0.02 below the public forecast's), **Calibrated** (within ±0.02 of it) or **Learning** (more than 0.02 behind it). A rolling chart shows your mean Brier over your last 10 resolved forecasts.

Calibration means that things you call 70 % happen about 70 % of the time. The game pools every (probability, outcome) pair into **ten bins** (0–10 %, 10–20 %, …) and shows each bin's observed frequency with a **Wilson 95 % interval**. A wide interval means too few forecasts in that bin to judge.

Forecast skill is never mixed into chess or financial prestige. A good forecast is not a profitable trade. Trading profit depends on prices paid and fees, and is reported separately.

Illustrated example

You forecast a match **White 0.50 · Draw 0.30 · Black 0.20**. Computed with the game's `brier` and `logScore`:

Result	Brier	Log
White wins	$(0.5-1)^2 + 0.3^2 + 0.2^2 =$ **0.38**	$-\ln 0.5 =$ **0.693**
Draw	$0.5^2 + (0.3-1)^2 + 0.2^2 =$ **0.78**	$-\ln 0.3 =$ **1.204**
Black wins	$0.5^2 + 0.3^2 + (0.2-1)^2 =$ **0.98**	$-\ln 0.2 =$ **1.609**

For comparison: the "know-nothing" forecast (⅓, ⅓, ⅓) always scores **0.667**. A confident miss such as (0.90, 0.05, 0.05) when Black wins scores **1.715** (log 2.996). The same forecast when White wins scores **0.015**.

Step-by-step reasoning: why honesty wins

Suppose a promotion is truly 70 % likely. Your expected Brier score (two-state, as the game computes it):

- say **0.70**: $0.7 \times 0.18 + 0.3 \times 0.98 =$ **0.42**
- say **0.90** (overconfident): **0.50**
- say **0.50** (timid): **0.50**

Any distortion, in either direction, worsens the expected score ([Demonstrated] arithmetic). That is what "proper scoring rule" means: **your best strategy is to report what you actually believe.**

Common mistakes

- **Judging yourself on a handful of forecasts.** For an honest forecaster facing 60 % events, the standard deviation of the mean Brier is about **0.062 over 10 forecasts, 0.028 over 50 and 0.014 over 200** (computed). The ±0.02 band between "Sharp", "Calibrated" and "Learning" needs **well over 50 forecasts** to mean much.
- **Rounding to 0 or 1.** The log score clamps at 0.001, but a near-zero forecast that misses still scores $-\ln 0.001 \approx 6.9$.
- **Copying the market price.** You will match the market's Brier, which is safe, but you learn nothing about your own judgement.

Advanced insight

Calibration and sharpness are different. A forecaster who always says (⅓, ⅓, ⅓) is perfectly calibrated in a trivial sense but scores 0.667 every time. Skill is being calibrated **and** confident when the evidence allows. The bins show calibration, and the mean Brier against the public forecast shows whether you add anything ([Counsel] on interpretation; the measures are [Demonstrated]).

Worked example

Five resolved forecasts on match markets:

Your forecast (W/D/B)	Result	Brier
0.50 / 0.30 / 0.20	White	0.38
0.40 / 0.35 / 0.25	Draw	0.16 + 0.4225 + 0.0625 = 0.645
0.60 / 0.25 / 0.15	White	0.16 + 0.0625 + 0.0225 = 0.245
0.30 / 0.30 / 0.40	White	0.49 + 0.09 + 0.16 = 0.74
0.45 / 0.30 / 0.25	Black	0.2025 + 0.09 + 0.5625 = 0.855

Mean = (0.38 + 0.645 + 0.245 + 0.74 + 0.855)/5 = **0.573**. That beats the uniform 0.667. Whether it beats the public forecast depends on *its* scores on the same five markets, and five forecasts are far too few to say (see above).

Summary

The game scores paper forecasts with the Brier score (sum over all states, 0–2) and the log score, both lower-is- better and both proper: honesty minimises your expected score. The calibration chart and the tiers need many forecasts before they mean anything. Forecast skill is separate from trading profit.

Sizing the Stake: Kelly as a ceiling, risk as a discipline

Explanation

The **Kelly criterion** gives the fraction of your bankroll that maximises long-run growth *if your probability is exactly right*. For buying a $1 claim at price p when your belief is q (ignoring fees and slippage):

f* = (q − p) / (1 − p)

Three facts make full Kelly dangerous ([Counsel], widely accepted in forecasting practice):

1. **Your q is an estimate.** If it is too high, Kelly over-bets, and over-betting by a factor of two already gives zero long-run growth.
2. **Swings are large** even when q is right.
3. **In this game, prices move while you buy,** so the real edge shrinks as size grows (Chapter VI.2).

So use **fractional Kelly (½ or ¼) as a ceiling, never as a target,** and cap it further by liquidity and by fixed exposure limits. **Never stake your whole bankroll on one market.** Keep cash for entry fees and running costs: your career pays bills too. The economy simulations in docs/V8_SOVEREIGN_STANDARD.md show strategies that ran cash below $100 for many rounds ([Demonstrated] that it happens; the rule is [Counsel]).

The game's own NPC traders obey similar discipline ([Demonstrated], src/exchange/npc.ts): each trades at most a fixed share of its wallet per market (2 %–12 %), and their **open cost basis may not exceed 60 % of the wallet**.

Illustrated example

Bankroll **$1,000**. White is at **$0.40**, and you believe **0.50**.

- f* = (0.50 − 0.40)/(1 − 0.40) = **0.1667**, so full Kelly is $167, half Kelly $83 and quarter Kelly $42.
- **Half Kelly in this market:** $83 buys **148 shares** at an average of $0.551 and pushes the price to $0.696. The EV at your belief is **−$8.37**. You paid more per share than you think the share is worth!
- **The liquidity limit:** EV peaks at about 45–50 shares, roughly **$20–23** (Chapter VI.2).

In a b = 120 market, **liquidity binds long before Kelly.** The right size here is about $20, or 2 % of the bankroll, not $83 or $167.

Step-by-step reasoning: a sizing routine

1. Compute f* and halve or quarter it. That is the **ceiling**.
2. Find the size at which the marginal price reaches your belief minus the fee. That is the **liquidity limit**.
3. Apply your **per-market cap**, for example 2–5 % of your bankroll ([Counsel]).
4. Check your **total exposure** across all open markets, for example at most 25 % ([Counsel]).
5. Take the **smallest** of these four numbers.

Common mistakes

- **Doubling up after a loss** ("chasing"). The next market is independent of your last result. Your bankroll is smaller, so your size should be too.

- **Treating a closed market as cash.** A closed market cannot be traded. Until settlement, the game values it only at its **cancellation vector** (`liquidationValue`).
- **Ignoring cancellation risk.** If you bought 100 White shares in a fresh match market for $43.66 and the game is cancelled (for example by the 400-ply playing limit), you receive **$33.33**.
- **All-in on a "sure thing".** In a chess game the favourite at 0.82 still fails nearly one time in five.

Advanced insight

Correlated positions count as one. Positions that depend on the same competitor, say a tournament-winner claim on them and "they win" in two of their featured match markets, rise and fall together. For sizing, add them up as a single exposure. Chapter VI.5 shows how.

Worked example

Bankroll $600. A promotion market (b = 80) shows Yes at $0.30, and you believe 0.40.

- f* = (0.40 − 0.30)/(0.70) = 0.143. Quarter Kelly is about **$21**.
- Liquidity: the marginal price reaches $0.396 (your belief minus the 1 % fee) after **34 shares**, which cost **$11.91** in total (average $0.347). Beyond that, extra shares lose EV. 50 shares would cost $18.70 and move Yes to $0.445, with EV down to +$1.30.
- A per-market cap of 3 % gives **$18**.
- The smallest limit is liquidity: buy about **34 shares for $11.91**, with EV = 0.40 × $34 − $11.91 = **+$1.69** (all computed with the game's functions). Check the live quote before you confirm.

Summary

Kelly f* = (q − p)/(1 − p) is a ceiling, not a target. Use ½ or ¼ of it at most. In the Crown Exchange, slippage usually limits size first: stop where the marginal price meets your belief minus the fee. Add correlated positions together, cap per-market and total exposure, keep cash for your career, and never go all-in.

Tournament Forecasting: correlated outcomes and hedging

Explanation

Market families. [Demonstrated] src/exchange/families.ts `RULES`:

Family	States	Trading closes	Settles on	Cancellation pays
Match	White / Draw / Black	the match starts (live: after each published move)	the result	1/3 per share
Winner (Swiss Open, Championship)	every entrant	first round processed or first game started	the champion (joint: shares split)	1/n
Promotion	Yes / No (top two)	last league round starts	the promotion list	½
Qualification	Yes / No (Crown Championship)	last league round starts	the Championship field	½
Season title	every division member	last league round starts	the title holder (league order)	1/16
Bracket	8 final-four paths	the semifinals start	the realised path	|S|/8 per share of claim S

The public models use public ratings and recorded results only: the Davidson match model, and Monte Carlo simulations of the remaining league rounds, of Swiss events and of the bracket (src/exchange/families.ts). Markets close **when their closing round starts**, so no result is ever traded after it is known (src/app/markets.ts).

Correlated outcomes move together. A competitor who wins their games raises their price in *every* market they appear in at once. Markets on the same competitor, or on competitors in the same league, are linked. In the bracket market, claims are subsets of the same 8 paths, so "champion", "finalist" and "beats Y in the final" are nested. A conjunction claim pays nothing if its condition fails. It is **not refunded**.

Hedging means reducing the spread of your possible outcomes, usually by selling part of a position after the price has moved. In this game you cannot short, so you hedge by **selling what you hold** (or, in a multi-state market, by buying other states).

Illustrated example

You bought **100 White shares** at $0.40 for **$50.78** (fee included). Later, other traders move the market to **White $0.60 · Draw $0.20 · Black $0.20**. Computed with the game's functions (b = 120):

Choice	Cash now	If White wins	If not
Hold all 100	$0	**+$49.22**	**−$50.78**
Sell 50 (nets $27.17, price falls to $0.497)	$27.17	**+$26.39**	**−$23.61**
Sell all 100 (nets $49.23, price falls to $0.395)	$49.23	**−$1.55**	**−$1.55**

Selling everything at the new price still *loses* money: selling 100 shares pushes the price back down (slippage), and you pay the fee twice. **Hedging has a price.** It is worth paying only if your belief has changed or your exposure is too large.

Step-by-step reasoning: a tournament forecast

1. **Start from the public model,** and know what it uses: public ratings and recorded results.
2. **Ask what it cannot know:** a high-RD rating is a weak signal (Book V, Chapter V.3), and the remaining schedule matters (who still plays whom).
3. **Write your probability vector** as a paper forecast before trading.
4. **Map your exposure:** list every market in which this competitor appears.
5. **Trade only the difference** between your belief and the price, sized by Chapter VI.4.

Common mistakes

- **Thinking you are diversified when you are not.** Three markets on one competitor are one bet.
- **Buying a conjunction expecting a refund.** "X beats Y in the final" pays $0 if Y never reaches the final.
- **Trading a winner market you might enter.** Your entry cancels it at 1/n, so your position returns 1/n per share, whatever you paid.
- **Forgetting the conflict rule.** If you sponsor a competitor, the game refuses your buy orders in their markets.

Advanced insight

In the bracket market every claim is priced from the same 8 path prices, so you can check coherence yourself: the champion claim must equal the sum of its two paths, and the finalist claim the sum of its four. If you believe one path is mispriced, the cheapest way to act is the **narrowest claim containing it**. Broader claims carry the other paths' prices too ([Counsel], following directly from the [Demonstrated] claim structure).

Worked example

A 16-player division title market. The public model gives the leader 0.35 with four rounds to go. The leader's remaining opponents are the three strongest players in the division, so you estimate 0.28. The leader's price is $0.34. **Do not buy the leader, and do not "short" it either (impossible).** Your options are a paper forecast of 0.28, or buying a rival you think is underpriced, *if* your belief about that rival is also above its price. A low opinion of one competitor does not by itself make any other one cheap.

Summary

Know each family's states, closing time, settlement and cancellation. Public models use only public information. Linked markets are one exposure. Hedge by selling what you hold, but remember that slippage and fees make hedging costly. Trade only the gap between a written belief and the price.

Live Odds and the Uncertain Board

Explanation

In-play trading. [Demonstrated] src/app/federation.ts, src/ui/SpectatorScreen.tsx. When you watch a featured Federation game live, its match market can be traded **between moves**. After each move is published, then before the next move is computed, you see what every trader sees: the moves, the engine's own evaluation, and the price history. **Future moves do not exist while you can trade.** Each move is computed only after the trading window closes, and a trade queued while a move is being computed is rejected once it is published. **Move grades stay hidden until the market is settled or cancelled,** and Match Analysis runs only then.

The live model (crown-odds-2) [Demonstrated] src/exchange/odds.ts, docs/ODDS_MODEL.md. It is a forecast, never the price. It is an *ordered logit*: one latent number x measures White's advantage, a draw width $\theta \geq 0$ sits on both sides of it, and with $\sigma(z) = 1/(1 + e^{-z})$:

$$P(\text{White}) = \sigma(x - \theta),\ P(\text{Black}) = \sigma(-x - \theta),\ P(\text{Draw}) = 1 - P(\text{White}) - P(\text{Black}).$$

Before the game, x comes from the **public rating gap** (deflated by both RDs, as in Glicko) plus a small fitted first-move edge for White, and θ from the players' **level** (stronger pairs draw more). After each published move n, with the published evaluation e (White's view, in units of 100 cp, clamped at ±16):

$$x = a(n)\cdot x_0 + b(n)\cdot e,\quad \theta = \theta_0 \cdot \exp(d(n) + m\cdot(\text{material traded})).$$

a, b and d were *fitted* on 3,200 Federation-style engine games (held-out validated; the report is in docs/ODDS_MODEL.md). The rating prior keeps a decaying weight: a = 0.83 at ply 10, 0.59 at ply 40, 0.15 at ply 160. The evaluation's weight b rises to a peak near ply 40. The draw width grows with the ply count and with traded material, because long, simplified games are drawish. At ply 0 the live model is exactly the pre-game forecast.

Two NPC desks act in-play: *Kestrel Tactical Desk* trades on this model, and *Tidewater Momentum* buys whatever has risen since the last window (price history only).

A fact worth noticing. [Demonstrated] Up to v0.8 the live model used *only* the evaluation, so at 0 cp it gave about 0.35 / 0.30 / 0.35 whatever the ratings (bug #21 in docs/BUG_REGISTER.md). crown-odds-2 fixed this: at an equal evaluation the favourite stays the favourite, by a margin that fades as the game goes on. How much the ratings should still count late in a game is where your own judgement as a forecaster comes in.

Illustrated example

A featured game between public ratings **1850 (White) and 1750**, both RD 70. Pre-game model: **0.660 / 0.107 / 0.233**. Live model after ply 20 with all pieces still on the board, computed with the game's own `liveWdl`, next to an equal pairing (1800 v 1800) for comparison:

Evaluation (White)	1850 v 1750: W / D / B	1800 v 1800: W / D / B
0 cp	0.578 / 0.172 / 0.251	0.462 / 0.190 / 0.347
+50 cp	0.690 / 0.139 / 0.170	0.584 / 0.170 / 0.246
+150 cp	0.855 / 0.073 / 0.072	0.788 / 0.102 / 0.109
+300 cp	0.962 / 0.020 / 0.018	0.942 / 0.031 / 0.028
−150 cp	0.240 / 0.168 / 0.592	0.166 / 0.137 / 0.697

And the same 1850 v 1750 game at a level 0 cp as the ply count grows (all pieces on): ply 1 **0.662 / 0.107 / 0.231**, ply 10 **0.613 / 0.133 / 0.253**, ply 40 **0.495 / 0.259 / 0.246**, ply 80 **0.341 / 0.447 / 0.212**, ply 160 **0.230 / 0.587 / 0.184**. A level position late in the game is mostly a draw.

Step-by-step reasoning: how certain is this board?

An evaluation is an estimate from a search of limited depth, and its certainty varies a lot.

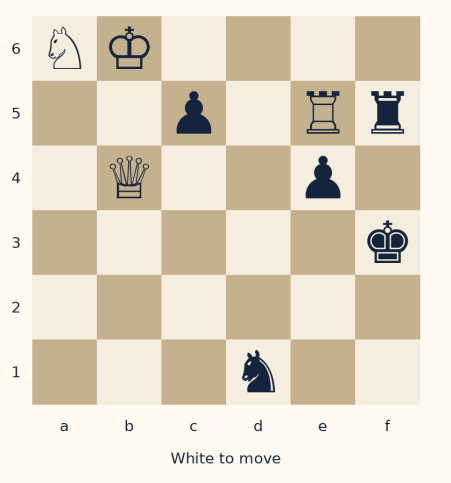


Figure 110: White to move. Here the result is certain: there is a forced mate in two.

Verified line: Qxe4+

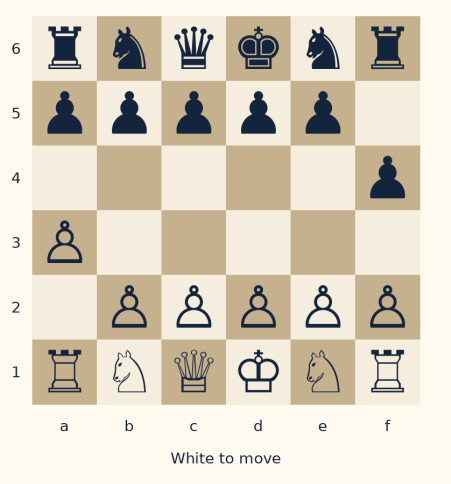


Figure 111: After 1.a3 f4, one of the book's dead-level openings (0 cp at Fairy-Stockfish depth 10). Most of the game is still to be played.

Verified line: a3 f4

1. **Forced mate on the board:** the result is settled (the first board is proven by exhaustive search). A forecast should be near 1, and only the remaining time on the clock adds doubt.

2. **A large but unforced edge** (+300 cp): the map says about 0.93 in engine games. But the map was fitted on *engine* games, and a weaker or time-pressed player converts less reliably ([Counsel]).
3. **A balanced position** (the second board): the evaluation says little. The draw rate and the players' strengths dominate, which is why the live model keeps a share of the rating prior and widens the draw band as the game goes on.
4. **Volatile positions:** if the evaluation has swung several times, the search is struggling. Widen your uncertainty rather than chasing each swing ([Counsel]).

Common mistakes

- **Treating the evaluation as the probability.** It is an input to a model that maps it to an expected score.
- **Chasing momentum.** A price that rose in the last window carries no promise about the next one.
- **Trading on a stale quote.** It will be rejected. Read the new position first.
- **Expecting grades during the game.** They appear only after settlement.

Advanced insight

The logistic σ is steep near balance and flat near decisive positions, like the map of Book V, Chapter V.6. So a 100 cp swing around 0 moves the live model much more than a 100 cp swing around +500. When a featured game is balanced, small evaluation changes produce large live-model changes, and prices driven by the Tactical Desk can swing. That is where a forecaster who reads the position calmly can disagree most usefully ([Counsel]).

Worked example

The 1850 v 1750 game above. After move 18 (ply 36) the evaluation is **+150 cp** for White, and one rook pair has been exchanged (material 40 of 50). The live model says **0.852 / 0.098 / 0.050**. The price of White is **$0.80**. You have watched the game: White's advantage is a protected passed pawn, and Black has no counterplay. You believe **0.88**.

- Small-trade break-even: $1.01 \times 0.80 = 0.808 < 0.88$, so the trade is in your favour.
- Size it with Chapter VI.4: a match market has $b = 120$, so at \$0.80 each share moves the price about $0.80 \times 0.20/120 \approx 0.0013$. Stop well before the price reaches about 0.87.
- Write the paper forecast as well. It is scored even if you do not trade.

Summary

Live trading happens between published moves, on public information only. Grades stay hidden until settlement. The live model (crown-odds-2) combines a fading share of the rating gap with the evaluation through an ordered logit, with a draw band that widens in long and simplified games. Judge how certain the board really is: proven mate, large edge, balanced, volatile.

The Crowd and the Mirror: NPC traders, overconfidence and the long run

Explanation

The NPC traders (crown-npc-traders-2). [Demonstrated] src/exchange/npc.ts. All eight trade through the same LMSR as you, long-only, with funded wallets. They use **only** information available when they decide: the published model (public ratings), the live engine evaluation during play, and public price history. They **never** see hidden strength or any future move or result. **No trader is guaranteed to profit.**

Trader	Behaviour	Noise	Margin	Budget per market	Acts
Old Harbour Fund	conservative value; trades only large mispricings	0.05	0.06	2 %	pre-game, between rounds
Crown Supporters Club	overconfident fans of the favourite (+0.06 on the favourite)	0.08	0.02	5 %	pre-game, live
Lantern Contrarians	back the underdog (+0.06)	0.08	0.04	3 %	pre-game, between rounds
First-Time Forecasters	noisy novices	0.14	0.02	3 %	pre-game, live
Tidewater Momentum	buys what has risen since the last window	—	0.03	2 %	live, between rounds
Summit Tournament Partners	tournament specialist (season and tournament markets)	0.03	0.02	6 %	pre-game, between rounds
Kestrel Tactical Desk	trades in-play on the public evaluation	0.03	0.03	4 %	live
Meridian Analytics	calibrated fundamental trader on the public model	0.03	0.02	12 %	pre-game, between rounds

Noise is the standard deviation added to the public belief, and *margin* is the minimum edge a value trader demands. A value trader buys the claim with the largest (belief − price − margin) up to a target price of (belief − margin). Biased and noisy traders act **first** and the informed ones **last**, so prices usually end near the public consensus, with residual noise. Wallets are topped up each season to at most $4,000, and open exposure is capped at 60 % of the wallet.

What this means for you ([Counsel], reasoned from the [Demonstrated] design): the price you see usually reflects the *public model* closely. If you have an edge at all, it comes from **better judgement** than the public ratings contain: about rating reliability (RD), about the remaining schedule, and about what the board shows in live games. It does not come from tricks. If you have no view, do not trade. A paper forecast is free.

Illustrated example: overconfidence

A forecaster says 0.9 to every "favourite wins" question. The favourites truly win 70 % of the time. Expected Brier (two-state, game formula): honest 0.70 gives **0.42**, while the overconfident 0.90 gives **0.50**. In trading, overconfidence makes Kelly-sized stakes far too large, and in this game it also makes you push prices past your true break-even (Chapter VI.4).

Step-by-step reasoning: luck versus skill

Suppose every trade is like the Chapter VI.2 example: 50 shares bought at $0.40 for **$22.78**, in an outcome that is truly 50 %. EV = **+$2.22** per trade. The outcome alone has a standard deviation of $25 per trade. With independent trades (computed with a normal approximation):

Trades	Expected profit	Standard deviation	Chance of being behind anyway
1	$2	$25	46.5 %
10	$22	$79	38.9 %
100	$222	$250	18.7 %
400	$889	$500	3.8 %
1,000	$2,222	$791	0.2 %

A genuine edge can lose for a long time, and a bad trader can be ahead after ten trades. Results tell you about skill only over hundreds of comparable decisions. Your Brier score against the public forecast is usually a faster, cleaner signal than your profit.

Common mistakes

1. **Outcome bias:** "I lost, so it was a bad trade." Judge EV at the time of the trade.
2. **Overconfidence:** probabilities too close to 0 or 1.
3. **Anchoring** on the price: write your belief first.
4. **Chasing** losses, or doubling up after a win.
5. **Ignoring the fee and slippage,** especially on round trips.
6. **Mistaking correlation for diversification.**
7. **Trading without a view:** when you have no view, the fee is your only certainty.
8. **Letting the Exchange take over your career.** Forecasting skill is not chess skill, and cash lost on the Exchange is cash you no longer have for entry fees and club costs.

Advanced insight

Base rates first. Before any specific reasoning, start from the base rate (for example the ≈ 31 % draw rate between equals in the result model), then adjust for specifics. Forecasters who start from the specifics tend to overreact to vivid details such as a dramatic last game ([Counsel], a well-established finding in forecasting practice, not measured in this game).

Worked example: a forecaster's journal entry

Market: season title, 4 rounds left. *Price* of competitor K: $0.22. *Public model:* 0.24. *My belief:* 0.30. *Reason:* K's rating is provisional (RD 180) after a strong run of results. I think K is better than the rating says, and K's remaining opponents are the weakest four. *Exposure on K elsewhere:* none. *Size:* quarter Kelly = (0.30 − 0.22)/0.78/4 ≈ 2.6 % of the bankroll, capped to 2 %. *Paper forecast:* recorded. *Review date:* after settlement. Check the Brier score and whether the provisional rating moved the way I expected.

Summary

The NPC traders trade on public information only, with documented biases, noise and limits, and no one is guaranteed a profit. Prices therefore tend to sit near the public model. Any edge you have must come from better judgement, and a paper forecast is the free way to test it. Beware overconfidence, outcome bias and correlation. Skill shows only over large samples, and the Brier score shows it faster than profit.

Execution, settlement and risk

All currency is fictional. LMSR marginal price is not the average cost of a sizeable order. Fees are 1% of executed cash on buys and sells, rounded upward in microcrowns; buy cost rounds up and sale proceeds down. A basic winning share pays one crown, but net profit subtracts the amount paid. Cancellation uses a published payout vector, not a refund of your purchase price.

Live result markets can trade between published moves; length markets close at the start. The 400-ply playing limit is a censored chess result: result-market VOID and length-market Over are different settlement rules. Do not count a cap as a draw. Decisive overlaps the White-win and

Black-win states; it is not an independent fourth outcome. A bracket conjunction can lose when its condition fails rather than refunding.

Track executable exit value, cost basis, realised P&L, unrealised P&L and pending closed positions separately. Test correlated scenarios and the VOID vector. Make paper forecasts before trading and score them on the same events. A lucky payout is not proof of calibrated judgement.

PART VIII · The Twelve-Week Training Program

A training schedule for Crown & Calculation that uses Books I–VI, the Academy, the Lab and the Federation.

This program turns the six books into twelve weeks of practice. Each week has the same parts: **objective, reading, Academy work, practice games, skills to demonstrate, review questions and a progress check.** Three adjustments (Beginner, Intermediate, Advanced) and four daily lengths (10, 20, 30 and 60 minutes) let you fit it to your level and your time.

How the program uses the game

[Demonstrated] All of the following come from the game's code (src/academy/*, src/ui/AcademyScreen.tsx, src/engine/levels.ts, src/games/types.ts):

- **Academy modes:** *Tactical Academy* (themed puzzles; choose a theme button, or let difficulty adapt), *Puzzle Rush* (3 minutes, three strikes; any solve with a hint or a wrong move counts as a strike), *Calculation Chamber* (puzzles needing two or more of your moves; 72 in the practice pool), *Defensive Survival* (251), *Endgame Workshop* (116; solve, then play the position out against the engine), *Replay Your Mistakes* (moves from your analysed games that lost ≥ 0.12 expected score), *Adaptive Training* (your weakest themes) and *Held-out Assessment* (10 puzzles you have never seen).
- **Puzzle pools by theme** (practice split, retired puzzles excluded): Defensive resource 173, Endgame technique 116, Hanging piece 109, Calculation 83, Position evaluation 76, Fork 44, Skewer 34, Pin 30, Discovered attack 13, Promotion race 10, Mate threat 9, Checkmate 1. A puzzle can carry several themes.
- **Assessments are a scarce resource.** There are 81 held-out puzzles, and each is shown **at most once**, so you have about **eight** ten-puzzle assessments in total. This program uses **four**: weeks 1, 4, 8 and 12. Do not spend them casually. Each one is a measurement, not training.
- **Opponents:** engine levels L1 Novice (800) to L8 Crown (2926) and L9 Sovereign Unbound (3038, derived), plus Federation competitors via *Random Federation Quick Match* or your league fixtures. These ratings are on the Crown scale, not FIDE.
- **Time controls:** Blitz 3+2, Rapid 5+3 (used for Federation games), Classical 15+10, Untimed. **Untimed games are never rated.** Each timed control has its own rating pool.
- **Review:** Analysis (Quick, Standard and Deep presets), *Retry from This Position (Unrated)*, bookmarks, the Lab (with 95 % intervals and "insufficient evidence" flags) and Paper Forecasts with Brier scoring.

Keep one analysis preset (Standard) for your record. The Lab combines only reports of one methodology (Book V, Chapter V.6).

Level adjustments

	Beginner	Intermediate	Advanced
Who	new to chess, or rated below about 1200 (Crown)	about 1200–1900	about 1900 and above
Main opponents	L1–L3	L3–L5, Federation 1300–1900	L5–L8, Federation 1900+
Main time control	Untimed, then Rapid 5+3	Rapid 5+3, Classical 15+10	Classical 15+10, Rapid 5+3
Puzzle difficulty	let the Academy adapt from 1–2	adapt from 2–3	adapt from 3–5
Reading	every chapter fully, all exercises	skim chapters you know; do every exercise	read Books V–VI fully; Books I–IV only for review
Book VI strand	paper forecasts only, no trading	paper forecasts; small trades (≤ 2 % of cash) from week 8	full strand, sized by Book VI, Chapter VI.4

Daily session variants

Pick the length you can keep up **every day**. Consistency beats length ([Counsel]).

Daily time	Shape of a session
10 minutes	5 themed puzzles in the week's theme, or one Puzzle Rush, then 2 minutes rereading one *Summary* section. Play one game per week at the weekend.
20 minutes	10 themed puzzles + one Rush; *or* one Rapid 5+3 game with a 5-minute review of its first turning point.
30 minutes	10 puzzles (week theme) + one Rapid game + a review using Retry from This Position on your worst move.
60 minutes	15 minutes reading and exercises, 15 minutes Academy (week theme + one Rush), one Classical 15+10 *or* two Rapid games, 10 minutes Analysis review, and one paper forecast.

The weekly targets below assume the **30-minute** session. With 10 minutes, halve the puzzle and game counts. With 60 minutes, use the upper numbers.

Week 1 — The Rules and the Board

- **Objective:** play legal, complete games without hesitation, and know every Crown rule that differs from orthodox chess.
- **Read:** Book I, Chapters 1–3 (*The Board and Its Coordinates*; *The Pieces I: Rook, Queen and King*; *The Pieces II: Knight and Pawn*) and Chapter 7 (*Stalemate and the Other Draws*). Rulebook §§3–5 and §10.
- **Academy:** take **Assessment 1** (baseline: 10 held-out puzzles; write down your score). Then 20 Tactical Academy puzzles at the adaptive difficulty.
- **Practice games:** 5 Untimed games against L1 (Beginner), L2 (Intermediate) or L4 (Advanced). Every game ends in checkmate, a draw rule or resignation. Note which rule ended it.
- **Demonstrate:** no illegal-move attempts; correctly predict when a position is **bare kings**, **stalemate** or a **claimable** threefold repetition.

- **Review questions:** (1) Why is there no double pawn step or en passant? (2) Which draws are automatic, and which must be claimed? (3) If you flag on time while your opponent has a single pawn left, what is the result?
- **Progress check:** a written baseline: Assessment 1 score, and your first five game results by termination type.

Week 2 — Piece Safety and Basic Tactics

- **Objective:** stop losing pieces for nothing, and win the ones your opponent leaves hanging.
- **Read:** Book I, Chapter 4 (*Capturing and Piece Safety*) and Chapter 5 (*Check and Escaping Check*). Book II, *The Duellist*: its chapters on double attacks, forks and pins.
- **Academy:** 40 puzzles: *Hanging piece* (20), *Fork* (10), *Pin* (10). Two Puzzle Rush runs (record the best).
- **Practice games:** 5 Rapid 5+3 games: L2 (Beginner), L3–L4 (Intermediate), L5 (Advanced). Analyse each game with Standard.
- **Demonstrate:** no **Blunder** grades caused by a hanging piece in your last three games. Use the coach's *Hanging pieces* topic as the check.
- **Review questions:** (1) What three questions do you ask before every move? (2) Why are knight forks especially common on a 6×6 board? (3) What does the coach mean by "threat awareness"?
- **Progress check:** Rush best score compared with week 1, and the number of hanging-piece notes per game.

Week 3 — Checkmates and Defence

- **Objective:** deliver the basic mates quickly, and survive attacks by finding the only move.
- **Read:** Book I, Chapter 6 (*Checkmate: King and Queen, King and Rook*) and Chapter 8 (*Simple Winning Patterns*). Book II: its chapters on mating patterns and defence. Book V, Chapter V.5 (*Critical Moments, Stronger Opponents and Predictable Mistakes*), "Explanation" section only.
- **Academy:** 30 *Defensive Survival* puzzles; all 9 *Mate threat* puzzles; the Grimoire's mate-in-1 and mate-in-2 boards in Book I. (The Academy's *Checkmate* pool has only one puzzle; the Grimoire boards are exhaustively verified substitutes.)
- **Practice games:** 4 Rapid games; in two of them, start a **training** game from a position where you have K+Q or K+R against K and mate within 20 moves (Untimed).
- **Demonstrate:** K+Q vs K and K+R vs K mates from any start position; one game saved with a defensive only-move.
- **Review questions:** (1) Why is the sixth rank (first rank for White) a mating wall? (2) How does a stalemate trap arise when you are mating with K+Q? (3) What is a "critical moment" in the game's Analysis?
- **Progress check:** your Defensive Survival first-try clean rate (Academy statistics) with its sample size.

Week 4 — Combinations

- **Objective:** chain two or three forcing moves into a combination, and verify it before playing it.
- **Read:** Book II's chapters on combinations, discovered attacks and skewers. Book V, Chapter V.1 (*The Pruned Tree: calculation and candidate discipline*), complete with exercises.
- **Academy:** 15 *Calculation Chamber* puzzles; *Skewer* (10) and *Discovered attack* (all 13). Then take **Assessment 2**.
- **Practice games:** 4 Rapid games against an opponent 100–200 points above you (a level or a Federation competitor).
- **Demonstrate:** solve Book V's exercises V.1.1–V.1.3 without hints; in a game, play one combination that you had calculated to the end.
- **Review questions:** (1) Why do you need only **one** good defence to reject a candidate, but must refute **every** defence to accept one? (2) Why is the quiet move Rb5 in Book V's worked

example easy to miss? (3) Did your Assessment 2 score change from Assessment 1, and is a ten-puzzle difference meaningful?
- **Progress check:** Assessment 2 compared with Assessment 1. Remember that 10 puzzles is a small sample. The Lab combines all assessments and shows the uncertainty.

Week 5 — Openings

- **Objective:** have a small, sound repertoire as White and a reply to each of White's ten first moves as Black.
- **Read:** Book III, *The Strategist*: its opening-principles chapter. Book V, Chapter V.7 (*The Sixty-Four Gates: building a repertoire*). Book VI, Chapter VI.1 (*The Language of Chance*).
- **Academy:** 20 puzzles of your choice, plus Adaptive Training (10).
- **Practice games:** 6 Rapid games, three as White with your chosen first move, and three as Black with your prepared replies. Opponents: L3 (Beginner), L5 (Intermediate), L6–L7 (Advanced).
- **Oracle strand:** record **3 paper forecasts** on featured match markets (Book VI, VI.1).
- **Demonstrate:** a one-sentence plan for each of your repertoire lines; no opening-phase Mistake in three games.
- **Review questions:** (1) What does "balanced" mean for the 64 book openings? (2) Why is memorising beyond move 3 of little use here? (3) What is the implied probability of a $0.31 Draw price?
- **Progress check:** the Lab's *Opening understanding* rate (or "insufficient evidence"; it needs at least 10 opportunities from 3 games).

Week 6 — Pawn Structure

- **Objective:** read pawn structures and play for the faster majority.
- **Read:** Book III's chapters on pawn structure, passed pawns and weak squares. Book I, Chapter 9 (*The Pawn's Road: Four Steps to the Crown*). Book V, Chapter V.2 (*Complexity, Imbalance and the Art of Simplifying*).
- **Academy:** all 10 *Promotion race* puzzles; *Position evaluation* (15).
- **Practice games:** 4 Rapid or 2 Classical games. In each, write at move 12 which side has the faster pawn majority, and check your note against the evaluation graph afterwards.
- **Oracle strand:** 3 more paper forecasts. Read Book VI, Chapter VI.2 (*The Engine of Prices*).
- **Demonstrate:** win or hold one game by pawn play; solve Book V exercises V.2.1–V.2.2.
- **Review questions:** (1) Why are passed pawns more dangerous on 6×6 than on 8×8? (2) When should you trade into a pawn ending? (3) How much does the price move when you buy 30 shares in a fresh match market?
- **Progress check:** your Promotion-race solve rate and the coach's *Promotion races* note count per game.

Week 7 — Endgames

- **Objective:** convert won endings and hold drawn ones, with the Crown draw and timeout rules in mind.
- **Read:** Book IV, *The Master*: its endgame chapters. Rulebook §10 (draws) and §13 (clocks).
- **Academy:** 25 *Endgame Workshop* puzzles. For five of them, use **play out against the engine** after solving.
- **Practice games:** 4 Rapid games. In two of them, use *Retry from This Position (Unrated)* from an endgame you mishandled.
- **Oracle strand:** 3 paper forecasts; read Book VI, Chapter VI.3 (*Measuring the Oracle*).
- **Demonstrate:** convert one winning ending against L4+; hold one inferior ending to a draw (repetition, bare kings or stalemate).
- **Review questions:** (1) What happens on time-out when your opponent has only a king? (2) When is a 100-halfmove draw automatic, and when must it be claimed? (3) Compute the Brier score of a forecast (0.5, 0.3, 0.2) that ends in a draw.
- **Progress check:** the Lab's *Endgame technique* rate with its interval.

Week 8 — Calculation

- **Objective:** calculate forcing lines three moves deep, accurately and in reasonable time.
- **Read:** Book V, Chapter V.1 again (redo the guided exercise from memory) and Chapter V.5. Book VI, Chapter VI.4 (*Sizing the Stake*).
- **Academy:** 25 *Calculation Chamber* puzzles; 3 Puzzle Rush runs. Then take **Assessment 3**.
- **Practice games:** 3 Classical 15+10 games. Before each critical-looking move, write your candidate list with one named refutation each.
- **Oracle strand:** Intermediate and Advanced may start **small trades** (Intermediate ≤ 2 % of cash per market). Beginners keep to paper forecasts.
- **Demonstrate:** in a Classical game, a correct three-move forcing line. Assessment 3 completed.
- **Review questions:** (1) In what order do you list candidates? (2) What is the Kelly fraction for q = 0.5, p = 0.4, and why is half of it still too much in a b = 120 market? (3) Did your clean rate in the Calculation Chamber rise compared with week 4?
- **Progress check:** Assessment 3 compared with Assessments 1–2; the Lab's *Calculation* dimension.

Week 9 — Positional Play and Defence

- **Objective:** make good quiet moves, and defend worse positions patiently.
- **Read:** Book III's chapters on planning and piece activity. Book IV's chapter on defence. Book V, Chapter V.3 (*Reading the Opponent: profiling and style adaptation*).
- **Academy:** 20 *Position evaluation* puzzles; 20 *Defensive Survival* puzzles; Adaptive Training (10).
- **Practice games:** 4 games against **Federation competitors** (Quick Match or league fixtures). Before each, write a profile: rating, RD, style (is it validated?), and your plan.
- **Oracle strand:** 3 paper forecasts on Federation markets; read Book VI, Chapter VI.5 (*Tournament Forecasting*).
- **Demonstrate:** one game with no Mistake in its quiet (non-critical) moves; a written opponent profile that correctly states which styles affect play.
- **Review questions:** (1) Which four styles are validated, and what does each change? (2) What does a high RD tell you about a rating? (3) Why are three markets on one competitor a single exposure?
- **Progress check:** the Lab's *Positional understanding* and *Defensive awareness* rates.

Week 10 — Tournaments and Time

- **Objective:** manage the clock and the standings across an event.
- **Read:** Book V, Chapter V.4 (*The Tournament Hall: Swiss, leagues and knockouts*). Rulebook §§16–17. Book VI, Chapter VI.6 (*Live Odds and the Uncertain Board*).
- **Academy:** 3 Puzzle Rush runs (speed under pressure); 15 mixed puzzles.
- **Practice games:** play the league/Swiss rounds that fall in your season, or simulate a 5-round Swiss: five Rapid 5+3 games against opponents within ±150 of you. Add 3 Blitz 3+2 games to practise knockout tie-breaks.
- **Oracle strand:** watch one featured game live. Write a paper forecast before the game and a second one at move 20, and compare both with the live model (Book VI, VI.6).
- **Demonstrate:** no time forfeits; compute your event performance rating and check it against the Squire norm.
- **Review questions:** (1) What is the Swiss tiebreak order? (2) How is a level knockout match resolved, and what changes in the final? (3) At an evaluation of +150 cp, what does the live model give?
- **Progress check:** the Lab's *Time management* dimension; your performance rating for the event.

Week 11 — Engine Analysis and Preparation

- **Objective:** turn every loss into training, and prepare for specific opponents.
- **Read:** Book V, Chapter V.6 (*The Engine's Mirror*) and Chapter V.8 (*The Unbound Challenge*). Book VI, Chapter VI.7 (*The Crowd and the Mirror*).
- **Academy: Replay Your Mistakes**: every puzzle it offers. Then 15 Adaptive Training puzzles.
- **Practice games:** 4 games against L7 (Advanced: L8), Classical or Rapid. Analyse each one with the full loss routine from Chapter V.6, and use *Retry from This Position (Unrated)* at the first turning point.
- **Oracle strand:** review all your scored forecasts. Which tier does the game show, and how many forecasts is it based on?
- **Demonstrate:** for two losses, a written explanation in words of the better move at the first turning point.
- **Review questions:** (1) Convert +100 cp and −200 cp into expected scores. (2) Why are Brilliant labels and Played Like not to be trusted yet? (3) What does a 3.8 % chance of being behind after 400 good trades tell you about short-term results?
- **Progress check:** count recurring patterns (5 occurrences in 3 games) and check that each one has assigned training.

Week 12 — Review and the Sovereign Challenge

- **Objective:** measure your progress honestly, and challenge Sovereign Unbound with realistic goals.
- **Read:** the *Summary* section of every chapter in Books V and VI. Book V, Chapter V.8 again.
- **Academy: Assessment 4**. Then one Rush and 10 Adaptive Training puzzles.
- **Practice games:** 2 casual games against **L9 at the Fast (1 s) budget** to learn it, then **2 rated games against L9 (always 10 s per move)** at Classical 15+10. Beginners: replace L9 with the strongest level against which you scored at least 25 % this program, and play L9 only casually.
- **Oracle strand:** write a final forecasting review: number of forecasts, mean Brier against the public forecast, and the calibration bins with their intervals.
- **Demonstrate:** survive the opening and the first critical moment against L9 in at least one game. (A draw is a triumph: the calibration model gives a 2000-rated player about 0.25 % expected score per game against it.)
- **Review questions:** (1) What do the budgets 1/5/10/15/30/60 s mean, and how does the clock limit them?
  (2) How was L9's 3038 rating obtained, and what does its RD of 171 mean? (3) Compare Assessments 1–4. What changed, and how sure can you be?
- **Progress check (final):** see below.

Final progress assessment

Fill in this table at the end of week 12. Every row has a source in the game, so no number has to be guessed.

Measure	Week 1	Week 12	Source
Held-out assessment score (of 10)			Academy (Assessments 1 and 4)
Puzzle Rush best			Academy statistics
First-try clean rate (with n)			Academy statistics
Rapid rating (state whether provisional)			Passport; provisional while games < 20 or RD > 100

Measure	Week 1	Week 12	Source
Lab: Tactical vision, Calculation, Endgame technique (rate and 95 % interval)			Lab
Recurring coach patterns			Lab / coach notes
Paper forecasts: n, mean Brier, public-forecast Brier			Forecasting screen
Best result against L7 / L8 / L9			Game history

How to read it ([Counsel]): look for changes larger than the intervals. Ten-puzzle assessments and fewer than 50 forecasts are noisy. If a measure did not move, choose that topic’s week and repeat it, before you move on to the next cycle. The program is meant to be run more than once, at a higher level each time.

Four workloads, the same discipline

Beginner: twenty minutes, accurate legal moves and a safety scan. Intermediate: forty minutes, short lines with the opponent’s reply. Advanced: sixty minutes, candidate comparison and accurate final boards. Elite study: seventy-five minutes with breaks, blind calculation and complete defence trees. These are workload suggestions, not rating predictions.

Each day: scan, solve without answers, study the week’s concept, review one of your own decisions, write one correction. Each week: one unseen assessment at a fixed budget and one review day. Record confidence before checking, move, defence, final board and error type. Separate a guess from a calculated solution. Retesting a familiar task measures retention; unseen tasks measure transfer.

PART IX · Original Puzzle Workbook

One hundred new composed mate exercises. Difficulty is editorial, not Elo. Mate in N means force checkmate within N attacking moves, against every legal defence; longer groups have no shorter forced mate. Write a move and the strongest reply before opening the separate answer key.

Puzzle 001: Foundation

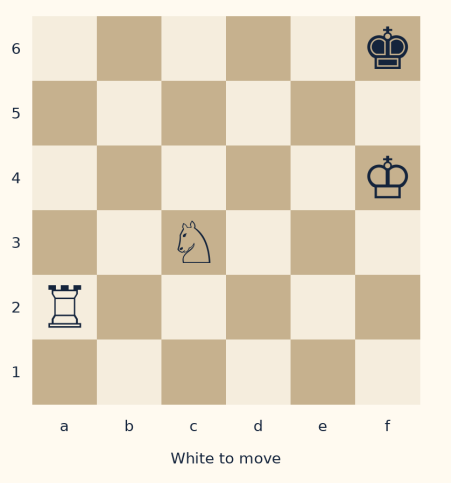


Figure 112: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `5k/6/5K/2N3/R5/6 w - - 0 1`

Puzzle 002: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/4N1/2R3/4K1/6/5k w - - 0 1`

Puzzle 003: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/6/1Q4/6/3p2/K1k3 w - - 0 1`

Puzzle 004: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `k1KQ2/6/6/4p1/6/6 w - - 0 1`

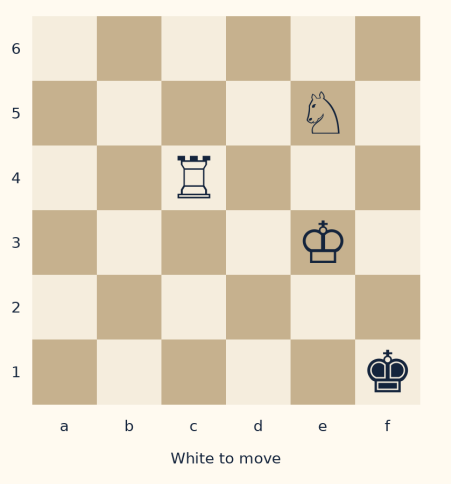


Figure 113: White to move.

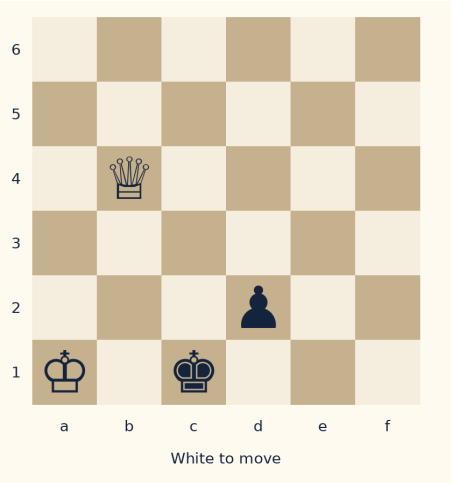


Figure 114: White to move.

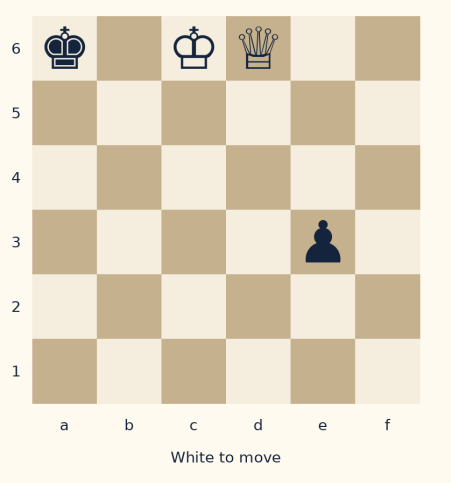


Figure 115: White to move.

Puzzle 005: Foundation

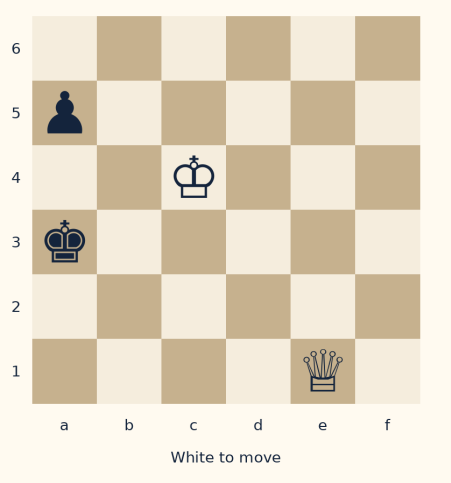


Figure 116: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/p5/2K3/k5/6/4Q1 w - - 0 1`

Puzzle 006: Foundation

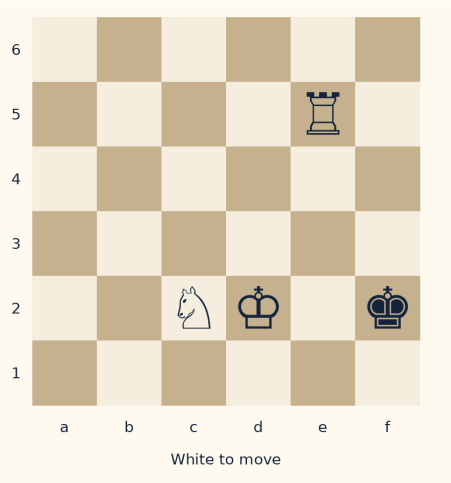


Figure 117: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/4R1/6/6/2NK1k/6 w - - 0 1`

Puzzle 007: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `k5/6/K1N3/6/6/3R2 w - - 0 1`

Puzzle 008: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `5k/6/4K1/6/5p/1Q4 w - - 0 1`

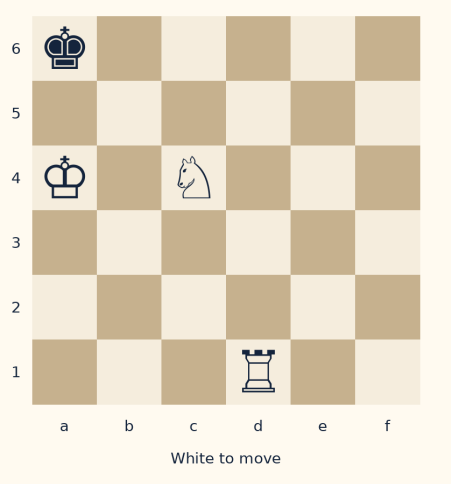


Figure 118: White to move.

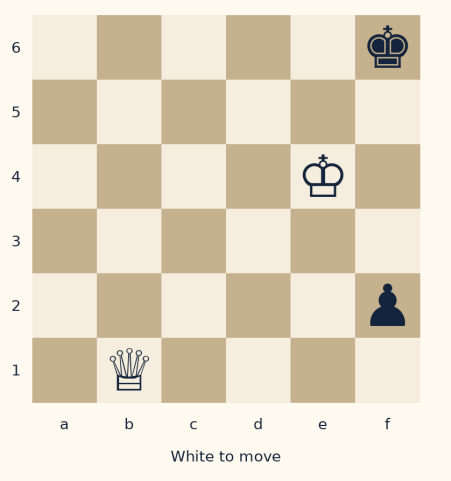


Figure 119: White to move.

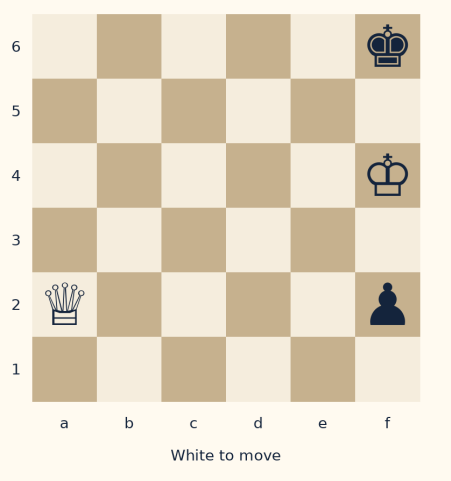


Figure 120: White to move.

Puzzle 009: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `5k/6/5K/6/Q4p/6 w - - 0 1`

Puzzle 010: Foundation

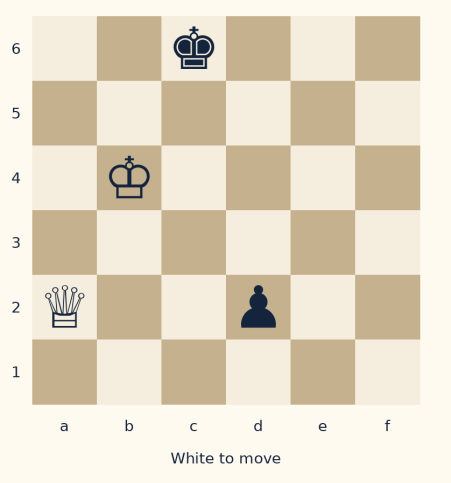


Figure 121: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `2k3/6/1K4/6/Q2p2/6 w - - 0 1`

Puzzle 011: Foundation

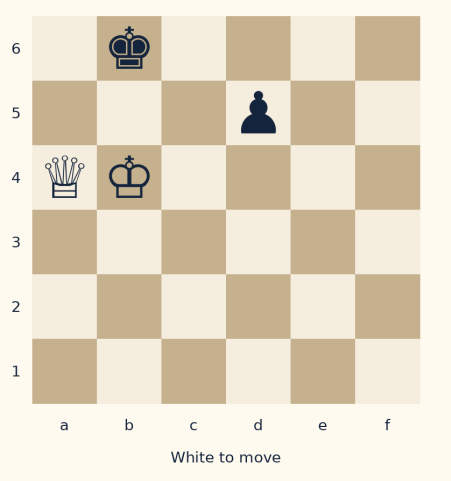


Figure 122: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `1k4/3p2/QK4/6/6/6 w - - 0 1`

Puzzle 012: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/1N4/6/1R4/6/k1K3 w - - 0 1`

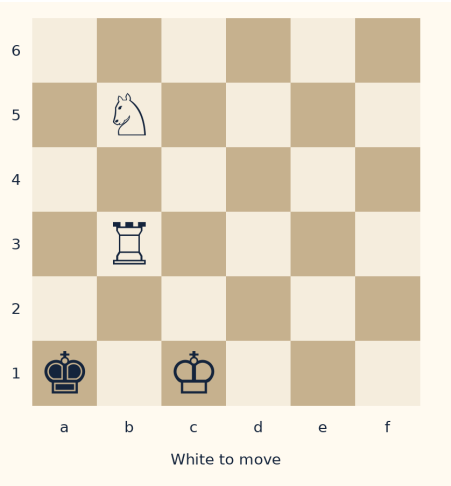


Figure 123: White to move.

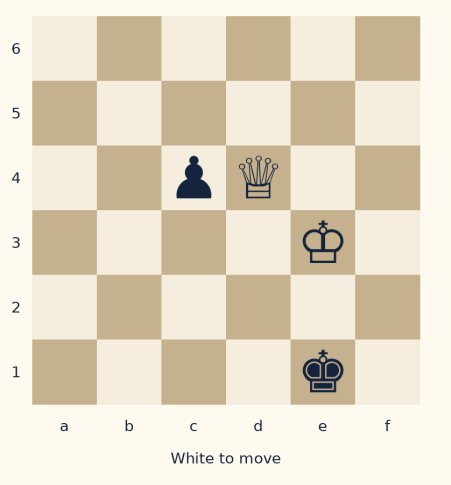


Figure 124: White to move.

Puzzle 013: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/6/2pQ2/4K1/6/4k1 w - - 0 1`

Puzzle 014: Foundation

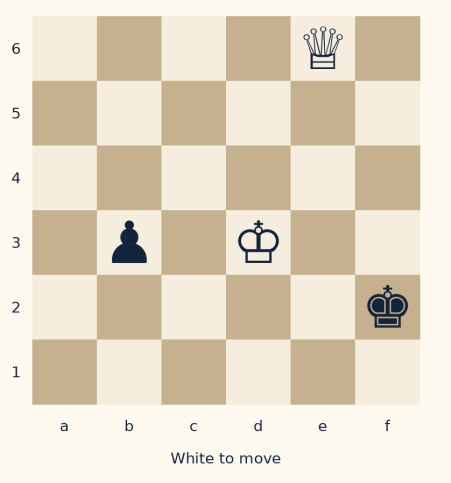


Figure 125: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `4Q1/6/6/1p1K2/5k/6 w - - 0 1`

Puzzle 015: Foundation

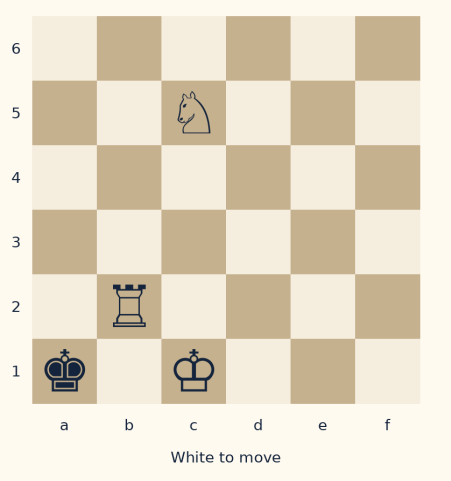


Figure 126: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/2N3/6/6/1R4/k1K3 w - - 0 1`

Puzzle 016: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `2K3/k5/p1Q3/6/6/6 w - - 0 1`

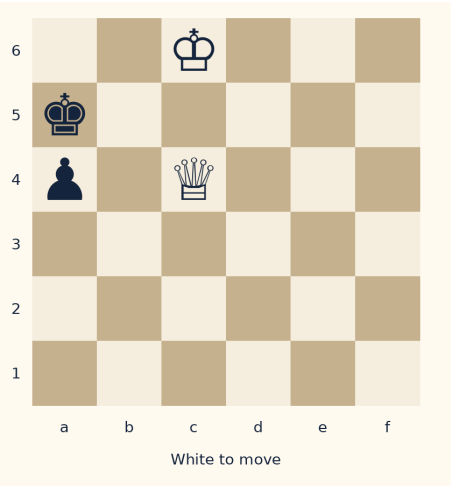


Figure 127: White to move.

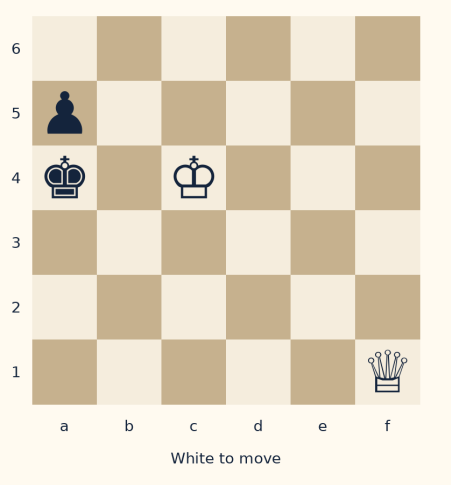


Figure 128: White to move.

Puzzle 017: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/p5/k1K3/6/6/5Q w - - 0 1`

Puzzle 018: Foundation

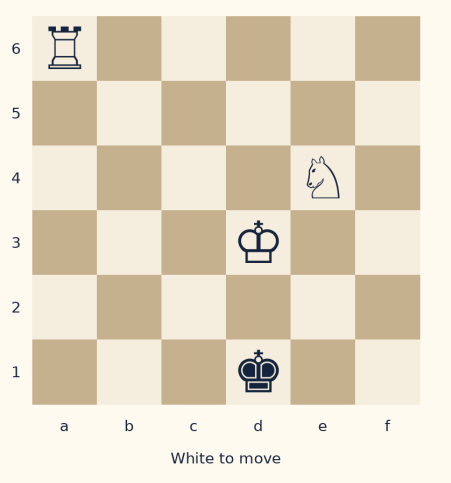


Figure 129: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `R5/6/4N1/3K2/6/3k2 w - - 0 1`

Puzzle 019: Foundation

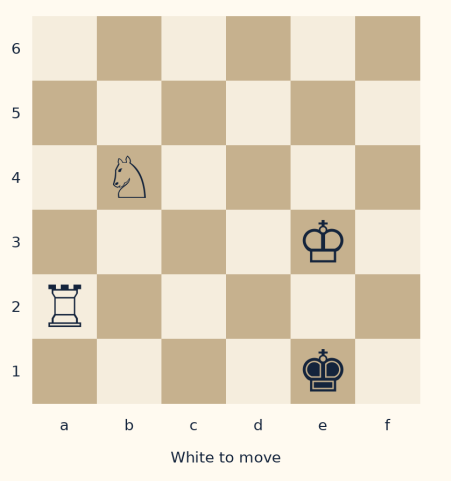


Figure 130: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/6/1N4/4K1/R5/4k1 w - - 0 1`

Puzzle 020: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/Q5/p5/6/3K1k/6 w - - 0 1`

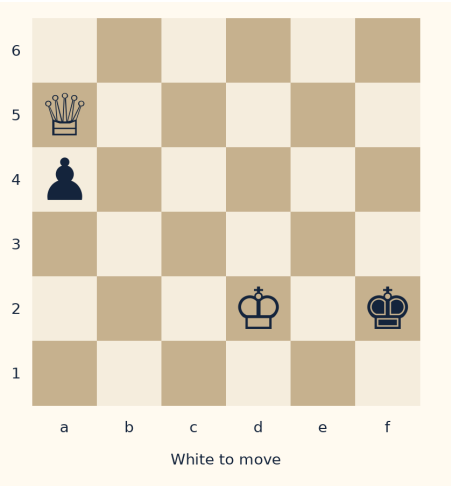


Figure 131: White to move.

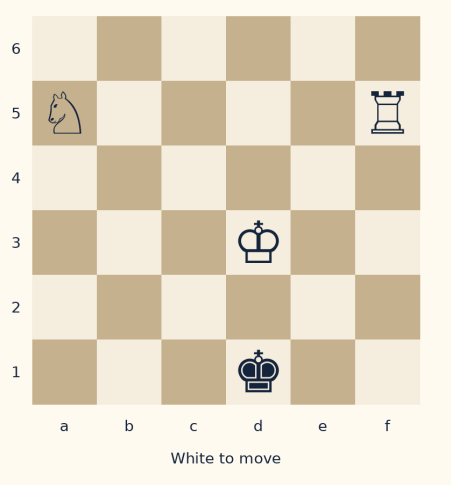


Figure 132: White to move.

Puzzle 021: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/N4R/6/3K2/6/3k2 w - - 0 1`

Puzzle 022: Foundation

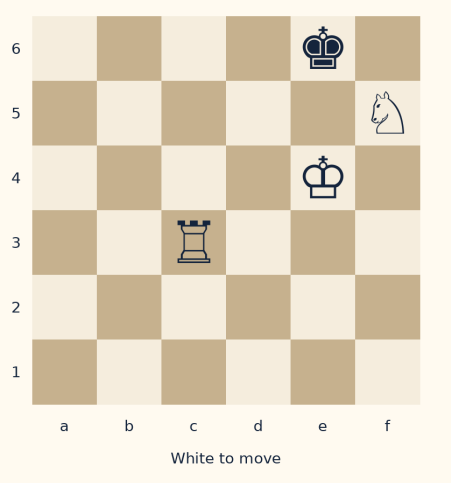


Figure 133: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `4k1/5N/4K1/2R3/6/6 w - - 0 1`

Puzzle 023: Foundation

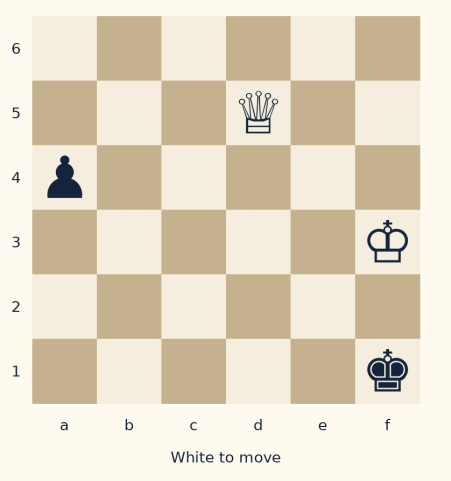


Figure 134: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/3Q2/p5/5K/6/5k w - - 0 1`

Puzzle 024: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `k5/6/1KR2N/6/6/6 w - - 0 1`

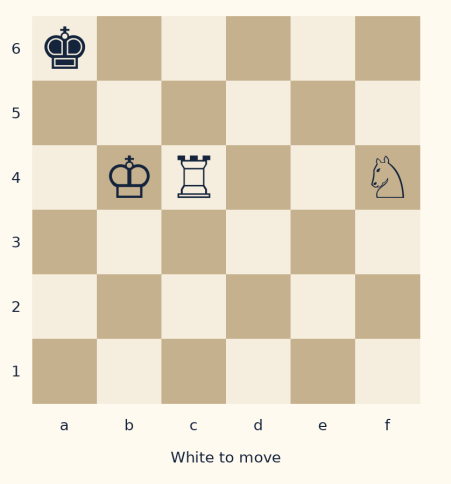


Figure 135: White to move.

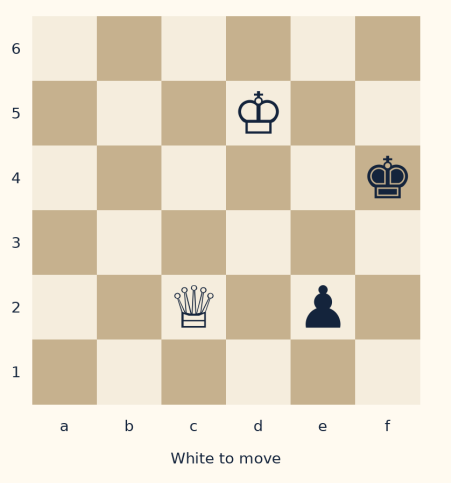


Figure 136: White to move.

Puzzle 025: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/3K2/5k/6/2Q1p1/6 w - - 0 1`

Puzzle 026: Foundation

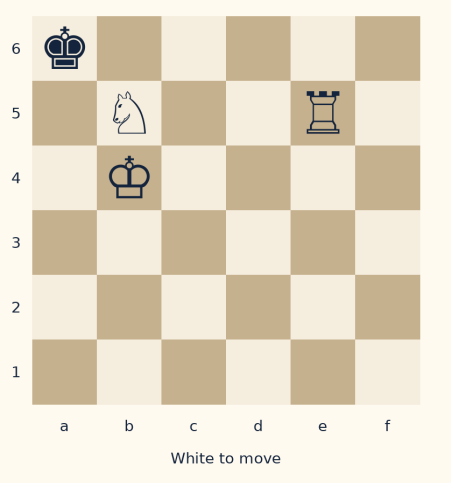


Figure 137: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `k5/1N2R1/1K4/6/6/6 w - - 0 1`

Puzzle 027: Foundation

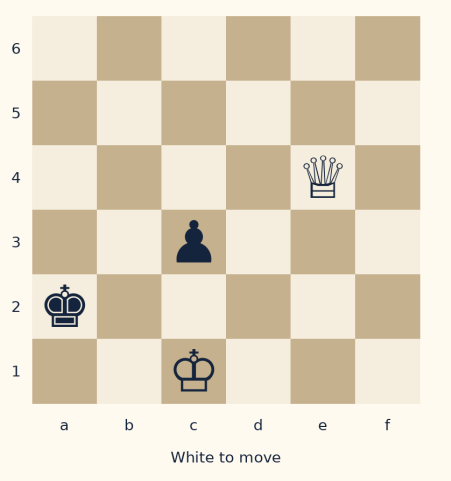


Figure 138: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `6/6/4Q1/2p3/k5/2K3 w - - 0 1`

Puzzle 028: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `6/k5/2K3/6/3p2/1Q4 w - - 0 1`

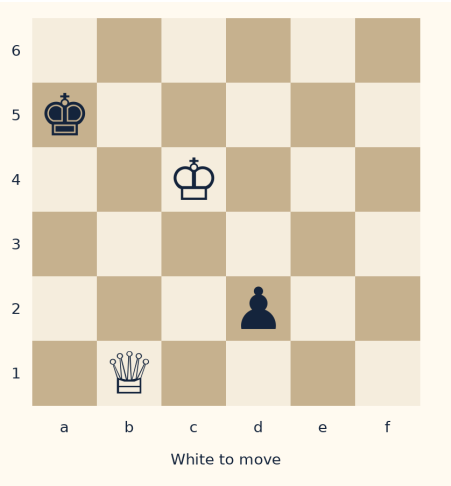


Figure 139: White to move.

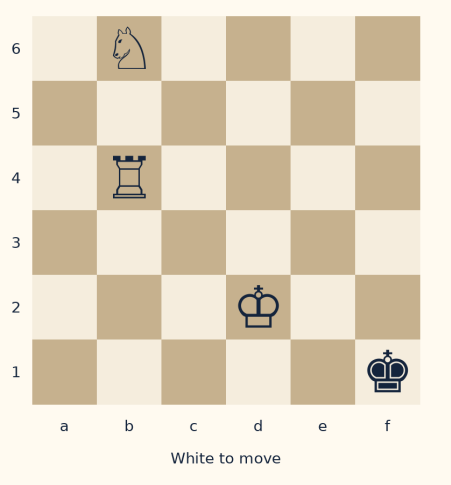


Figure 140: White to move.

Puzzle 029: Foundation

Objective: mate in 1. Record your full line. Answer.

Position: `1N4/6/1R4/6/3K2/5k w - - 0 1`

Puzzle 030: Foundation

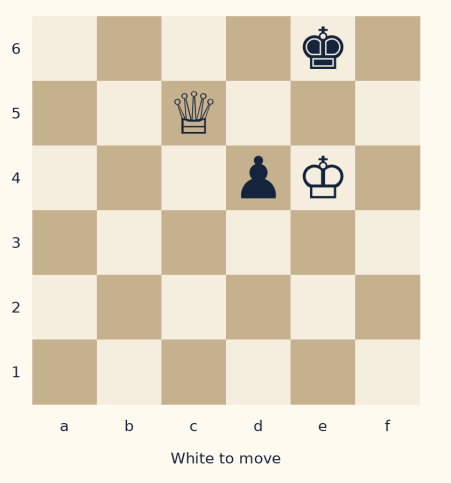


Figure 141: White to move.

Objective: mate in 1. Record your full line. Answer.

Position: `4k1/2Q3/3pK1/6/6/6 w - - 0 1`

Puzzle 031: Calculation

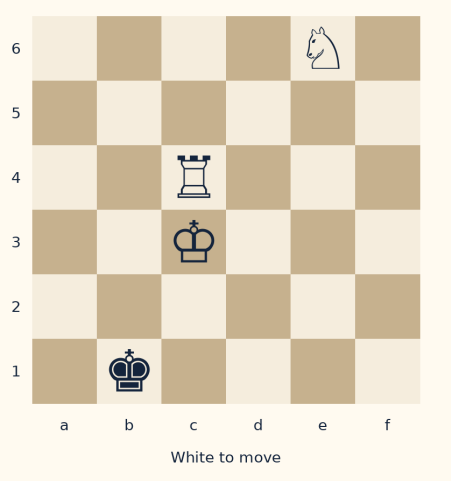


Figure 142: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `4N1/6/2R3/2K3/6/1k4 w - - 0 1`

Puzzle 032: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `kN4/6/2K3/6/6/1R4 w - - 0 1`

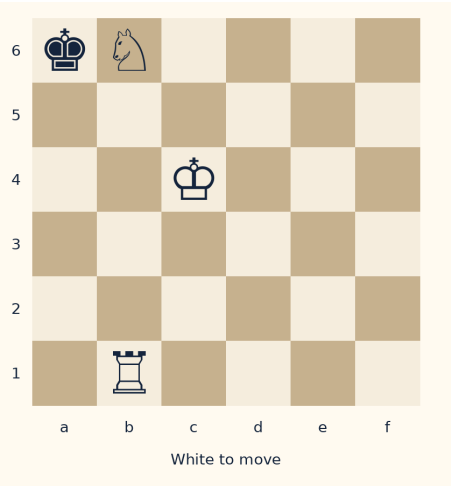


Figure 143: White to move.

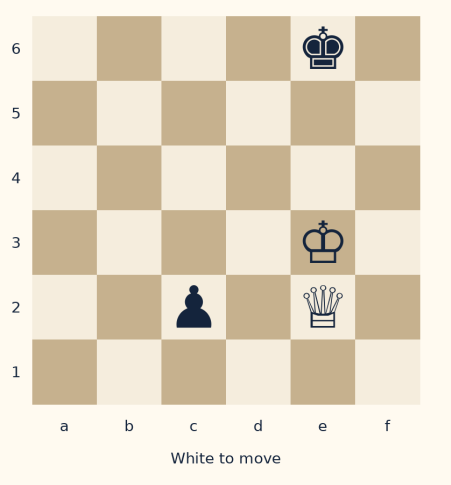


Figure 144: White to move.

Puzzle 033: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `4k1/6/6/4K1/2p1Q1/6 w - - 0 1`

Puzzle 034: Calculation

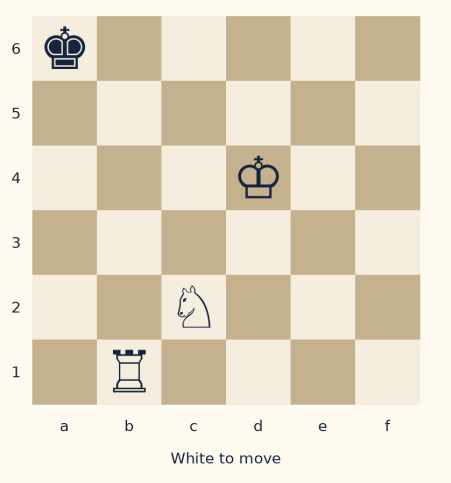


Figure 145: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `k5/6/3K2/6/2N3/1R4 w - - 0 1`

Puzzle 035: Calculation

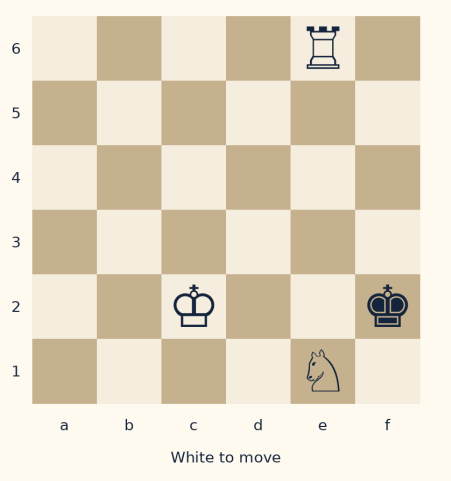


Figure 146: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `4R1/6/6/6/2K2k/4N1 w - - 0 1`

Puzzle 036: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `1k4/6/6/1K4/6/R1N3 w - - 0 1`

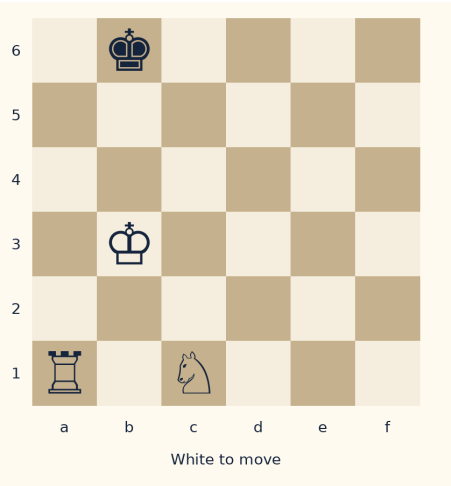


Figure 147: White to move.

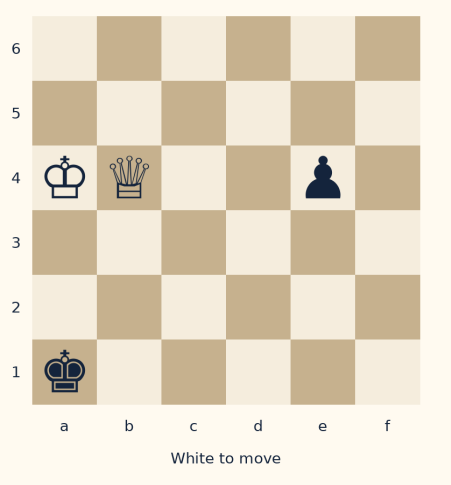


Figure 148: White to move.

Puzzle 037: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/KQ2p1/6/6/k5 w - - 0 1`

Puzzle 038: Calculation

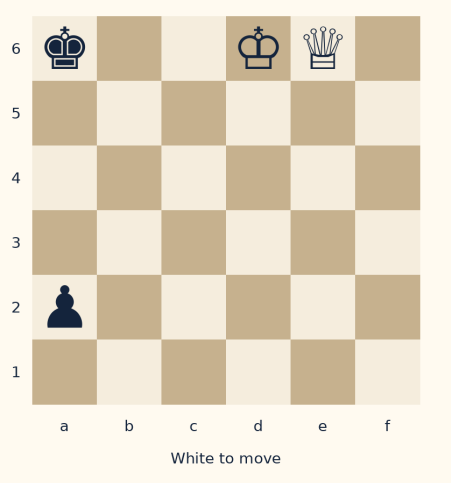


Figure 149: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `k2KQ1/6/6/6/p5/6 w - - 0 1`

Puzzle 039: Calculation

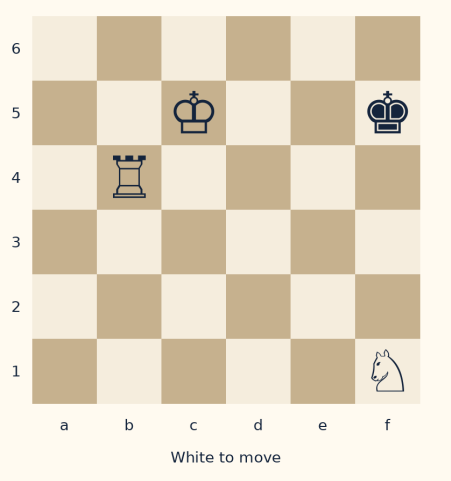


Figure 150: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/2K2k/1R4/6/6/5N w - - 0 1`

Puzzle 040: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `5k/6/6/2K2p/6/4Q1 w - - 0 1`

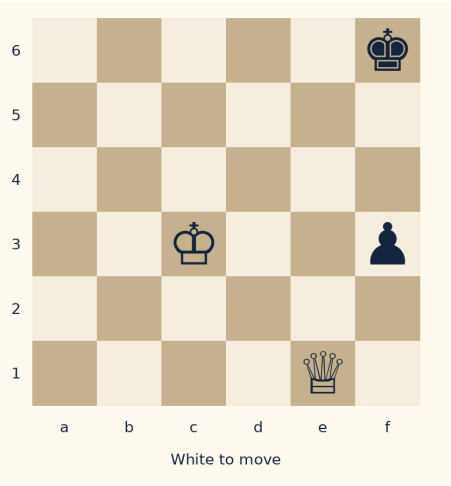


Figure 151: White to move.

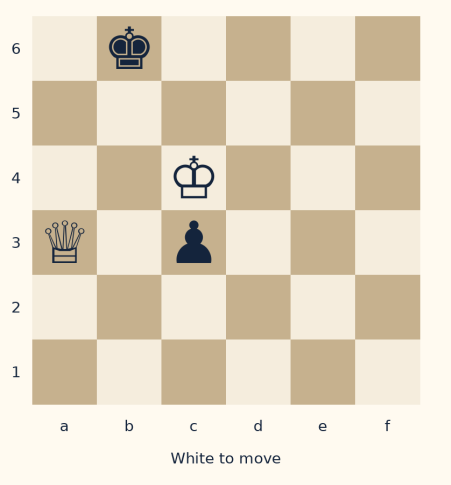


Figure 152: White to move.

Puzzle 041: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `1k4/6/2K3/Q1p3/6/6 w - - 0 1`

Puzzle 042: Calculation

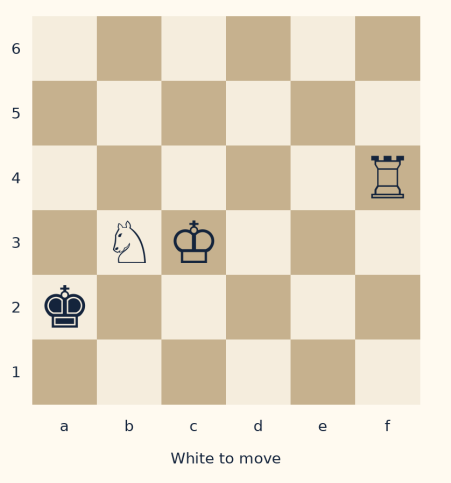


Figure 153: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/5R/1NK3/k5/6 w - - 0 1`

Puzzle 043: Calculation

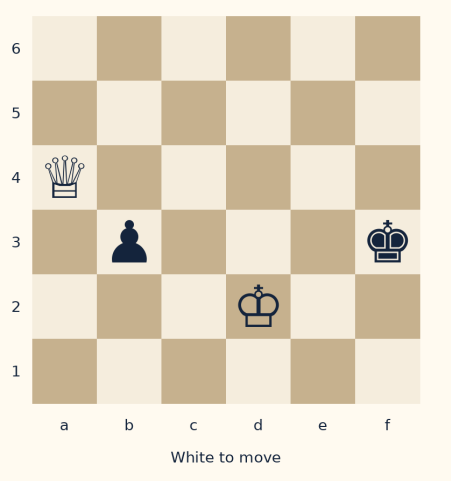


Figure 154: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/Q5/1p3k/3K2/6 w - - 0 1`

Puzzle 044: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/RN4/1K4/6/1k4 w - - 0 1`

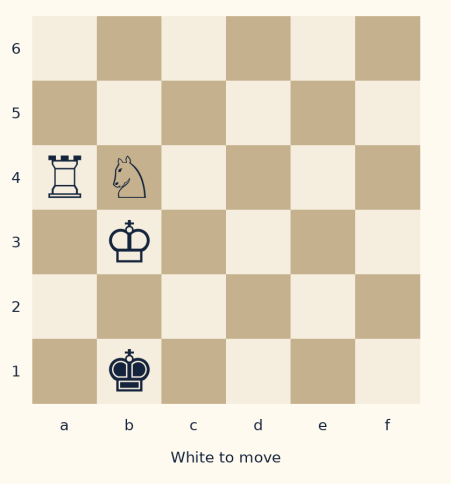


Figure 155: White to move.

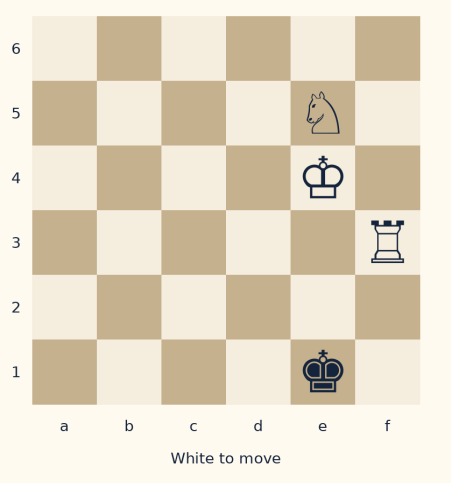


Figure 156: White to move.

Puzzle 045: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/4N1/4K1/5R/6/4k1 w - - 0 1`

Puzzle 046: Calculation

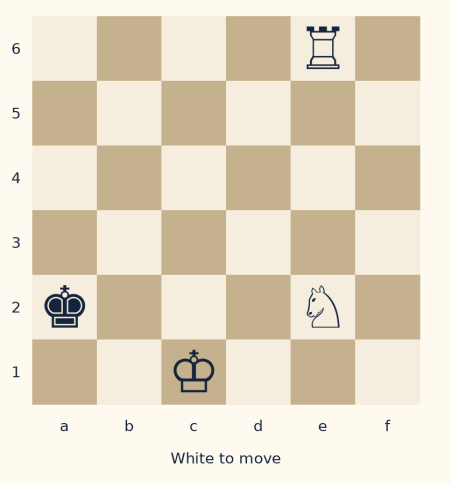


Figure 157: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `4R1/6/6/6/k3N1/2K3 w - - 0 1`

Puzzle 047: Calculation

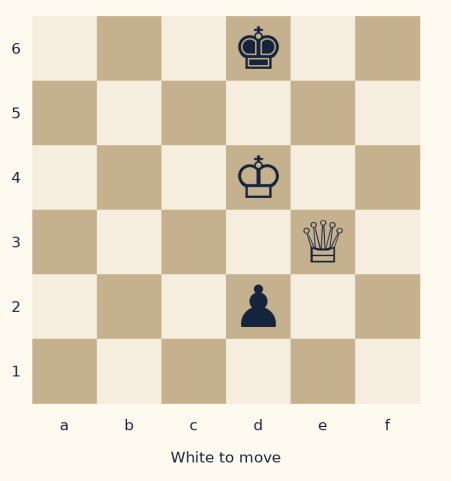


Figure 158: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `3k2/6/3K2/4Q1/3p2/6 w - - 0 1`

Puzzle 048: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `2k3/1p4/1QK3/6/6/6 w - - 0 1`

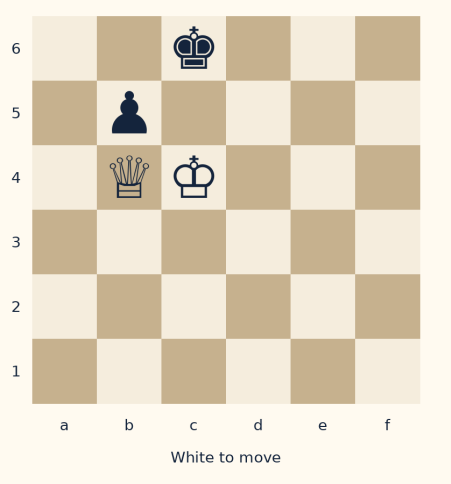


Figure 159: White to move.

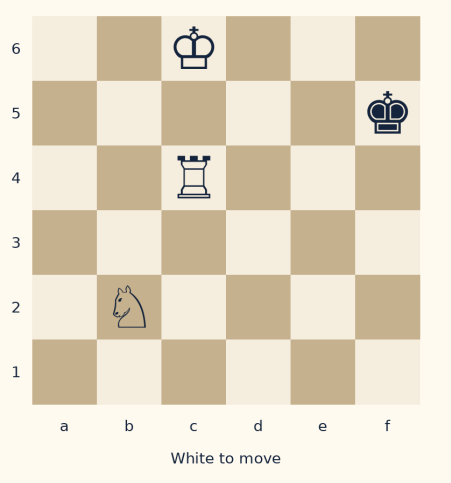


Figure 160: White to move.

Puzzle 049: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `2K3/5k/2R3/6/1N4/6 w - - 0 1`

Puzzle 050: Calculation

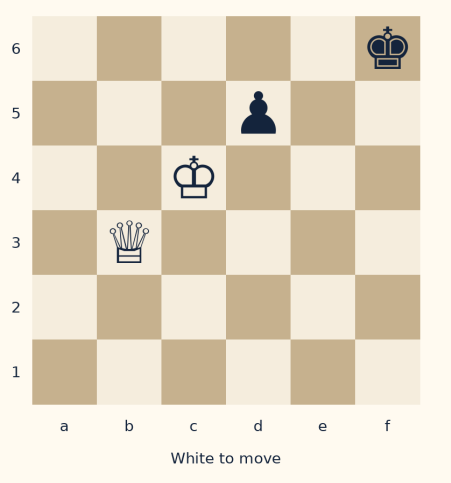


Figure 161: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `5k/3p2/2K3/1Q4/6/6 w - - 0 1`

Puzzle 051: Calculation

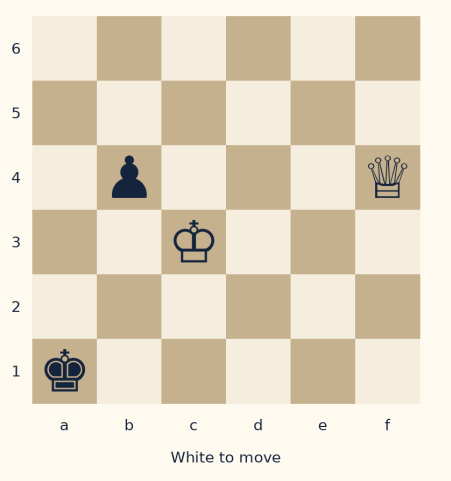


Figure 162: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/1p3Q/2K3/6/k5 w - - 0 1`

Puzzle 052: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/4Q1/6/2K3/4p1/k5 w - - 0 1`

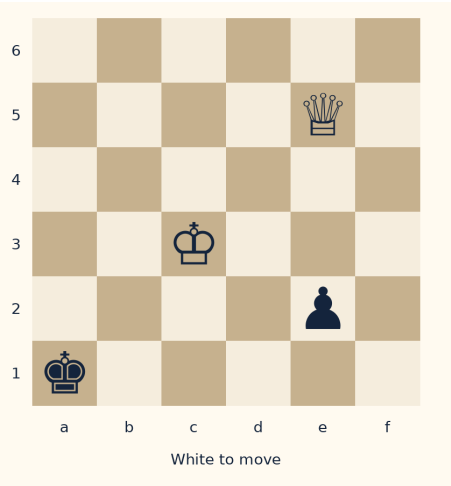


Figure 163: White to move.

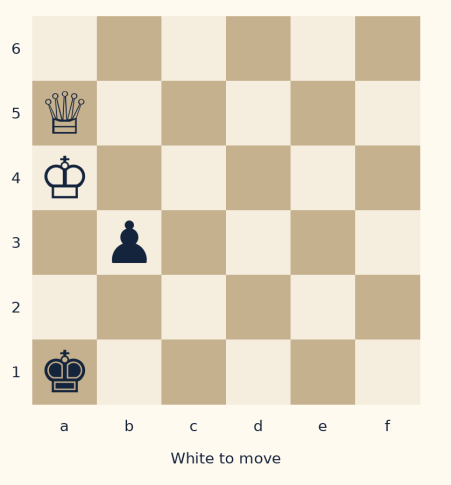


Figure 164: White to move.

Puzzle 053: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/Q5/K5/1p4/6/k5 w - - 0 1`

Puzzle 054: Calculation

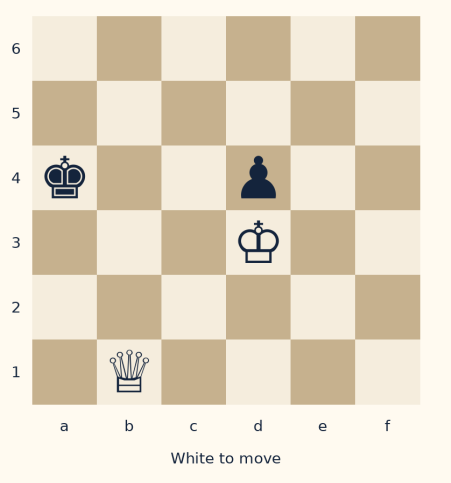


Figure 165: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/k2p2/3K2/6/1Q4 w - - 0 1`

Puzzle 055: Calculation

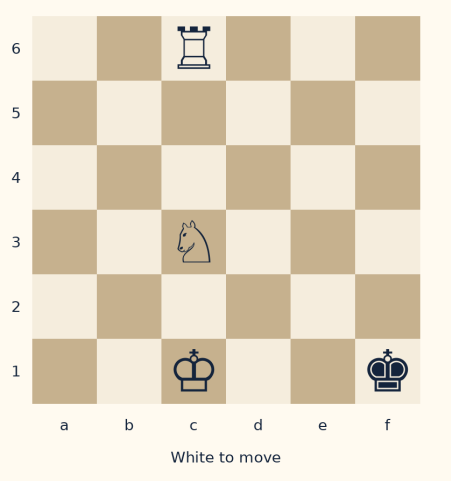


Figure 166: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `2R3/6/6/2N3/6/2K2k w - - 0 1`

Puzzle 056: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/5k/2K3/6/2p1Q1/6 w - - 0 1`

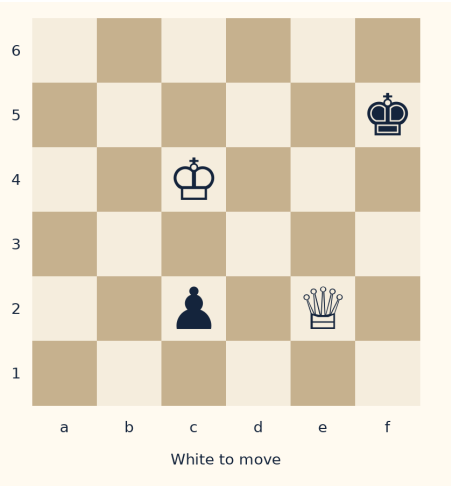


Figure 167: White to move.

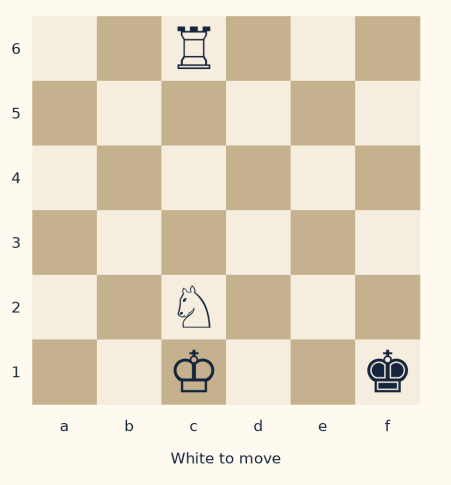


Figure 168: White to move.

Puzzle 057: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `2R3/6/6/6/2N3/2K2k w - - 0 1`

Puzzle 058: Calculation

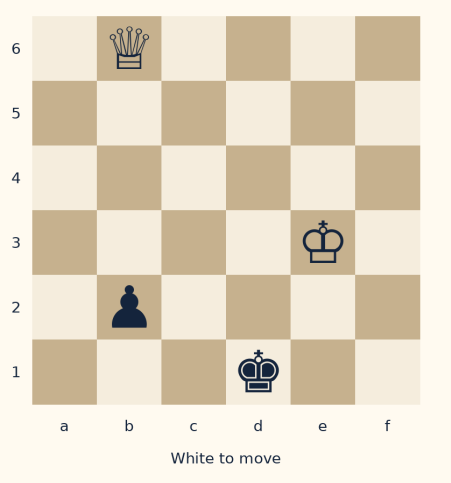


Figure 169: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `1Q4/6/6/4K1/1p4/3k2 w - - 0 1`

Puzzle 059: Calculation

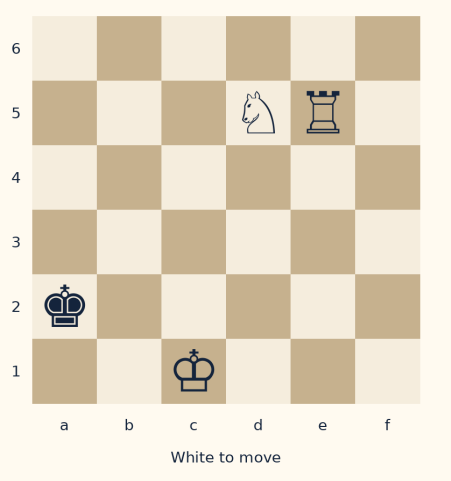


Figure 170: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/3NR1/6/6/k5/2K3 w - - 0 1`

Puzzle 060: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `1N3k/6/5K/1R4/6/6 w - - 0 1`

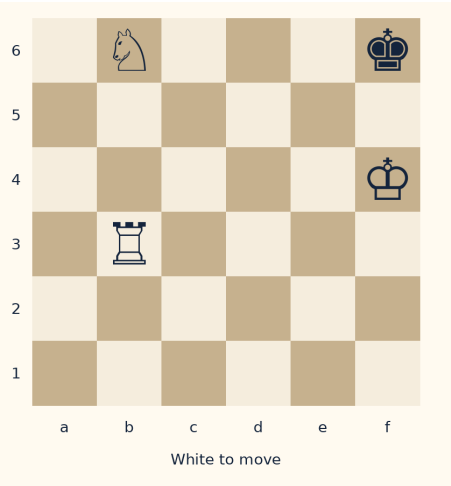


Figure 171: White to move.

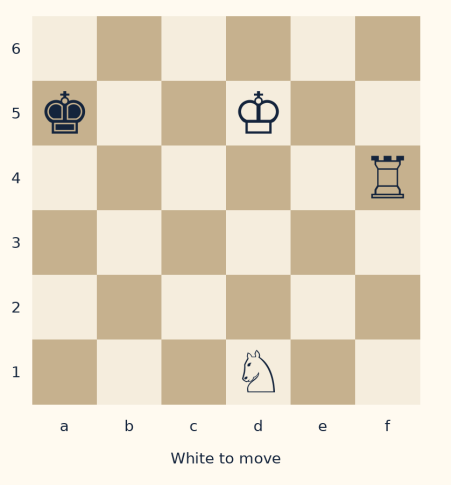


Figure 172: White to move.

Puzzle 061: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/k2K2/5R/6/6/3N2 w - - 0 1`

Puzzle 062: Calculation

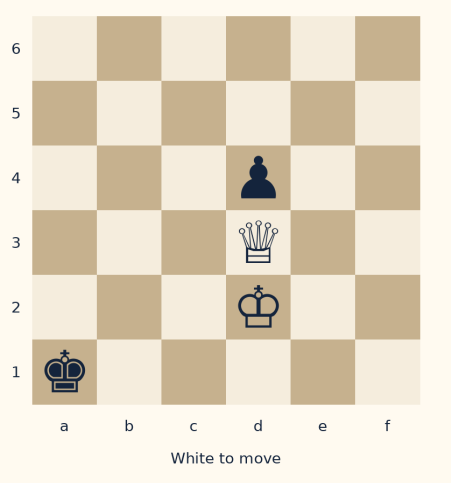


Figure 173: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/3p2/3Q2/3K2/k5 w - - 0 1`

Puzzle 063: Calculation

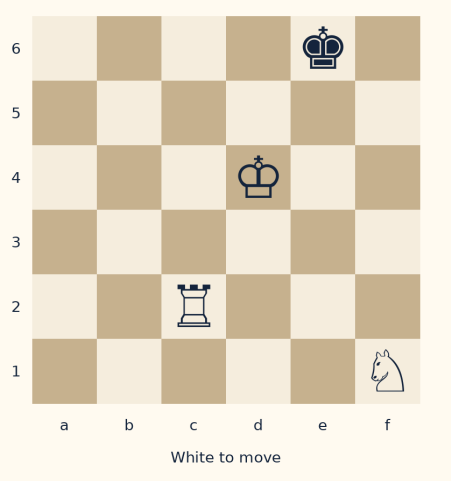


Figure 174: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `4k1/6/3K2/6/2R3/5N w - - 0 1`

Puzzle 064: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/2K3/5k/p2Q2/6/6 w - - 0 1`

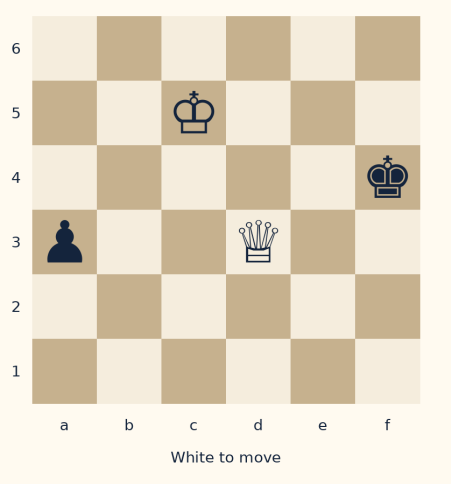


Figure 175: White to move.

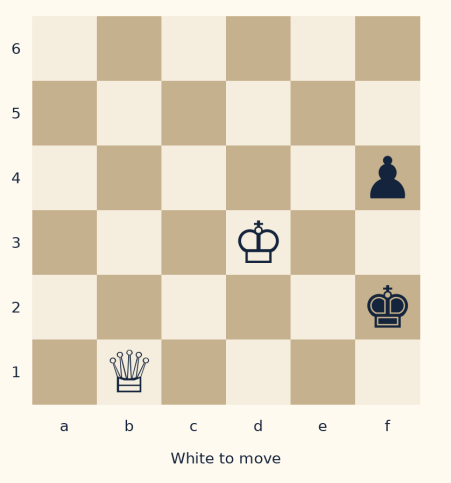


Figure 176: White to move.

Puzzle 065: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/5p/3K2/5k/1Q4 w - - 0 1`

Puzzle 066: Calculation

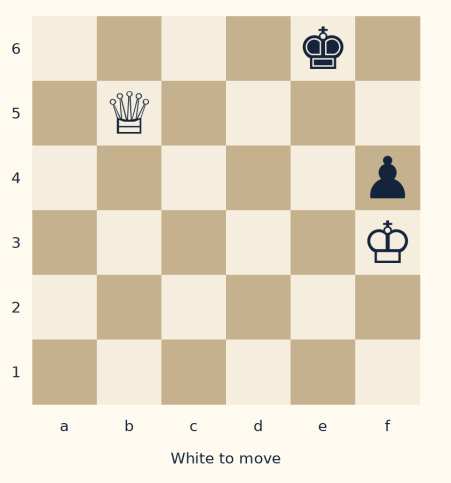


Figure 177: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `4k1/1Q4/5p/5K/6/6 w - - 0 1`

Puzzle 067: Calculation

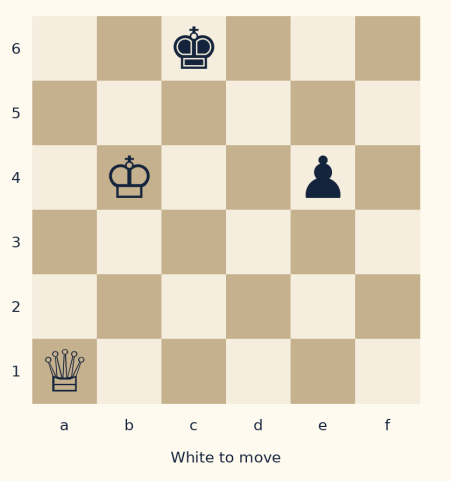


Figure 178: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `2k3/6/1K2p1/6/6/Q5 w - - 0 1`

Puzzle 068: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/3p1k/2K3/6/3Q2/6 w - - 0 1`

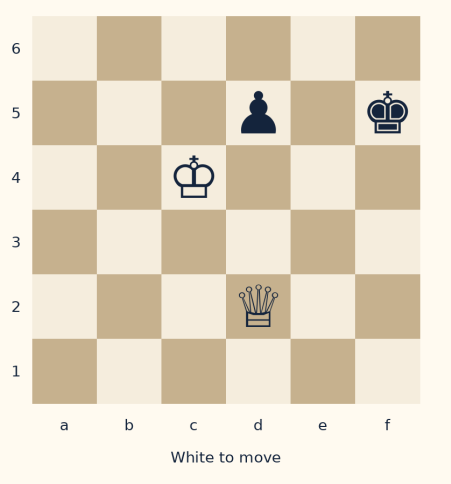


Figure 179: White to move.

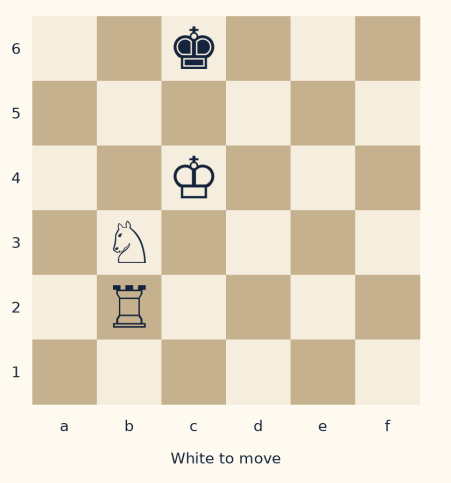


Figure 180: White to move.

Puzzle 069: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `2k3/6/2K3/1N4/1R4/6 w - - 0 1`

Puzzle 070: Calculation

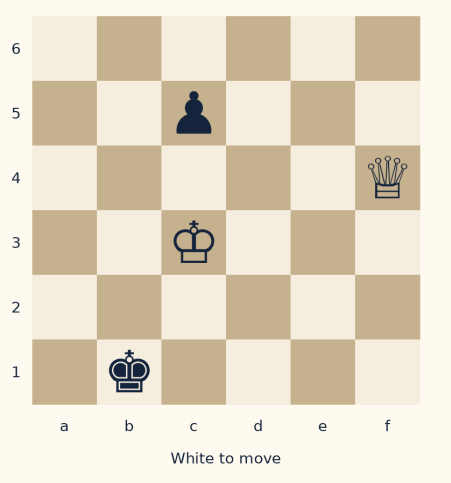


Figure 181: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/2p3/5Q/2K3/6/1k4 w - - 0 1`

Puzzle 071: Calculation

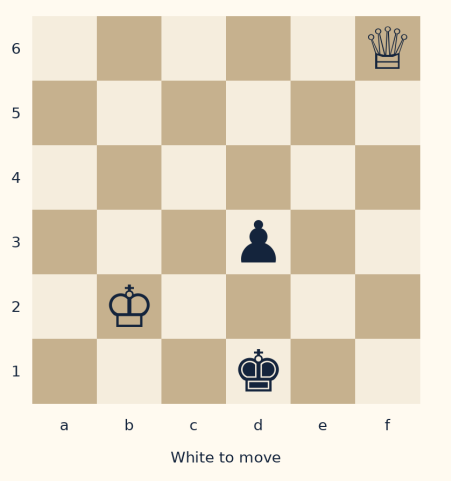


Figure 182: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `5Q/6/6/3p2/1K4/3k2 w - - 0 1`

Puzzle 072: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `k5/6/2K3/1R3N/6/6 w - - 0 1`

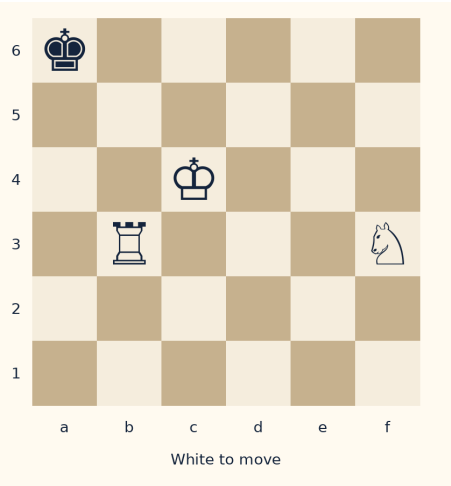


Figure 183: White to move.

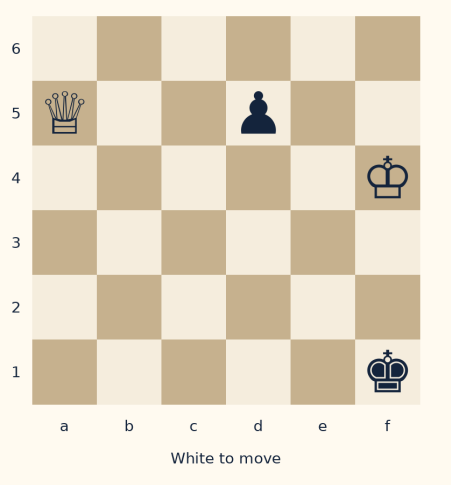


Figure 184: White to move.

Puzzle 073: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/Q2p2/5K/6/6/5k w - - 0 1`

Puzzle 074: Calculation

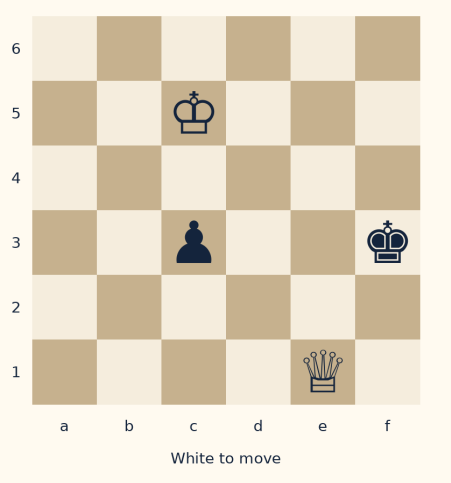


Figure 185: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/2K3/6/2p2k/6/4Q1 w - - 0 1`

Puzzle 075: Calculation

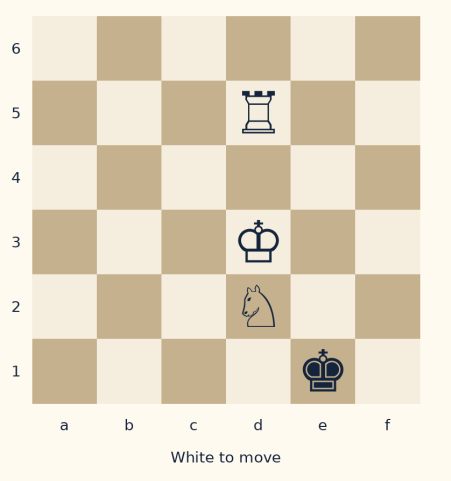


Figure 186: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/3R2/6/3K2/3N2/4k1 w - - 0 1`

Puzzle 076: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `k3N1/6/2K3/6/6/1R4 w - - 0 1`

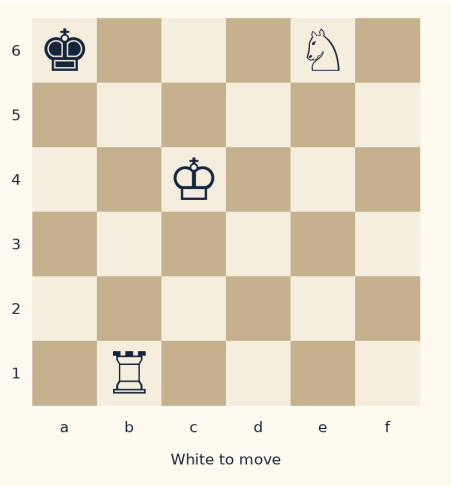


Figure 187: White to move.

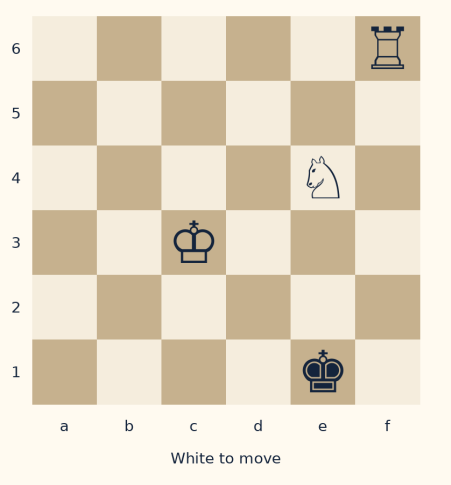


Figure 188: White to move.

Puzzle 077: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `5R/6/4N1/2K3/6/4k1 w - - 0 1`

Puzzle 078: Calculation

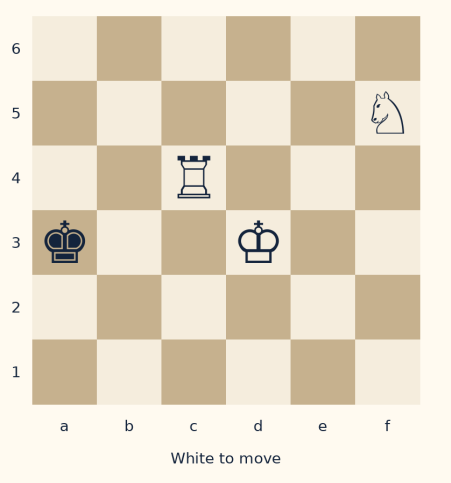


Figure 189: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `6/5N/2R3/k2K2/6/6 w - - 0 1`

Puzzle 079: Calculation

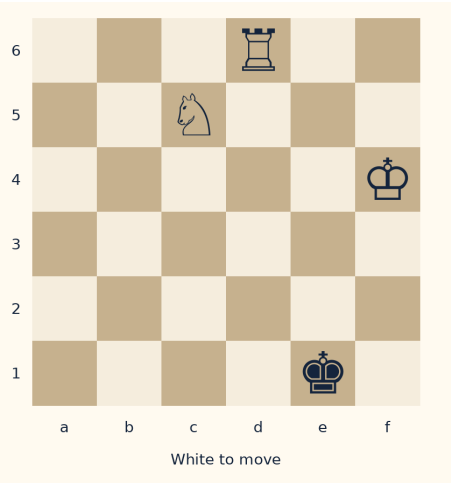


Figure 190: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `3R2/2N3/5K/6/6/4k1 w - - 0 1`

Puzzle 080: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `2K3/6/k2N2/1R4/6/6 w - - 0 1`

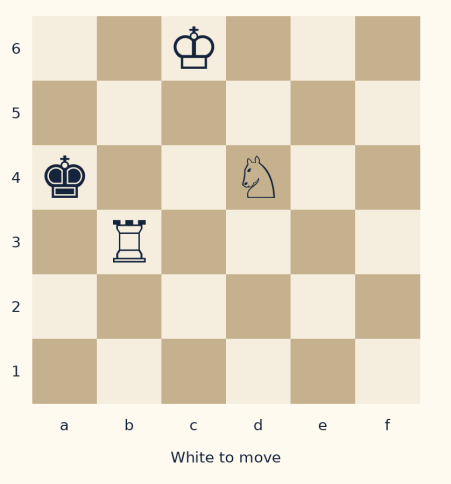


Figure 191: White to move.

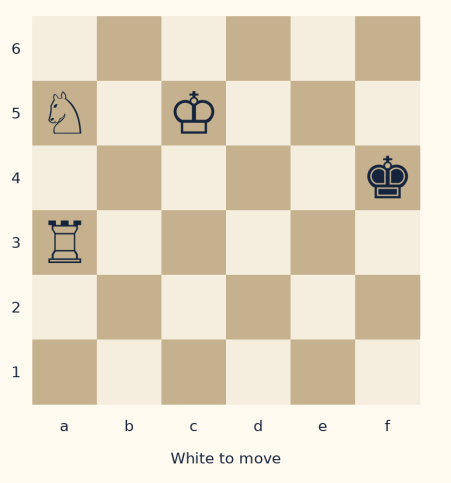


Figure 192: White to move.

Puzzle 081: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/N1K3/5k/R5/6/6 w - - 0 1`

Puzzle 082: Calculation

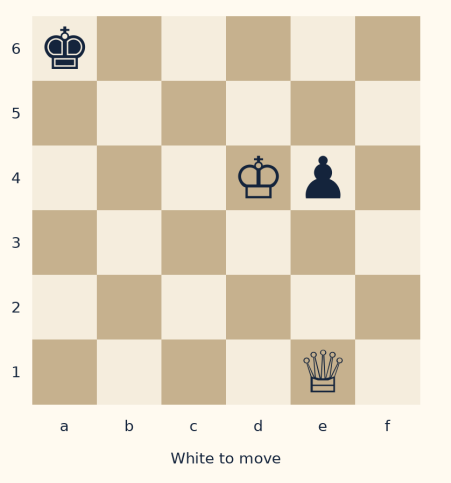


Figure 193: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `k5/6/3Kp1/6/6/4Q1 w - - 0 1`

Puzzle 083: Calculation

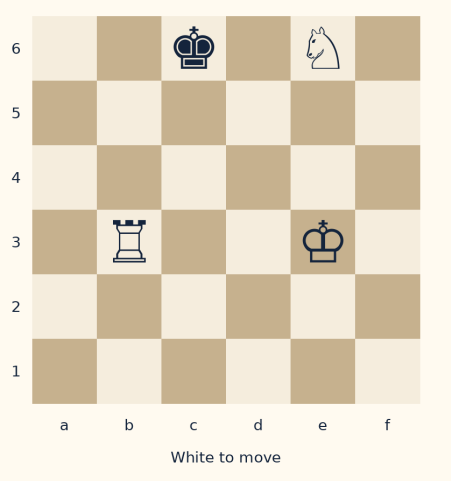


Figure 194: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `2k1N1/6/6/1R2K1/6/6 w - - 0 1`

Puzzle 084: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/6/1N4/6/2K2k/1R4 w - - 0 1`

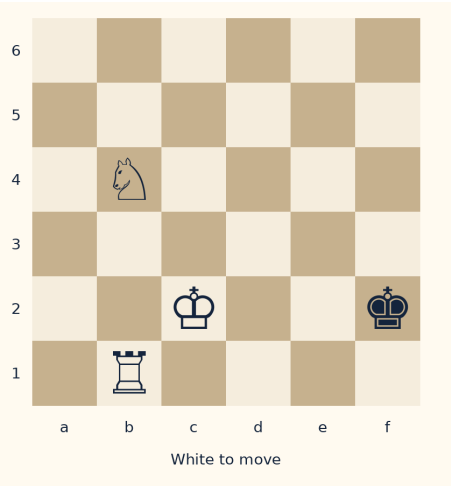


Figure 195: White to move.

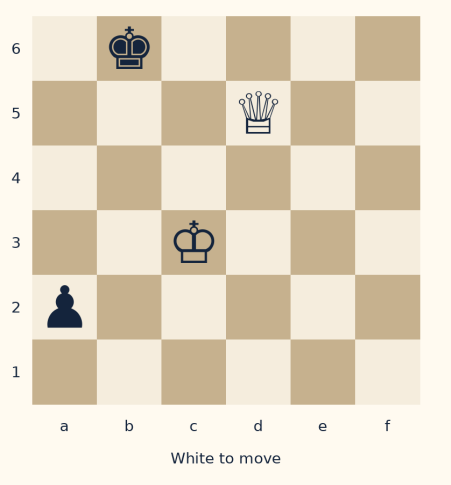


Figure 196: White to move.

Puzzle 085: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `1k4/3Q2/6/2K3/p5/6 w - - 0 1`

Puzzle 086: Calculation

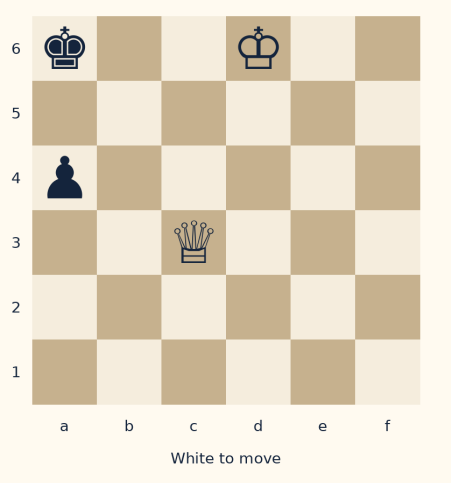


Figure 197: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `k2K2/6/p5/2Q3/6/6 w - - 0 1`

Puzzle 087: Calculation

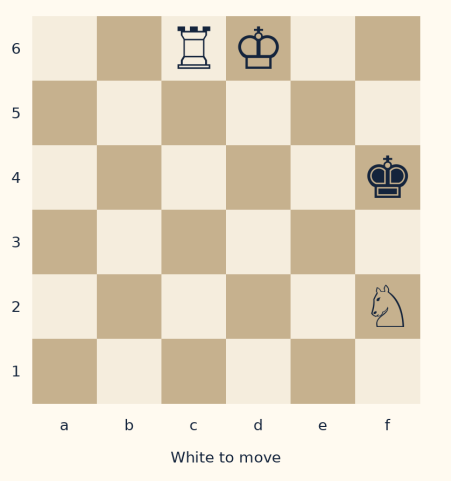


Figure 198: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `2RK2/6/5k/6/5N/6 w - - 0 1`

Puzzle 088: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/2K3/6/1p3k/6/4Q1 w - - 0 1`

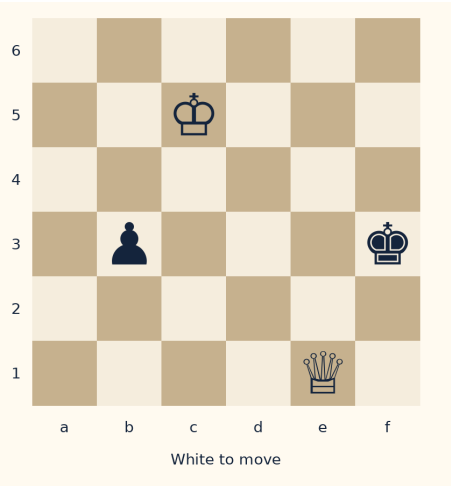


Figure 199: White to move.

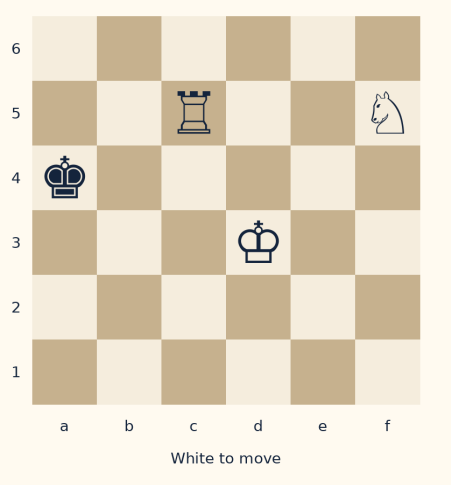


Figure 200: White to move.

Puzzle 089: Calculation

Objective: mate in 2. Record your full line. Answer.

Position: `6/2R2N/k5/3K2/6/6 w - - 0 1`

Puzzle 090: Calculation

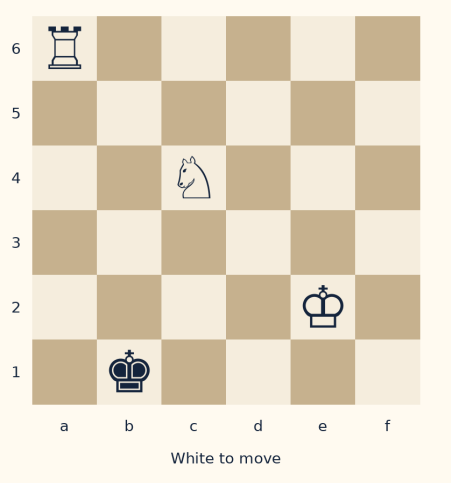


Figure 201: White to move.

Objective: mate in 2. Record your full line. Answer.

Position: `R5/6/2N3/6/4K1/1k4 w - - 0 1`

Puzzle 091: Expert calculation

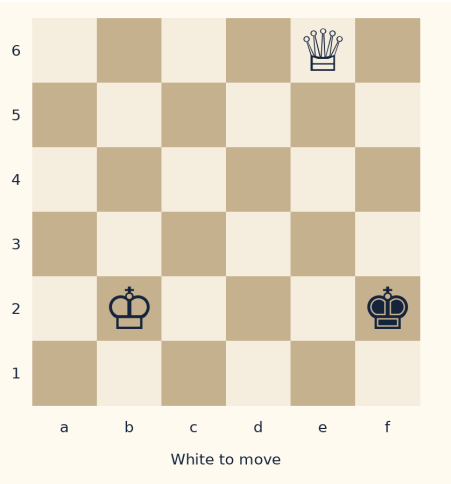


Figure 202: White to move.

Objective: mate in 3. Record your full line. Answer.

Position: `4Q1/6/6/6/1K3k/6 w - - 0 1`

Puzzle 092: Expert calculation

Objective: mate in 3. Record your full line. Answer.

Position: `6/3k2/6/1K2Q1/6/6 w - - 0 1`

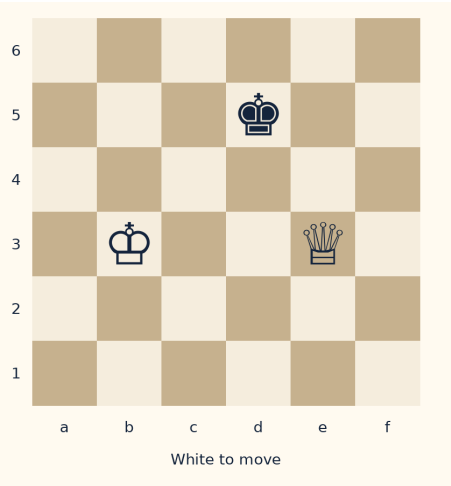


Figure 203: White to move.

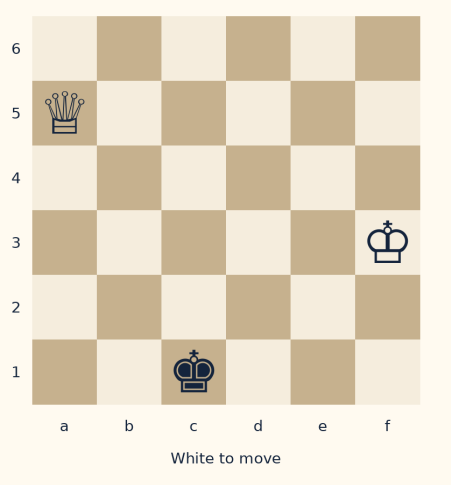


Figure 204: White to move.

Puzzle 093: Expert calculation

Objective: mate in 3. Record your full line. Answer.

Position: `6/Q5/6/5K/6/2k3 w - - 0 1`

Puzzle 094: Expert calculation

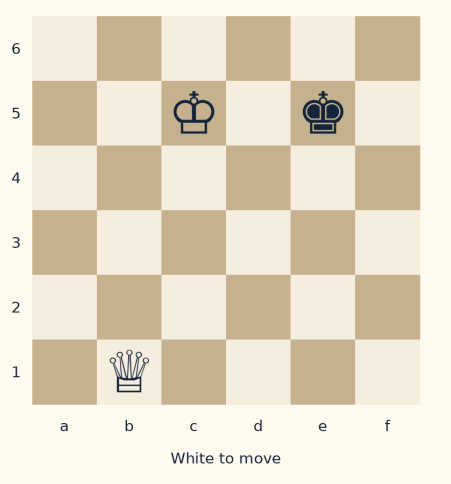


Figure 205: White to move.

Objective: mate in 3. Record your full line. Answer.

Position: `6/2K1k1/6/6/6/1Q4 w - - 0 1`

Puzzle 095: Expert calculation

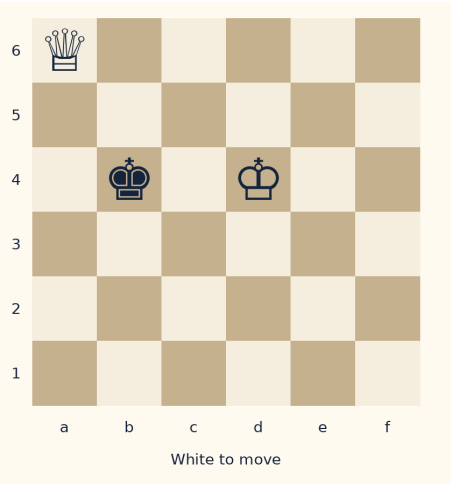


Figure 206: White to move.

Objective: mate in 3. Record your full line. Answer.

Position: `Q5/6/1k1K2/6/6/6 w - - 0 1`

Puzzle 096: Expert calculation

Objective: mate in 3. Record your full line. Answer.

Position: `6/4Q1/K5/6/6/2k3 w - - 0 1`

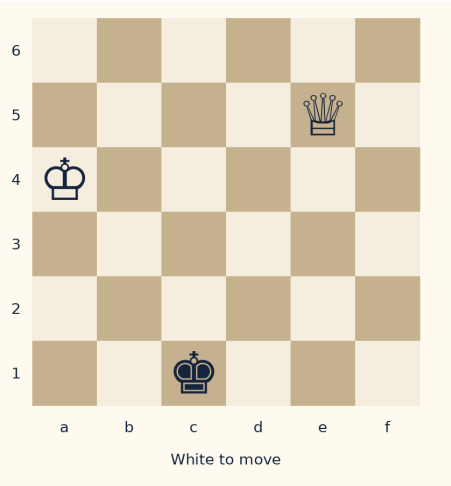


Figure 207: White to move.

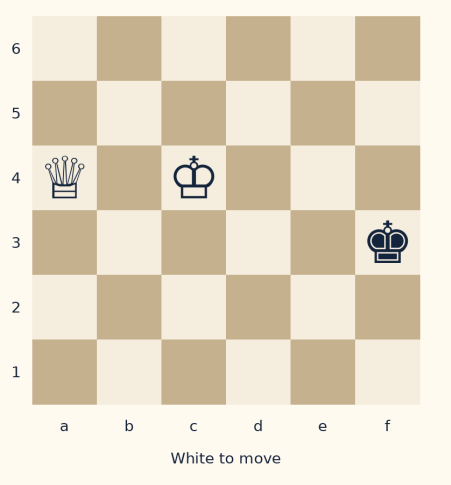


Figure 208: White to move.

Puzzle 097: Expert calculation

Objective: mate in 3. Record your full line. Answer.

Position: `6/6/Q1K3/5k/6/6 w - - 0 1`

Puzzle 098: Expert calculation

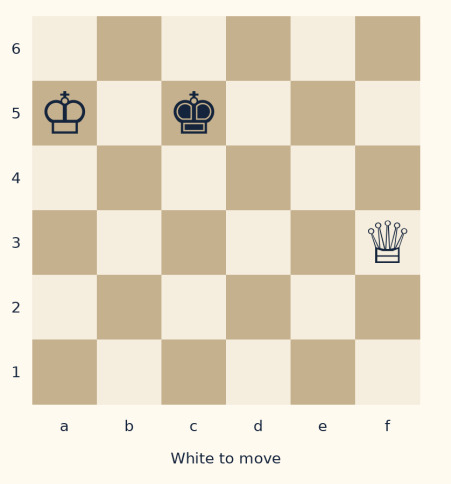


Figure 209: White to move.

Objective: mate in 3. Record your full line. Answer.

Position: `6/K1k3/6/5Q/6/6 w - - 0 1`

Puzzle 099: Expert calculation

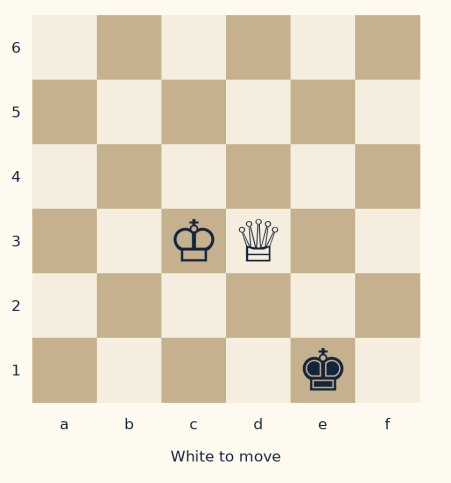


Figure 210: White to move.

Objective: mate in 3. Record your full line. Answer.

Position: `6/6/6/2KQ2/6/4k1 w - - 0 1`

Puzzle 100: Expert calculation

Objective: mate in 3. Record your full line. Answer.

Position: `4Q1/6/1K4/6/6/5k w - - 0 1`

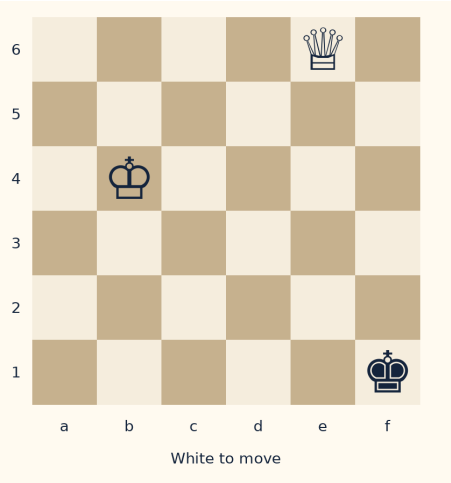


Figure 211: White to move.

Workbook Answer Key

The accepted move sets are exhaustive for the stated horizon. Representative lines show a legal defence; the proof checked all defences, not only that printed line.

Answer 001

Return to puzzle. **Accepted:** Ra6#. **Verified line:** Ra6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 002

Return to puzzle. **Accepted:** Rc1#. **Verified line:** Rc1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 003

Return to puzzle. **Accepted:** Qb1#. **Verified line:** Qb1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 004

Return to puzzle. **Accepted:** Qa3#. **Verified line:** Qa3#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 005

Return to puzzle. **Accepted:** Qa1#. **Verified line:** Qa1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 006

Return to puzzle. **Accepted:** Rf5#. **Verified line:** Rf5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 007

Return to puzzle. **Accepted:** Rd6#. **Verified line:** Rd6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 008

Return to puzzle. **Accepted:** Qb6#. **Verified line:** Qb6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 009

Return to puzzle. **Accepted:** Qa6#. **Verified line:** Qa6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 010

Return to puzzle. **Accepted:** Qe6#. **Verified line:** Qe6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 011

Return to puzzle. **Accepted:** Qb5#. **Verified line:** Qb5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 012

Return to puzzle. **Accepted:** Ra3#. **Verified line:** Ra3#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 013

Return to puzzle. **Accepted:** Qa1#. **Verified line:** Qa1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 014

Return to puzzle. **Accepted:** Qe2#. **Verified line:** Qe2#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 015

Return to puzzle. **Accepted:** Nb3#. **Verified line:** Nb3#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 016

Return to puzzle. **Accepted:** Qb5#. **Verified line:** Qb5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 017

Return to puzzle. **Accepted:** Qa1#. **Verified line:** Qa1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 018

Return to puzzle. **Accepted:** Ra1#. **Verified line:** Ra1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 019

Return to puzzle. **Accepted:** Ra1#. **Verified line:** Ra1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 020

Return to puzzle. **Accepted:** Qf5#. **Verified line:** Qf5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 021

Return to puzzle. **Accepted:** Rf1#. **Verified line:** Rf1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 022

Return to puzzle. **Accepted:** Rc6#. **Verified line:** Rc6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 023

Return to puzzle. **Accepted:** Qd1#. **Verified line:** Qd1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 024

Return to puzzle. **Accepted:** Rc6#. **Verified line:** Rc6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 025

Return to puzzle. **Accepted:** Qe4#. **Verified line:** Qe4#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 026

Return to puzzle. **Accepted:** Re6#. **Verified line:** Re6#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 027

Return to puzzle. **Accepted:** Qa4#. **Verified line:** Qa4#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 028

Return to puzzle. **Accepted:** Qb5#. **Verified line:** Qb5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 029

Return to puzzle. **Accepted:** Rf4#. **Verified line:** Rf4#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 030

Return to puzzle. **Accepted:** Qe5#, Qc6#, Qb6#. **Verified line:** Qe5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 031

Return to puzzle. **Accepted:** Kb3, Ra4. **Verified line:** Kb3 Ka1 Rc1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 032

Return to puzzle. **Accepted:** Kc5, Nd5. **Verified line:** Kc5 Ka5 Ra1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 033

Return to puzzle. **Accepted:** Kd4+. **Verified line:** Kd4+ Kf6 Qe5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 034

Return to puzzle. **Accepted:** Kc5. **Verified line:** Kc5 Ka5 Ra1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 035

Return to puzzle. **Accepted:** Kd2, Kd1. **Verified line:** Kd2 Kf1 Rf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 036

Return to puzzle. **Accepted:** Kc4. **Verified line:** Kc4 Kc6 Ra6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 037

Return to puzzle. **Accepted:** Ka3, Kb3. **Verified line:** Ka3 e3 Qb2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 038

Return to puzzle. **Accepted:** Kc5+. **Verified line:** Kc5+ Ka5 Qxa2#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 039

Return to puzzle. **Accepted:** Kd5, Kd6. **Verified line:** Kd5 Kf6 Rf4#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 040

Return to puzzle. **Accepted:** Kd4. **Verified line:** Kd4 f2 Qe5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 041

Return to puzzle. **Accepted:** Qa4, Qb4+, Qc5+. **Verified line:** Qa4 c2 Qb5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 042

Return to puzzle. **Accepted:** Ra4+, Rf1. **Verified line:** Ra4+ Kb1 Ra1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 043

Return to puzzle. **Accepted:** Qd4. **Verified line:** Qd4 b2 Qe3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 044

Return to puzzle. **Accepted:** Kc3, Na2. **Verified line:** Kc3 Kc1 Ra1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 045

Return to puzzle. **Accepted:** Kd3. **Verified line:** Kd3 Kd1 Rf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 046

Return to puzzle. **Accepted:** Re3. **Verified line:** Re3 Ka1 Ra3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 047

Return to puzzle. **Accepted:** Qe5+. **Verified line:** Qe5+ Kc6 Qc5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 048

Return to puzzle. **Accepted:** Qxb5+. **Verified line:** Qxb5+ Kd6 Qd5#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 049

Return to puzzle. **Accepted:** Kd6, Kd5. **Verified line:** Kd6 Kf6 Rf4#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 050

Return to puzzle. **Accepted:** Kxd5. **Verified line:** Kxd5 Kf5 Qf3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 051

Return to puzzle. **Accepted:** Kb3, Qxb4. **Verified line:** Kb3 Kb1 Qf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 052

Return to puzzle. **Accepted:** Kb3+, Kc2+, Qxe2. **Verified line:** Kb3+ Kb1 Qb2#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 053

Return to puzzle. **Accepted:** Kxb3+. **Verified line:** Kxb3+ Kb1 Qe1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 054

Return to puzzle. **Accepted:** Kc4. **Verified line:** Kc4 Ka5 Qb5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 055

Return to puzzle. **Accepted:** Kd2. **Verified line:** Kd2 Kf2 Rf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 056

Return to puzzle. **Accepted:** Kd4, Kd5. **Verified line:** Kd4 c1=Q Qe5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 057

Return to puzzle. **Accepted:** Kd1, Kd2. **Verified line:** Kd1 Kf2 Rf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 058

Return to puzzle. **Accepted:** Qxb2. **Verified line:** Qxb2 Ke1 Qe2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 059

Return to puzzle. **Accepted:** Re3. **Verified line:** Re3 Ka1 Ra3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 060

Return to puzzle. **Accepted:** Rc3. **Verified line:** Rc3 Ke6 Rc6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 061

Return to puzzle. **Accepted:** Kc6, Kc5. **Verified line:** Kc6 Ka6 Ra4#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 062

Return to puzzle. **Accepted:** Kc2. **Verified line:** Kc2 Ka2 Qa6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 063

Return to puzzle. **Accepted:** Rf2. **Verified line:** Rf2 Kd6 Rf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 064

Return to puzzle. **Accepted:** Kd5, Kd4. **Verified line:** Kd5 a2 Qe4#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 065

Return to puzzle. **Accepted:** Qb2+, Qc2+, Qa2+. **Verified line:** Qb2+ Kf3 Qe2#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 066

Return to puzzle. **Accepted:** Ke4. **Verified line:** Ke4 f3 Qe5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 067

Return to puzzle. **Accepted:** Qd4, Qe5. **Verified line:** Qd4 e3 Qc5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 068

Return to puzzle. **Accepted:** Kxd5. **Verified line:** Kxd5 Kf6 Qf2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 069

Return to puzzle. **Accepted:** Rd2. **Verified line:** Rd2 Kb6 Rd6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 070

Return to puzzle. **Accepted:** Qf2, Qd2. **Verified line:** Qf2 Kc1 Qc2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 071

Return to puzzle. **Accepted:** Qf2. **Verified line:** Qf2 d2 Qf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 072

Return to puzzle. **Accepted:** Kc5. **Verified line:** Kc5 Ka5 Ra3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 073

Return to puzzle. **Accepted:** Kf3. **Verified line:** Kf3 d4 Qa1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 074

Return to puzzle. **Accepted:** Kd4. **Verified line:** Kd4 c2 Qf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 075

Return to puzzle. **Accepted:** Rf5. **Verified line:** Rf5 Kd1 Rf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 076

Return to puzzle. **Accepted:** Kc5, Nc5+. **Verified line:** Kc5 Ka5 Ra1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 077

Return to puzzle. **Accepted:** Kd3. **Verified line:** Kd3 Kd1 Rf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 078

Return to puzzle. **Accepted:** Kc2. **Verified line:** Kc2 Ka2 Ra4#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 079

Return to puzzle. **Accepted:** Ke3, Kf3. **Verified line:** Ke3 Kf1 Rd1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 080

Return to puzzle. **Accepted:** Kc5. **Verified line:** Kc5 Ka5 Ra3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 081

Return to puzzle. **Accepted:** Kd5. **Verified line:** Kd5 Kf5 Rf3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 082

Return to puzzle. **Accepted:** Kc5. **Verified line:** Kc5 e3 Qa1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 083

Return to puzzle. **Accepted:** Ke4, Kd4. **Verified line:** Ke4 Kd6 Rb6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 084

Return to puzzle. **Accepted:** Kd3. **Verified line:** Kd3 Kf3 Rf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 085

Return to puzzle. **Accepted:** Kc4, Kb4. **Verified line:** Kc4 a1=Q Qb5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 086

Return to puzzle. **Accepted:** Kc6, Kc5. **Verified line:** Kc6 a3 Qxa3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 087

Return to puzzle. **Accepted:** Rc3. **Verified line:** Rc3 Kf5 Rf3#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 088

Return to puzzle. **Accepted:** Kd4. **Verified line:** Kd4 b2 Qf1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 089

Return to puzzle. **Accepted:** Kc3. **Verified line:** Kc3 Ka3 Ra5#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 090

Return to puzzle. **Accepted:** Kd3. **Verified line:** Kd3 Kc1 Ra1#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 091

Return to puzzle. **Accepted:** Kc2, Kc3. **Verified line:** Kc2 Kf3 Kd3 Kf4 Qf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 092

Return to puzzle. **Accepted:** Kb4. **Verified line:** Kb4 Kd6 Kc4 Kc6 Qe6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 093

Return to puzzle. **Accepted:** Qa2. **Verified line:** Qa2 Kd1 Qb2 Ke1 Qe2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 094

Return to puzzle. **Accepted:** Qe1+. **Verified line:** Qe1+ Kf5 Kd5 Kf6 Qf1#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 095

Return to puzzle. **Accepted:** Qb6+. **Verified line:** Qb6+ Ka4 Kc4 Ka3 Qb3#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 096

Return to puzzle. **Accepted:** Qe2. **Verified line:** Qe2 Kb1 Kb3 Kc1 Qc2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 097

Return to puzzle. **Accepted:** Kd3. **Verified line:** Kd3 Kf2 Qe4 Kf1 Qe2#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 098

Return to puzzle. **Accepted:** Qd3. **Verified line:** Qd3 Kc6 Kb4 Kb6 Qd6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 099

Return to puzzle. **Accepted:** Qe4+, Qb5, Qc2. **Verified line:** Qe4+ Kf1 Kd3 Kf2 Qe2#.

The first move checks: count all legal king moves, captures and interpositions. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

Answer 100

Return to puzzle. **Accepted:** Kc4, Kc3. **Verified line:** Kc4 Kf2 Kd3 Kf3 Qf6#.

The first move is quiet: locate the escape square or line it changes. At the final position the enemy king is attacked and has no legal reply. Reconstruct which piece controls each escape and protects the mating piece. Any unlisted first move misses this mate horizon; it is not necessarily losing the whole game.

PART X · Cheat Sheets

Rules

- 6×6; RNQKNR; no bishops, castling, double-step or en passant.
- Promote to Q/R/N. Mate wins; stalemate draws.
- Bare kings automatic; threefold/100 claimable; fivefold/150 automatic; mate takes precedence.

Opening

- Develop with purpose. Check your uncastled king before opening files.
- A stored shortlist is not proof. Calculate the opponent’s central capture.

Tactics

- Fork, pin, skewer, discovery: name the targets and legal replies.
- Deflection moves a defender; interference blocks its line.
- A sacrifice needs calculation, not a decorative label.

Calculation

- Checks, captures, threats and a quiet candidate.
- Remove captured pieces from the mental board.
- Reject with a refutation; accept only after the defence test.

King safety

- Trace open files, diagonals and knight jumps.
- Recount escapes after your proposed move.

Endgames

- Activate the king safely. Count promotion steps and interception squares.
- Check the clock and history; DTM does not override draw policy.

Tournament

- Confirm policy, format, tiebreak and outstanding fixtures.
- Separate sporting points, Career Elo, rank and titles.

Engine study

- Record build, budget, perspective and variation.
- Search scores are provisional. Keep assistance out of official play.

Exchange

- Read claim, payout and VOID. Include slippage and both fees.
- Expected score = P(win) + ½P(draw), not P(win).
- Review correlated losses and executable exit value.

Glossary

Terms are defined **as Crown & Calculation uses them** (variant crown-la6-v1, Federation crown-federation-1, Exchange crown-exchange-2). Where a term means something different in orthodox chess or in real-world markets, the entry says so. Pointers: **RB §n** = docs/RULEBOOK.md section n; **I.n** = Book I, Chapter n; **V.n / VI.n** = Book V or VI, Chapter n; **P-Wk n** = Twelve-Week Program, week n.

100-halfmove rule. If 100 consecutive halfmoves (50 by each side) pass without a capture or a pawn move, either player may **claim** a draw. At **150** halfmoves the draw is automatic. → RB §10; I.7.

Accuracy. 100·exp(−4 × mean expected-score loss) over a player's graded moves (crown-accuracy-v2), always between 0 and 100. The constant 4 is an uncalibrated placeholder, so accuracy is labelled experimental. → V.6.

Adaptive Training. Academy mode that picks puzzles from the themes where your first-try record is weakest, or from repeated engine-verified errors in your games. → P-Wk 5.

Agreement, draw by. A draw both players accept. Allowed only after each side has made at least ten moves. → RB §10.

Analysis presets. Quick (depth 5), Standard (depth 7, default) and Deep (depth 9, four candidate lines). The Lab combines only reports of one preset and methodology. → V.6.

Assessment (Held-out Assessment). Ten puzzles from the 81-puzzle held-out split, each shown to a player at most once and never used in practice. It measures; it does not train. → V.6; P-Wk 1.

b (liquidity parameter). The LMSR constant that sets how far prices move per share: match markets 120, tournament winner 150, promotion and qualification 80, season title and bracket 100, Sandbox 60. → VI.2.

Bare kings. Only the two kings remain. This is an **automatic draw**, the only insufficient-material case drawn automatically in Crown. → RB §10; V.5.

Blitz / Rapid / Classical / Untimed. Time controls 3+2, 5+3, 15+10 (minutes + seconds per move) and none. Each timed control has its own rating pool. Untimed games are never rated. Federation games use Rapid 5+3. → RB §13.

Bookmark / Memorable move. Analysis markers that save a ply of a game for later study. → V.6.

Bounded selection. How engine levels L1–L4 choose: at random (softmax with a temperature) among the moves whose score is within a fixed margin of their own best (L1 320 cp … L4 70 cp). It is never used when a mate is found. → V.3.

Bracket market. An Exchange family over the 8 possible final-four paths of the Crown Championship. Claims (champion, finalist, "X beats Y in the final") are sets of paths. A conjunction claim pays $0 if its condition fails. → VI.5.

Brier score. The game's forecast score: $B = \Sigma$ over all states of $(p - y)^2$, from 0 (perfect) to 2 (worst); lower is better. For a two-state market this is twice the usual binary Brier. → VI.3.

Brilliant / Great. Special grades for a best move that is clearly better than every alternative (Great), and that also invests verified material (Brilliant). **Not validated**; Brilliant-move achievements wait for validated grading. → V.6.

Buchholz / Buchholz Cut-1. Swiss tiebreaks: the sum of your opponents' scores, and that sum minus the lowest one. → V.4; RB §16.

Calculation Chamber. Academy mode with puzzles needing two or more of your own moves against best defence. → V.1.

Calibration. Agreement between stated probabilities and observed frequencies. The game shows ten probability bins, each with a Wilson 95 % interval. (*Calibration* also names the engine-strength measurement in src/engine/calibration.json.) → VI.3.

Cancellation (market). Resolving a market without a result. Holders receive the published cancellation vector: 1/n per share for most families, |S|/n for a claim on S states, and the opening prices for Sandbox markets. Cancelled markets are not scored as forecasts. → VI.1; RB §20.

Candidate move. A move you seriously consider. Order: checks, captures, threats, then quiet moves. → V.1.

Centipawn (cp). One hundredth of a pawn: the unit of engine evaluation, from the side to move. → V.6.

Checkmate. The side to move is in check and has no legal move. It takes precedence over every draw rule. → RB §8; I.6.

Circuit Grade. A rank of the 16-grade feudal ladder (crown-circuit-grades-1, names proposed), each with sublevels III → II → I: 48 steps from Commoner III to High Sovereign I, two grades per division. Set by your division and by the standing score at two checkpoints a season (after round 8 and at season end), with hysteresis. Your highest rank ever is permanent. Not a rating, not a Federation title. → V.4.

Claim (draw). A request by a player for a draw that is not automatic: threefold repetition or the 100-halfmove rule. → RB §10.

Claim (Exchange). A tradable set of a market's states. A share of the claim pays the sum of the payouts of its states. → VI.2.

Closed (market). No trading is possible; the market awaits settlement or cancellation. Positions in a closed market are valued only at the cancellation vector. → VI.4.

Coach note. An evidence-based explanation attached to an analysed move (coach crown-coach-0), under topics such as Hanging pieces, Knight forks, Mating patterns, Threat awareness, Promotion races and Calculation. → V.5.

Conflict of interest (Exchange). Official markets never list your own games. While you sponsor a competitor, you cannot buy in markets involving them. Entering an event cancels its winner market at 1/n first. → VI (introduction); RB §20.

Critical moment. A position where the best move is at least 0.10 expected score better than the second best, or where a mate is available (crown-critical-1). The summary counts how many you *handled* (loss ≤ 0.05). → V.5.

Crown scale. The engine-strength rating scale, anchored at L1 = 800 by design. It is not FIDE. → V.3; V.8.

Davidson model. The Federation result model: P(W) ∝ π_w, P(D) ∝ ν√(π_wπ_b), P(B) ∝ π_b, with π = 10^(rating/400) and ν = 0.9 (crown-result-model-1). Used on public ratings for forecasts. → VI.1.

Defensive Survival. Academy mode: find the only move that holds. → V.5.

Division. One of 8 ordered tiers of the Federation: 1 Crown, 2 Grand, 3 Premier, 4 Masters, 5 Senior, 6 District, 7 Club, 8 Open. Each division has 16 regional leagues. → V.4; RB §17.

Endgame Workshop. Academy mode of endgame puzzles, which can then be played out against the engine. → P-Wk 7.

Expected score. The average result (win 1, draw ½, loss 0) a position or pairing is worth. For positions the game uses the map crown-outcome-map-v2 (for example +100 cp ≈ 0.68). For pairings it uses the Davidson model. → V.6; VI.1.

Expected value (EV). The average profit of a trade at your probability: belief × $1 × shares − total paid. → VI.1.

Exposure. The total cost basis at risk in open or closed markets. Correlated positions count together. → VI.4.

Featured game. A Federation engine game chosen for a match market and live spectating. → VI.6.

Fee. 1 % of the executed amount (rounded up) on every buy and every sell. → VI.1.

Federation. The Six-Square Federation: 2,048 AI competitors in 16 regions × 8 divisions × 16-player leagues, with 19 rounds per season. → V.4; RB §§14–17.

Federation title. A permanent honour earned by performance norms: Squire, Knight, Paladin and Sovereign (crown-titles-1). Not a FIDE title, and distinct from rating titles. → V.4; RB §21.

Fivefold repetition. The same position (board and side to move) for the fifth time: an **automatic draw**. → RB §10.

fullBudget. An engine option (since v0.8) for L8 and L9: keep starting search iterations until 62.5 % of the move's time budget is used. Measured gain: +23 [6, 41] Elo. → V.8.

Glicko-2. The rating system for players and competitors (rating, RD, volatility). New players start at 1500, RD 350. → RB §15.

Grade. The label for a move's expected-score loss (crown-grade-v3-experimental): Best ≤ 0.002, Excellent ≤ 0.02, Good ≤ 0.05, Inaccuracy ≤ 0.12, Mistake ≤ 0.25, Blunder above. The bins are experimental. → V.6.

Hedging. Reducing the spread of possible outcomes, in this game by selling shares you hold (no short selling). It costs fees and slippage. → VI.5.

Highest rank ever achieved. The best Circuit Grade (with sublevel) you have held, with the season it was first reached. Only a genuine step up raises it; demotion, relegation and rating losses never lower it. → V.4.

Implied probability. The probability a price expresses: a $0.40 share implies 40 %. → VI.1.

Kelly criterion. The growth-optimal stake fraction f* = (q − p)/(1 − p) for a $1 claim at price p with belief q. Use fractional Kelly only as a ceiling, never as a target. → VI.4.

Knockout. Crown Championship format: two-game matches, then a rapid pair, then a blitz pair. If still level, the higher seed advances, except in the final, which gives joint champions. → V.4; RB §16.

Lab (Personal Chess Intelligence Lab). Skill dimensions measured as success rates with denominators and 95 % intervals (crown-lab-1). They are performance summaries, not psychometric scores. → V.6.

Latent strength. A competitor's hidden playing strength, which chooses their engine profile. It is never shown, and never used by forecasts or NPC traders. → V.3.

Level (L1–L9). Engine strength profiles (crown-levels-6): L1 Novice … L8 Crown, L9 Sovereign Unbound. → V.3.

Live model. The in-play W/D/B forecast computed from the published engine evaluation and the pre-game draw probability. It is never the price. → VI.6.

LMSR (logarithmic market scoring rule). The automated market maker: $C(q) = b \cdot \ln \Sigma\, e^{(q_i/b)}$, $p_i = e^{(q_i/b)} / \Sigma\, e^{(q_j/b)}$, and a trade costs $C(q') - C(q)$. → VI.2.

Log score. $-\Sigma\, y \cdot \ln \max(p, 0.001)$; lower is better. → VI.3.

Long-only. You can sell only shares you hold. There is no short selling. → VI.5.

Market family. A kind of Exchange market: match, winner, promotion, qualification, season, bracket, sandbox. → VI.5.

Market revision. A counter that advances with every trade and every published live move. A quote from an older revision is rejected. → VI.2; VI.6.

Norm. An event of at least 5 games whose performance rating and average opposition reach a title tier's thresholds. → V.4.

NPC trader. One of eight simulated traders (crown-npc-traders-2) that trade on public information only, with documented noise, bias, margins and budgets. → VI.7.

Opening book (benchmark). The 64 two-ply openings in tools/bench/openings.json, each within ±80 cp at Fairy-Stockfish depth 10. → V.7.

Paper forecast. A free probability vector over a market's states, locked at market close and scored on settlement. → VI.3.

Performance rating. Average opponent rating + dp, with dp = $-400 \cdot \log_{10}(1/p - 1)$ for score fraction p (±800 cap). → V.4.

Played Like. An estimate of the rating at which a game was played. Shown as **"Calibrating"**: the model failed its release gate (no better than a naive baseline). → V.6.

Playing-length limit. Featured engine games stop at 400 plies. Such a game is *censored*: its market is cancelled, and the standings record an adjudicated draw, labelled as such. → VI.6.

Promotion (pawn). A pawn reaching the far rank becomes a queen, rook or knight (there are no bishops). It is only four steps from the starting rank. → RB §11; I.9.

Promotion and relegation (league). At season end, the top two of each regional league move up a division and the bottom two move down. → V.4; RB §17.

Provisional rating. A rating with fewer than 20 games or RD > 100. It is excluded from peaks and title collections. → RB §15.

Puzzle Rush. Academy mode: 3 minutes, three strikes. Any solve that is not clean counts as a strike. → P-Wk 2.

Rating pool. A separate rating per timed control (Blitz, Rapid, Classical). → RB §15.

Rating title. A descriptive band of 100 rating points (31 bands, crown-rating-titles-1), from *Wandless Wanderer* (0–99) to *Eternal Sovereign* (3000+). It is not a Federation title. → V.4; RB §15.

RD (rating deviation). Glicko-2's uncertainty in a rating. The higher it is, the less the rating means. → V.3.

Regional league. One of the 16 leagues of 16 competitors within each division; a 15-round round robin. → V.4.

Replay Your Mistakes. Academy mode built from your analysed moves that lost ≥ 0.12 expected score. → V.6.

Repertoire (style). The opening preference of a validated style: a choice among book moves within 30 cp of the best at depth 7, for the first six plies only. → V.3.

Retry from This Position (Unrated). Starts an untimed training game from the position before an analysed move, against the same opponent. → V.6.

Sandbox Exchange. Practice markets with separate practice money on your own casual quick matches. You may only back yourself, and trading closes at the first move. → VI (introduction); RB §20.

Settlement. Resolving a market on its result: each winning share pays $1 (joint champions split). → VI.1.

Share. A unit of a claim that pays $1 if the claim's state occurs. → VI.1.

Slippage. The gap between the price shown and the average price paid, caused by your own trade moving the price. → VI.2.

Sonneborn–Berger. A tiebreak: the sum of the scores of the opponents you beat, plus half the scores of those you drew. → V.4.

Sovereign Unbound (L9). The canonical engine with no handicap and a chosen budget of 1/5/10/15/30/60 s per move (rated: 10 s). Rated 3038 ± 171 (derived). → V.8.

Stalemate. The side to move is not in check and has no legal move: a draw. → RB §9; I.7.

Standing score. 0–100 at a Circuit Grade checkpoint: 45 % league position, 25 % score in sanctioned rated games, 30 % rating percentile in your division. → V.4.

Style. A competitor's label (Universal, Tactical, Positional, Defensive, Endgame, Aggressive). Only validated styles change play: Tactical, Positional, Defensive and Endgame, through a whole-game styled evaluation (Tactical and Positional also through an opening repertoire). Aggressive is not validated and plays as Universal. → V.3.

Sublevel. The III / II / I part of a Circuit Grade (III lowest). Shown by one, two or three chevrons under the grade's insignia. → V.4.

Subsidy. The maker's worst-case loss b·ln n, funded before a market opens (match market: $131.83). → VI.2.

Swiss. A pairing system matching players on similar scores who have not met (crown-swiss-1, simplified). → V.4; RB §16.

Threefold repetition. The same position for the third time. The draw is **claimable**, not automatic. → RB §10.

Timeout rule. If you flag, you lose if your opponent has any non-king piece or pawn; otherwise the game is a draw. → RB §13; V.2.

Training game. An unrated game (for example a retry). It is excluded from the Lab's game evidence. → V.6.

Turning point. A ply where the evaluation crosses between better (> 0.65), balanced and worse (< 0.35). → V.5.

Zugzwang. A position where any move worsens the mover's position. → V.1.

Topic index

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Edition and verification notes

Rebuilt Academy edition, 6 November 2026. Edited source: the owner’s Sovereign’s Grimoire and published crown-la6-v1 code at commit 5126c1262a002260116a8bdcaf60bbe1d46a22cc. The composition records and verification logs accompany the editable source ZIP. This is a replacement edition, not recovery of the lost earlier file. No human rating or financial return is promised.